Exogenous Features in Intraday Forecasts of Closing-Half-Hour Variance

Bullet-point draft of the full paper

James Chen Mihai Cucuringu Christopher S. Jones Austin Pollok

Working draft, 23 October 2026 — not for citation

Abstract

We investigate whether exogenous predictors improve intraday forecasts of a futures-based (E-mini) proxy for the realized variance of the S&P 500 over the closing half-hour interval (15:30–16:00 ET). Specifically, we estimate pooled models on the 13 regular-hours half-hour bars of each trading day with a rolling window of 3,000 trading days and evaluate the forecast issued at 15:30 for the closing half hour over the out-of-sample period from 3 January 2020 to 30 April 2024 ($T_{\text{oos}} = 1{,}082$ trading days), and the forecasts of every regular-hours bar from 2 January 2014 to 30 April 2024 (2,579 trading days). We compare a benchmark heterogeneous autoregressive (HAR) predictor set against an augmented predictor set that incorporates exogenous series. Under the QLIKE loss, scored on every regular-hours bar, the full predictor set yields a lower average out-of-sample loss than the HAR benchmark in all 6 intraday model classes, each reduction statistically significant at the 5% level. For the forecast of the closing half hour, it lowers the loss in 3 of the 6 classes, the three tree ensembles, none of these reductions statistically significant at the 5% level, and raises it for the three penalized linear models. Furthermore, although every intraday specification remains less accurate under QLIKE than a HAR-type model in the spirit of Corsi (2009), estimated by OLS on the closing-interval observations of the target's trailing averages over $k \in \{1, 5, 25, 125, 625, 3125\}$ samples plus calendar columns, incorporating the exogenous predictors narrows this gap in 3 of the 6 model classes, the tree ensembles. In an economic application to zero-days-to-expiration (0DTE) S&P 500 straddles on expiration days, a sign-based trading strategy achieves a higher Sharpe ratio under the full predictor set than under the HAR benchmark in every one of the 6 intraday model classes, in point estimates; the gain is statistically significant at the 5% level in 1 of them.

Notes. Colored notes are draft comments, not paper text.
Red: two live statements disagree.
Orange: a sentence does not match the computation behind a printed number.
Blue: a reader cannot recover the input or the rule.
The first line is the decision. The second line is the problem. The rest is the exposition.

E34. Decision: one design for the 12 pooled intraday forecasts (user, 2026-10-22), now the final rerun (decided 23 October 2026; writeup/final_run.py, generators switched with REHAUL_FINAL_RUN=1): the 13 regular-hours bars, the base-5 HAR horizons, the 564 columns of the full predictor set, a rolling window of 3,000 trading days ($W = 39{,}000$ bars) re-estimated at every bar, evaluation from 2 January 2014; the linear penalties chosen once on the last pre-sample fold (validation year 2013) and then frozen, and each tree method on its frozen configuration from the earlier Optuna search, not re-tuned. Inputs: the linear models from results/window_sweep_2014start_2026-10-22/w3000; LightGBM and XGBoost from results/pooled_trees_final_2014start_w3000; the random forest, fitted on its frozen configuration with Intel's oneDAL engine (Intel Extension for Scikit-learn), from results/pooled_rf_skx_2014start_w3000 (RULE_rf_skx.txt). The abstract, the introduction and the conclusion state counts over these forecasts (note N15c, resolved 23 October 2026).
Problem: the model set (the 112 forecasts of the master table) still holds earlier designs, the earlier draft's pooled 48-interval forecasts (the "paper" family, among them its LightGBM and XGBoost) and the master table's pooled linear runs on all 48 intervals with $W = 96{,}000$ intervals, so the statistics computed over the model set in the appendix include them. Section 7 no longer uses it: since 23 October 2026 the interval-shift test and the compounded wealth run on the 12 pooled forecasts (writeup/make_rehaul_pooled_extras.py; notes E26, E27, N15b resolved).
Robustness: the same runs at a window of 3,500 trading days are in Table 7. Pending: the pooled MinGRU (parked).

1 Introduction

- **Research question.** We investigate whether incorporating exogenous predictors improves intraday forecasts of a futures-based (E-mini) proxy for the realized variance of the S&P 500 over the closing half-hour interval.
  - Specifically, at the 15:30 ET forecast origin on trading day t, we seek to predict RV_t, which denotes the realized variance over the 15:30–16:00 closing interval (see Section 3).
  - Each pooled intraday model is estimated on the 13 regular-hours half-hour bars of the trading day, and we evaluate its out-of-sample predictions on the forecast issued at 15:30 for the closing half hour.
  - To isolate the contribution of the additional covariates, we compare the full predictor set against the HAR benchmark; the exogenous series constitute the exact difference between the two sets (see Section 4).
- **Motivation for the HAR baseline.** Following Corsi (2009), the standard HAR regression models realized volatility solely as a linear function of its own past lags. Consequently, it omits a rich set of exogenous variables available by 15:30, including higher-order realized moments, liquidity measures, the cross-section of constituent returns, news and social-media sentiment, option-implied volatility, volatility demand indicators, and scheduled macroeconomic and earnings announcements.
- **Empirical design.** For each intraday interval (t, i), we consider a single forecast mapping of the form
$$\hat{y}_{t,i} = f_{m,\mathcal{X},\mathcal{I}}(\boldsymbol{x}_{t,i}),$$
indexed by the model class m and the predictor set $\mathcal{X} \in \{\mathrm{HAR}, \mathrm{Full}\}$.
  - The primary empirical results reported in Section 6 focus on the pooled intraday estimation scheme, $\mathcal{I} = \{1, \ldots, M\}$.
  - As our primary benchmark forecast, we employ the ordinary least squares (OLS) regression on the HAR benchmark predictor set, estimated exclusively on the closing interval (the closing-interval HAR benchmark; see Sections 5.2 and 5.3).
  - We evaluate predictive accuracy via the average QLIKE loss over the out-of-sample

evaluation span $\mathcal{T}_{\text{oos}}$ from 3 January 2020 to 30 April 2024 (T_{oos} = 1,082 trading days; see Section 5.3); the accuracy of the pooled linear models and tree ensembles in Table 1 is scored on every regular-hours bar from 2 January 2014 to 30 April 2024 (2,579 trading days, a rolling window of 3,000 trading days).

- **Main findings.**
  - Scored on every regular-hours bar (Table 1), incorporating the full predictor set reduces the average out-of-sample QLIKE loss relative to the HAR benchmark in all 6 intraday model classes, each reduction statistically significant at the 5% level. For the forecast of the closing half hour, it reduces the loss in 3 of the 6 classes, the three tree ensembles, none of these reductions statistically significant at the 5% level, and raises it for the three penalized linear models, significantly so for ridge regression and the lasso (see Section 6.3).
  - Every pooled intraday forecast incurs a higher average QLIKE loss than the closing-interval HAR benchmark. Under the full predictor set, this performance gap is statistically significant at the 5% level for 3 of the three linear models and for 1 of the three tree ensembles. Augmenting the intraday models with the full predictor set narrows this gap in 3 of the 6 model classes, the tree ensembles, and widens it for the penalized linear models (see Section 6.4).
  - In the variable importance of the pooled models (Section 6.5), the non-standard series, those outside the HAR components, the futures' own realized moments and trading activity, and the VIX, hold 43.7–45.9% of the tree ensembles' exogenous attribution, less than in the penalized linear models (56.0–73.9%). The trees' attribution is concentrated: half of each tree's exogenous attribution sits in 4 series, against 11 for ridge regression. It also depends on the day: the trees' non-standard share is higher on FOMC statement days than on other days.
- **Economic application.** To assess the economic value of the forecasts on trading days with a same-day SPXW expiration, we construct a trading strategy that takes a long or short position in a zero-days-to-expiration (0DTE) at-the-money straddle based on the sign of the difference between the variance forecast and option-implied variance (see Section 7). Using the full predictor set increases the annualized Sharpe ratio of this strategy relative to the HAR benchmark in every one of the 6 intraday model classes, in point estimates, and significantly at the 5% level in 1 of them (see Section 7.3).
- **Paper organization.** The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and target construction, and Section 4 details the benchmark and exogenous predictors. Section 5 presents the forecasting models and rolling estimation protocol. Section 6 reports the out-of-sample forecasting results and, in Section 6.5, the variable importance of the pooled models. Section 7 evaluates the economic value of the forecasts in the 0DTE straddle strategy, and Section 8 concludes.

2 Related work

- **Realized variance and the HAR model.** Following Corsi (2009), the heterogeneous autoregressive (HAR) model specifies the conditional expectation of next-day realized variance as a linear function of past realized variances aggregated over daily, weekly, and monthly horizons: $\mathbb{E}_t[\mathrm{RV}_{t+1}^{(d)}] = \beta_0 + \beta_d \,\mathrm{RV}_t^{(d)} + \beta_w \,\mathrm{RV}_t^{(w)} + \beta_m \,\mathrm{RV}_t^{(m)}$, where $\mathbb{E}_t$ denotes the conditional expectation given information available at the forecast origin on day t, and $\mathrm{RV}_t^{(d)}$, $\mathrm{RV}_t^{(w)}$, and

$\mathrm{RV}_t^{(m)}$ denote the average daily realized variance over the preceding 1, 5, and 22 trading days, respectively. Because all three components represent daily realized variances aggregated at the daily, weekly, and monthly horizons, forecasting the realized variance of a specific intraday interval lies outside this standard daily specification.
See Corsi (2009), Andersen et al. (2003), Barndorff-Nielsen and Shephard (2004), Andersen et al. (2007), Corsi et al. (2008), Bollerslev et al. (2016), Bekierman and Manner (2018), Bennedsen et al. (2022), Hansen et al. (2012), and Andersen et al. (2006).

  - In a multivariate setting, the panel HAR framework of Bollerslev et al. (2018) estimates a single pooled regression jointly across a cross-section of assets.
  - For the theoretical foundations of realized variance, see Barndorff-Nielsen and Shephard (2002).

- **Diurnal pattern.** It is well documented that intraday return volatility exhibits a pronounced U-shaped pattern across the trading session. To account for this periodicity, standard practice scales intraday realized variance by a diurnal factor estimated from the corresponding clock-time interval on prior trading days.
See Andersen and Bollerslev (1997) on intraday periodicity, Engle and Gallo (2006) on intraday range and volatility measures as indicators of daily volatility, and Hizmeri and Bevilacqua (2025) on morning volatility uncertainty.

  - See also Taylor and Xu (1997) on diurnal volatility adjustments and Engle and Sokalska (2012) on multiplicative decompositions of intraday volatility.

- **0DTE options and the variance risk premium.** The variance risk premium, defined as the difference between option-implied variance and expected realized variance, $\mathbb{E}_t^{\mathbb{Q}}[\mathrm{RV}_t] - \mathbb{E}_t^{\mathbb{P}}[\mathrm{RV}_t]$, under the risk-neutral ($\mathbb{Q}$) and physical ($\mathbb{P}$) measures (where RV_t denotes the realized variance through expiration), is empirically positive on average.
See Hizmeri and Bevilacqua (2025), as well as the foundational option-pricing frameworks of Black and Scholes (1973) and Merton (1973).

  - On zero-days-to-expiration (0DTE) options, see Bandi et al. (2023), Beckmeyer et al. (2023), Brogaard et al. (2023), Dim et al. (2024), Vilkov (2026), and Almeida et al. (2025); on the variance risk premium more broadly, see Carr and Wu (2009) and Bollerslev et al. (2009).

- **Machine learning for volatility and returns.** A growing literature models conditional return and volatility dynamics via flexible nonlinear mappings $f(\boldsymbol{x}_t)$ of a high-dimensional predictor vector $\boldsymbol{x}_t$.
See Gu et al. (2020) for return prediction and Christensen et al. (2023) for realized volatility forecasting; for gradient-boosted decision trees, see Ke et al. (2017) (LightGBM) and Chen and Guestrin (2016) (XGBoost); for Bayesian hyperparameter optimization, see Akiba et al. (2019) (Optuna).

  - For ensemble methods, see Breiman (2001) on random forests and Friedman (2001) on gradient boosting; for machine-learning applications to realized volatility and intraday commonality, see Bucci (2020), Rahimikia and Poon (2020), and Zhang et al. (2024).

- **Shrinkage with many weak signals.** In high-dimensional linear regressions, penalized estimators regularize the coefficient vector via

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda(\alpha\|\boldsymbol{\beta}\|_1 + \tfrac{1-\alpha}{2}\|\boldsymbol{\beta}\|_2^2)$$

for a response vector $\boldsymbol{y}$, a predictor matrix $\boldsymbol{X}$, and an overall penalty strength λ, nesting

ridge regression ($\alpha = 0$) and the lasso ($\alpha = 1$) as special cases; multi-penalty ridge extensions assign distinct regularization weights across feature blocks via $\lambda \sum_j \omega_j \beta_j^2$, where the penalty weight ω_j is held constant within each block.
See Giannone et al. (2021) on dense versus sparse representations, Shen and Xiu (2025) on learning from many weak signals, Kozak et al. (2020) on shrinking the cross-section of returns, Hoerl and Kennard (1970) on ridge regression, Tibshirani (1996) on the lasso, Zou and Hastie (2005) on the elastic net, and van de Wiel et al. (2021) and van Wieringen (2023) on multi-penalty ridge estimation.

  - See also Zou (2006) on the adaptive lasso and feature-specific penalty weights, and Friedman et al. (2010) on coordinate-descent estimation under the elastic-net parametrization.

3 Data

- **Data sources.** We group the exogenous covariates into eight distinct categories. Specifically, the pooled intraday specifications employ the 46 series across all categories, the FOMC group included. A complete list and definition of each individual series is provided in Appendix Table 13.
  - **Realized moments** (7 series). We obtain intraday data on the front-month E-mini S&P 500 futures contract (defined as the most actively traded contract), which trades nearly 24 hours a day from 1998 through 2024, aggregated onto a 30-minute grid. Here, $t = 1, \ldots, T$ indexes trading days and $i = 1, \ldots, M$ indexes the half-hour intervals within day t, each ending on a half-hour timestamp; in particular, $i = c$ denotes the 15:30–16:00 ET closing interval. Each realized moment for interval (t, i) is computed by summing the corresponding one-minute quantity across all non-missing minutes within that half hour.
  - Let $r_{t,i,k}$ denote the one-minute log-price change of the front-month contract for the k-th non-missing minute in interval i of day t, where $k = 1, \ldots, N_{t,i}$ and $N_{t,i} \leq 30$. Following Andersen et al. (2003), we define the realized variance over interval (t, i) as

$$\mathrm{RV}_{t,i} = \sum_{k=1}^{N_{t,i}} r_{t,i,k}^2.$$

  - We construct the 7 exogenous realized-moment series as analogous one-minute sums over interval (t, i) (note that $\mathrm{RV}_{t,i}$ above serves as our forecast target and is not counted among the exogenous series):

$$\texttt{sumret}_{t,i} = \sum_k r_{t,i,k}, \quad \texttt{sumabsret}_{t,i} = \sum_k |r_{t,i,k}|, \quad \texttt{sumret3}_{t,i} = \sum_k r_{t,i,k}^3,$$
$$\texttt{sumret4}_{t,i} = \sum_k r_{t,i,k}^4, \quad \texttt{sumpret2}_{t,i} = \sum_k r_{t,i,k}^2 \, \mathbf{1}\{r_{t,i,k} > 0\},$$
$$\texttt{sumbipow}_{t,i} = \sum_{k \geq 2} |r_{t,i,k}| \, |r_{t,i,k-1}|, \quad \texttt{sumautocov}_{t,i} = \sum_{k \geq 2} r_{t,i,k} \, r_{t,i,k-1}.$$

    Specifically, the third sum corresponds to the unnormalized realized skewness of Amaya et al. (2015) (that is, prior to dividing by $\mathrm{RV}_{t,i}^{3/2}/\sqrt{N_{t,i}}$), and the fourth sum is the unscaled realized quarticity (without the factor $N_{t,i}/3$). Furthermore, $\texttt{sumpret2}_{t,i}$ denotes the positive realized semivariance of Barndorff-Nielsen et al. (2010), $\texttt{sumbipow}_{t,i}$

denotes the unscaled bipower variation of Barndorff-Nielsen and Shephard (2004) (without the $\pi/2$ normalization factor), and $\texttt{sumautocov}_{t,i}$ denotes the first-order realized autocovariance.

– **Liquidity** (14 series). From the front-month futures contract, we include total contract volume traded during the half hour as well as the number of observed one-minute intervals $N_{t,i}$ (`numobs`). In addition, using intraday data on S&P 500 constituents from 1993 to 2024, we construct both equal-weighted and value-weighted cross-sectional aggregates of share turnover, buyer-initiated turnover, seller-initiated turnover, the effective spread, the quoted spread, and order-flow imbalance. Here, constituent turnover is defined as the number of shares traded during the half hour divided by prior-day shares outstanding, while buyer- and seller-initiated turnover aggregate trades according to their estimated trade direction. The effective spread measures the average percentage deviation between the transaction price and the prevailing mid-quote, and the quoted spread measures the average percentage bid–ask spread. Equal-weighted aggregates take simple cross-sectional averages across constituents, whereas value-weighted aggregates weight each constituent by its prior-day market capitalization. For each weighting scheme (`ofi_ewstock` and `ofi_vwstock`), we define order-flow imbalance as

$$\mathrm{OFI}_{t,i} = \frac{\mathrm{Buy}_{t,i} - \mathrm{Sell}_{t,i}}{\mathrm{Buy}_{t,i} + \mathrm{Sell}_{t,i}} \in [-1, 1], \qquad \text{missing if } \mathrm{Buy}_{t,i} + \mathrm{Sell}_{t,i} = 0,$$

where $\mathrm{Buy}_{t,i}$ and $\mathrm{Sell}_{t,i}$ denote the aggregate buyer- and seller-initiated turnover in interval i of day t, respectively.

– **Equal-weighted market** (6 series) and **value-weighted market** (6 series). For each weighting scheme described above, we compute six cross-sectional constituent return moments: realized variance, realized skewness, realized quarticity, absolute variation, bipower variation, and positive semivariance. In contrast to the futures moments, which are constructed from one-minute log-price changes, these constituent moments are computed from 5-minute log returns. Moreover, the constituent moment groups omit signed returns and realized autocovariance, whereas the exogenous futures moment group omits realized variance (which serves as the target variable).

– **Sentiment** (3 series). Using StockTwits message streams beginning in 2009 for total posts and 2010 for user-tagged posts, we construct three half-hourly indicators: investor attention, defined as the total number of posts within the half hour; a sentiment score that equals 1 when all tagged posts are bullish and -1 when all tagged posts are bearish; and the sentiment count, defined as the number of posts carrying either a bullish or bearish tag.

E30. Decision: open, for the professor.
Problem: the text gives only the ± 1 endpoints of the sentiment score.
Question for the professor: what is the formula of the score between the endpoints, and what value does it take on intervals with no tagged posts? The repository receives the score precomputed (data/spy_and_sentiment.parquet, column stocktwits_sentiment, first observed in June 2010); src/data/loading.py:280 reads it as a net score in $[-1, 1]$, and no script in src/ or experiments/ computes it.

– **Implied volatility** (3 series). We include three Cboe volatility indices: the 30-day S&P 500 implied volatility index (VIX, available from 2006), the three-month implied volatility index (VIX3M, from 2009), and the implied volatility of the VIX (VVIX, from 2012), all extending through February 2024.

– **Volatility demand** (4 series). Using Cboe option-volume records from 2011 through

August 2023, we measure net vega purchased by non-market-maker participants across four series formed by crossing two option universes—options on the S&P 500 index and top S&P 500 exchange-traded funds versus all exchange-traded options—with two transaction subsets, namely all opening and closing customer trades versus opening customer trades only.

- **FOMC** (3 series; used by the pooled intraday model and by the closing-interval specifications of Appendix G). Based on scheduled Federal Open Market Committee announcement timestamps from 1993 through November 2023, we define 3 series: an indicator for whether an announcement occurs on day t and two proximity measures $1/(1+n^{+}_{t,i})$ and $1/(1+n^{-}_{t,i})$, where $n^{+}_{t,i}$ and $n^{-}_{t,i}$ denote the number of intervals until the next scheduled announcement and since the most recent scheduled announcement, respectively. Because the meeting schedule is announced in advance, these variables are known prior to the start of the interval.
- **Ends of the feeds.** Our data end on 30 April 2024, but three feeds stop earlier. The file of the implied volatility indices ends on 12 February 2024. For the remaining 54 trading days we splice in the hourly closes of the free Yahoo Finance feed for the same three indices, each written at the end of its hour and carried to the next half-hour stamp, on the stamps at which the file prints each index. On the stamps both sources carry, from December 2023 to 12 February 2024, the two print the same value to the second decimal on 96% (VIX), 88% (VIX3M), and 100% (VVIX) of the stamps. The FOMC release file ends with the statement of 1 November 2023; we add the three scheduled statements it omits before the end of our data, so no FOMC value is missing. The volatility-demand feed ends on 31 August 2023 and is not extended: from then on its four series are unobserved (see the data cleaning below).
- **Time stamps.** Every file stamps an interval by its end, in Eastern time, on a grid of half-hour stamps from 00:00 to 23:30: interval (t,i) holds the half hour that ends at its stamp, so the closing interval $i=c$ is stamped 16:00. *[name the vendor of each file, the price inside the one-minute log-price change (trade or quote), and the day whose volume selects the front-month contract.]*

  E2. Decision: name the vendor, the price field, the roll, and the bin edges.
  Problem (updated 2026-10-23): the bin edges are stated (item above: each stamp is the end of its interval, the column endbartime of every file; src/data/loading.py builds the half-hour grid and cuts Friday 20:00 to Sunday 18:30). Not documented anywhere in the repository, so still open, for the data provider: the vendor of every file except the Yahoo splice (no file or document names one; the builder of data/core_stats.parquet is not in the repository, live/close_signal/features.py), the price inside the one-minute log change (trade or quote), and the day whose volume picks the front month (data_explainer_v11 says only "front-month (i.e., the most active)").
  Also to confirm with the provider: data/core_stats.parquet carries the stamps 17:30 and 18:00 ET with thirty one-minute observations but almost no volume (15 March 2023: 71 and 205 contracts, against 8,624 at 17:00), inside CME's 17:00–18:00 ET maintenance halt.

- **Intraday grid and data cleaning.**
  - We construct the panel on a 30-minute grid starting from the first observed realized variance. An interval i on day t is retained in the interval grid $\mathcal{G}$ if and only if its realized variance is observed and it falls outside the weekend trading break:

$$(t,i)\in\mathcal{G} \iff \mathrm{RV}_{t,i} \text{ observed and } (t,i)\notin (\text{Fri } 20{:}00,\ \text{Sun } 18{:}30).$$

  - Lag operators step backward sequentially through the retained intervals in $\mathcal{G}$ in chronological order, spanning overnight sessions and trading-day boundaries; thus, $(t, i-j)$

refers to the j-th retained interval preceding (t, i), which may lie on an earlier trading day.

- Exchange holidays and early-close sessions are retained in the grid $\mathcal{G}$, on which the trailing averages of Section 4 are computed, whereas early-close days are excluded from the estimation and evaluation samples of the pooled models and from the option-trading evaluation sample.
- Before any other step, a print outside the domain of its series is set to missing: a nonpositive quoted or effective spread, total turnover, futures volume, observation count, or index level, and a negative value of a series that cannot be negative. This rule removes 21,943 prints, 21,852 of them zero futures volumes recorded on intervals that carry one-minute price changes.
- For each exogenous series u, we first record a binary availability (non-missingness) indicator prior to any imputation:

$$a_{t,i} = \mathbf{1}\{u \text{ observed in interval } (t, i)\} \in \{0, 1\}.$$

- Subsequently, $u_{t,i}$ is forward-filled, carrying the last observed value forward over at most 26 intervals, for the purpose of computing trailing baseline statistics, after which every interval with $a_{t,i} = 0$ is replaced by the causal median imputation described in step 4 of Section 4.
- A series whose feed stops is unobserved from its last print on, and its availability averages $\bar{a}_{t,i}^{(k)}$ fall to zero. Whenever a k-interval average holds no observed interval, its standardized value is set to zero (step 7 of Section 4), so the series enters the model at the neutral value. In the pooled models this rule acts, on at least one interval, on 134 of the 276 value averages of the full predictor set over the whole sample and on 113 of them within the evaluation span from 2 January 2014: on every horizon of the four volatility-demand series once their feed has stopped, and on the shorter horizons of the series that do not print on every interval (volatility demand, the implied volatility indices, the constituent aggregates, and the StockTwits series) where the averaging window falls in a gap between their prints.
- The closing-interval forecasts (Appendix G) were fitted before these data fixes and are reported as fitted: in their data no print is set to missing by the domain rule, the implied volatility indices are unobserved after 12 February 2024 and the FOMC series after 1 November 2023, the forward fill has no cap, every unobserved interval is imputed with the median of the series over the full sample, and no average is set to zero for want of observations or before a series starts.
- The diurnal adjustment factor $s_{t,i}^2$ and the winsorized target $\tilde{y}_{t,i}$ are formally defined in Section 5.1, with the corresponding rolling window lengths and quantile thresholds reported in Appendix B.1.

- **Estimation and evaluation samples.** Appendix Table 11 lists the samples. The estimation samples and the evaluation span of the forecasts are set out with the experiments (Section 6.1), the option-trading sample with the application (Section 7.1), and the closing-interval sample with the closing-interval models (Section G.5).

4 Predictors

- **Eight groups of exogenous predictors (46 series in total).** As listed in Appendix Table 13, the exogenous series span eight economic categories: realized moments (7), liquidity

(14), equal-weighted market moments (6), value-weighted market moments (6), sentiment (3), implied volatility (3), volatility demand (4), and FOMC announcements (3). Section 3 describes the construction of each category, and Appendix Table 13 provides the definition of each individual series:

$$46 = 7 + 14 + 6 + 6 + 3 + 3 + 4 + 3.$$

  - Specifically, the 3 FOMC series are defined as

$$\mathbf{1}\{\text{announcement on day } t\}, \quad 1/(1 + n^{+}_{t,i}), \quad 1/(1 + n^{-}_{t,i}),$$

  - where $n^{\pm}_{t,i}$ denotes the number of intervals until the next (+) and since the last (−) scheduled announcement.

- **Predictors constructed from each series.**
  - Following Corsi (2009), we construct HAR-type trailing averages of each transformed series $\tilde{u}$ over horizons of k samples (one sample is one 30-minute grid interval), lagged by one sample:

$$\bar{u}^{(k)}_{t,i} = \frac{1}{k} \sum_{j=1}^{k} \tilde{u}_{t,i-j}, \qquad k \in \mathcal{K},$$

    where the horizon set is

$$\mathcal{K} = \{1, 5, 25, 125, 625, 3125\} \quad \text{(base 5, 6 horizons)}.$$

  - Here, k counts samples, the retained intervals of the full intraday grid $\mathcal{G}$ (including overnight sessions), in the intraday and the closing-interval model alike. The closing-interval specification uses the same horizon set (Appendix G.2).
  - In addition, we compute the corresponding k-interval trailing averages of the availability indicator, denoted $\bar{a}^{(k)}_{t,i}$.
  - The intraday model does not use trailing averages of the activity indicator $\chi_{t,i}$ (step 5 of the transformations below). The closing-interval model does: $\bar{\chi}^{(k)}_{t,i}$, for its mode-inflated series (Appendix G.2).
  - For notational simplicity in our tables and figures, we suppress the overbar and the interval index i ($i = c$ for the scored forecast), writing $u^{(k)}_t$ for series u averaged over k samples; for instance, $\texttt{vix}^{(25)}_t$ denotes the transformed VIX series averaged over the 25 samples ending at 15:30 on day t. We refer to each exogenous series by its column name in the data (Table 13): the implied volatility series are `vix`, `vix3m`, and `vvix`; the suffixes `_ewstock` and `_vwstock` mark the equal- and value-weighted constituent aggregates (e.g., `ofi_ewstock`, `sumret2_vwstock`, and `spread_vwstock`); and the announcement predictors are `fomc_day`, `fomc_until_inv`, and `fomc_since_inv`, while writing $\bar{a}^{(k)}_t(u)$ and $\bar{\chi}^{(k)}_t(u)$ for the trailing averages of the availability and activity indicators.

- **HAR benchmark.** The HAR benchmark predictor set consists of the HAR components of the target, $\bar{y}^{(k)}_{t,i}$—constructed via the same trailing-average operator as $\bar{u}^{(k)}_{t,i}$ applied to the winsorized target (6 columns in the intraday model)—together with the calendar vector $\boldsymbol{D}_{t,i}$, which comprises five day-of-the-week dummy variables and the hour of the interval-end timestamp. The pooled models are estimated on the 13 regular-hours intervals (Section 6.1), which hold no overnight interval, so the calendar vector has no overnight session indicator. The closing-interval model, as fitted, has 16 candidate calendar columns (Appendix G.2).

- **Sequential feature transformations.** Each exogenous series undergoes the following seven transformations in sequence:
  1. *Diurnal adjustment:* we divide each raw series $u_{t,i}$ by a time-of-day scale factor $b_{t,i}^{(u)}$ estimated over the same prior-day set $\mathcal{D}_{t,i}$ as $s_{t,i}^2$ and bounded relative to its own trailing median:

$$\tilde{u}_{t,i} = \frac{u_{t,i}}{b_{t,i}^{(u)}}, \qquad b_{t,i}^{(u)} = \begin{cases} \mathrm{clip}(\mathrm{mean}_{t' \in \mathcal{D}_{t,i}}\, u_{t',i},\ 0.5\,\rho_{t,i},\ 2\rho_{t,i}), & u \geq 0, \\ \max\{\mathrm{clip}(\mathrm{sd}_{t' \in \mathcal{D}_{t,i}}\, u_{t',i},\ 0.5\,\rho_{t,i},\ 2\rho_{t,i}),\ 0.1\,\bar{\sigma}_{t,i}^{(u)}\}, & u \text{ signed}, \end{cases}$$

     - where $\rho_{t,i}$ denotes the median of the 250 most recent unbounded diurnal factors for interval i (requiring at least 20 observations),
     - $\bar{\sigma}_{t,i}^{(u)}$ denotes the median of the nonzero trailing standard deviations $\mathrm{sd}_{t'' \in \mathcal{D}_{t',i'}}\, u_{t'',i'}$ over all intervals (t', i') up to (t, i), every intraday interval pooled (before the first such standard deviation the adjusted value is undefined and is imputed as in step 4),
     - a series is signed when the quantity it measures can be negative, not when its sample has a negative value (Table 13),
     - $b_{t,i}^{(u)}$ is set to 1 whenever the factor of a nonnegative series is zero or undefined, and
     - no diurnal adjustment is applied to the implied volatility indices (VIX, VIX3M, and VVIX), the StockTwits sentiment series, or the FOMC series.
     - The closing-interval forecasts predate this floor and this sign rule: they take $\bar{\sigma}^{(u)}$ as the full-sample median of the same standard deviations, and call a series signed when any value in its full sample is negative, which signs the value-weighted quoted spread through a single invalid print (Table 13).
  2. *Variance-stabilizing transformation:* we apply a domain-specific mapping $\tilde{u}_{t,i} \leftarrow g(\tilde{u}_{t,i})$ according to the first matching rule in

$$g(v) = \begin{cases} \sqrt{v} & \text{squared returns, realized variance, turnover,} \\ & \text{bipower variation, effective spread,} \\ \mathrm{sign}(v)\,|v|^{1/2} & \text{autocovariance,} \\ v^{1/3} & \text{cubed returns,} \\ \mathrm{sign}(v)\,|v|^{1/4} & \text{fourth-power returns,} \\ v & \text{other signed series, absolute returns,} \\ \log \max(v, 10^{-12}) & \text{otherwise.} \end{cases}$$

     - A signed series without a specific power-root rule is routed at interval (t, i) to an inverse hyperbolic sine transformation of the raw series when its standard-deviation divisor sat at its floor on at least 50% of its intervals since its first print and before (t, i), at least 250 of them. This routes the volatility-demand series, on almost all of their intervals (Table 13), and no other series:

$$\tilde{u}_{t,i} = \mathrm{asinh}(u_{t,i}/\kappa_{t,i}), \qquad \kappa_{t,i} = \max\{e_{t,i-1},\ 0.01\,\mathrm{med}_{j \geq 1}\, e_{t,i-j}\},$$

     - where $e_{t,i}$ denotes an exponentially weighted moving average of $|u|$ computed over nonzero intervals ($u \neq 0$) with a half-life of 12,000 such intervals (nominally 250 days of 48 intervals).

- The FOMC statement-day dummy passes through unchanged, with no transformation and no winsorization (step 3).
- The closing-interval forecasts predate these rules: they route a series by the share over its full sample, the intervals before its first print included, which routes the same volatility-demand series on every interval; they fill $\kappa_{t,i}$ with its full-sample median where it is not yet defined; and they take the logarithm of the FOMC statement-day dummy and winsorize it.

3. *Winsorization:* each series is clipped at the same rolling quantile thresholds used for the target (see Section 5.1), evaluated over the series' own preceding intervals.
4. *Missing-value imputation:* any missing observation is imputed using the expanding median of the series' previously observed transformed values, while retaining the availability indicator $a_{t,i}$ as a separate predictor:

$$\tilde{u}_{t,i} \leftarrow \begin{cases} \tilde{u}_{t,i}, & a_{t,i} = 1, \\ \operatorname{med} \mathcal{A}_{t,i}, & a_{t,i} = 0,\ \mathcal{A}_{t,i} \neq \emptyset, \\ \tilde{u}_{(1)}, & a_{t,i} = 0,\ \mathcal{A}_{t,i} = \emptyset, \end{cases}$$

   - where $\mathcal{A}_{t,i} = \{\tilde{u}_{t',i'} : (t',i') \prec (t,i),\ a_{t',i'} = 1\}$, with $(t',i') \prec (t,i)$ indicating that (t',i') is a retained interval strictly prior to (t,i), and
   - $\tilde{u}_{(1)}$ denotes the transformed value on the series' first observed interval.
   - The closing-interval forecasts predate this rule: on every interval with $a_{t,i} = 0$ they impute the median of the series' observed transformed values over the full sample (Section 3).

5. *Activity indicators:* a series is mode-inflated at interval (t,i) when it has at least 250 observed prints strictly before (t,i) and their most frequent value, $m_{t,i}$, accounts for at least 20% of them. For each series that is mode-inflated at some interval, we define a binary activity indicator from its observed prints: on every interval at which the series is mode-inflated,

$$\chi_{t,i} = \begin{cases} \mathbf{1}\{u_{t,i} \neq m_{t,i}\}, & a_{t,i} = 1, \\ \chi^{\circ}_{t,i}, & a_{t,i} = 0, \end{cases}$$

   where $\chi^{\circ}_{t,i}$ is the more frequent value of χ on the series' earlier observed intervals (0 on a tie), and $\chi_{t,i} = 0$ on the other intervals.
   - The mode-inflated class and the mode are decided from earlier prints only, and the indicator is read from observed prints, not from the forward-filled series. Under this rule, our specification, 5 series are mode-inflated at some interval: the number of observations, the 3 StockTwits series (on part of the sample) and the FOMC statement-day dummy; the volatility-demand series never are.
   - The pooled intraday models of Table 1 carry no activity averages $\bar{\chi}^{(k)}$; the indicator enters the closing-interval models (Appendix G.2).
   - The closing-interval forecasts we report predate this rule and use a full-sample one: a series is mode-inflated when its full-sample mode accounts for at least 20% of all intervals, and $\chi_{t,i} = \mathbf{1}\{u_{t,i} \neq \text{mode}(u)\}$ is read on the forward-filled series. Under that rule the 6 series marked in Table 13 carry the indicator (the number of observations, the 4 volatility-demand series and the FOMC statement-day dummy) and, in the fitted forecasts, so does the FOMC statement-interval dummy, since dropped from the predictors.

– Read on the forward-filled series, the indicator of a series whose feed has stopped keeps the value of its last print. In the closing-interval forecasts, whose FOMC series end with the statement of 1 November 2023 and are filled without a cap (Section 3), the FOMC statement-day indicator therefore equals 1 on every later interval. The pooled intraday forecasts read the indicator from observed prints, and in their data no FOMC value is missing (Section 3).

6. *HAR aggregation:* we form the trailing averages $\bar{u}_{t,i}^{(k)}$, $\bar{a}_{t,i}^{(k)}$, $\bar{\chi}_{t,i}^{(k)}$ (closing-interval model only), and $\bar{y}_{t,i}^{(k)}$ across horizons $k \in \mathcal{K}$.
7. *Robust standardization:* under the pooled intraday scheme, we standardize the trailing averages of series values over the rolling estimation window using the median and interquartile range (with lower floors). Specifically, the median and interquartile range are computed exclusively over intervals with $a_{t',i'} = 1$ inside the rolling window $\mathcal{W}_{t,i}$ (see Appendix B.2), re-evaluated every 480 intervals and held constant until the next grid point. Let $M_{t,i}^{(k)}$ denote the maximum such interquartile range observed across all grid points up to (t, i). We apply a 0.01 relative floor in the denominator (note that $\mathrm{IQR}^{*(k)}$, defined in Appendix G.4, does not enter this pooled denominator):

$$z_{t,i}^{(k)} = \begin{cases} \dfrac{\bar{u}_{t,i}^{(k)} - \mathrm{med}\{\bar{u}_{t',i'}^{(k)} : (t',i') \in \mathcal{W}_{t,i},\ a_{t',i'} = 1\}}{\max\{\mathrm{IQR}_{t,i}^{(k)},\ 0.01\, M_{t,i}^{(k)}\}} & \text{if at least 512 such intervals,} \\ 0 & \text{otherwise,} \end{cases}$$

$$\mathrm{IQR}_{t,i}^{(k)} = \mathrm{IQR}\{\bar{u}_{t',i'}^{(k)} : (t',i') \in \mathcal{W}_{t,i},\ a_{t',i'} = 1\}.$$

   – On the initial estimation window, both statistics are computed using that window's own rows; an interquartile range below 10^{-12} is replaced by 1 on the initial window and, at subsequent grid points, retains the previous scale factor.
   – The availability averages $\bar{a}^{(k)}$ are left unstandardized.
   – The target HAR components and calendar variables do not use the availability-restricted denominator above; instead, their median and interquartile range are computed across all rows of $\mathcal{W}_{t,i}$ with $\mathrm{IQR}^{*(k)} = 0$ ($\mathrm{IQR}^{*(k)}$ is defined in Appendix G.4), subject to the same 0.01 floor relative to the running maximum interquartile range of that column.
   – For a series whose first print comes after the first day of the sample, $z_{t,i}^{(k)} = 0$ until every interval of the window $\mathcal{W}_{t,i}$ is live, an interval being live when its k-interval average holds no interval before the day of the series' first print.
   – $z_{t,i}^{(k)} = 0$ whenever the k-interval average holds no observed interval, $\bar{a}_{t,i}^{(k)} = 0$, so that the averages of a feed that has stopped are exactly neutral.
   – This step equalizes each column's interquartile range, not its variance. The ratio of a column's standard deviation to its interquartile range varies with its tails, its skewness and its share of zeros; the availability averages $\bar{a}^{(k)}$ keep their $[0, 1]$ scale; and the scale factors come from the rolling grid rather than from the sample on which a model is estimated. The step serves the data: it limits the influence of outliers and keeps each series comparable over time. The penalized linear models therefore also standardize every column within their own estimation window (Section 5.2), so that the penalty treats every column alike.

- **Column screening.** At each estimation origin, we screen out constant and duplicate predictors over the active estimation window $\mathcal{W}_t$ by retaining the index set

$$\mathcal{J}_t = \{j : \boldsymbol{x}_{\cdot j} \text{ not constant}, \ \boldsymbol{x}_{\cdot j} \neq \boldsymbol{x}_{\cdot j'} \ \forall\, j' < j, \ j' \in \mathcal{J}_t\}, \qquad \boldsymbol{x}_{\cdot j} = (x_{t'j})_{t' \in \mathcal{W}_t}.$$

  - Specifically, column equality is evaluated bitwise, and whenever two columns are identical on $\mathcal{W}_t$, the first column in order is retained.
- **Predictor set dimensions.** The column counts of the closing-interval specification are given in Appendix G.2.
  - In the pooled intraday specification, the full predictor set comprises 564 candidate columns: the 6 HAR components of the target, the calendar vector $\boldsymbol{D}_{t,i}$ (the weekday dummies and the hour), and the value and availability averages of 46 exogenous series at 6 horizons; at each estimation the column screens of this section and of Section 5.2 keep a subset of them:

$$564 = \underbrace{6}_{\bar{y}^{(k)}} + \underbrace{6}_{\boldsymbol{D}_{t,i}} + \underbrace{46 \times 6}_{\bar{u}^{(k)}} + \underbrace{46 \times 6}_{\bar{a}^{(k)}}.$$

    Here, the 46 intraday exogenous series encompass every category in Section 3, the FOMC group (3 series) included. Appendix Table 13 lists each individual series.
- **Causal construction and absence of look-ahead.** Every rolling statistic indexed by (t, i) is computed strictly from intervals prior to (t, i), ensuring that every predictor of interval (t, i) is in the information set at its forecast origin (15:30 ET for the scored closing interval $i = c$). This includes the HAR trailing averages, the diurnal factor $s^2_{t,i}$, the winsorization quantiles, the median baseline $\rho_{t,i}$, the standardization moments, and the active column set $\mathcal{J}_t$.
  - In the pooled intraday forecasts, the standard-deviation floor $\bar{\sigma}^{(u)}_{t,i}$ and the inverse hyperbolic sine routing are computed from earlier intervals, and the sign of each series is read from what it measures, not from the full sample (steps 1 and 2); the activity indicator does not enter them (step 5).
  - We disclose the full-sample quantities used in preprocessing the closing-interval forecasts, which predate those rules: the mode that defines $\chi_{t,i}$ under the rule those forecasts were built with (step 5), the standard-deviation floor $\bar{\sigma}^{(u)}$ and the sign of each series (step 1), the rule assigning series to the inverse hyperbolic sine transformation (step 2), and the median that imputes unobserved intervals (step 4).
  - As noted in Section 3, the FOMC proximity variables rely solely on the pre-announced meeting schedule, which is publicly known ahead of each interval.
  - Any missing interval prior to a series' initial observation is filled with that first observed print (step 4); under the late-start rule of step 7 the standardized averages that hold such intervals are set to 0.
  - Finally, standardization statistics on the first estimation window are computed exclusively from burn-in observations.

5 Methods

5.1 Problem formulation

- **Rolling-window out-of-sample exercise.** We conduct a rolling-window out-of-sample forecasting exercise in which each model is re-estimated every period using only predictors

in the information set $\mathcal{F}_t$ available at the 15:30 ET forecast origin. A period corresponds to a 30-minute intraday interval under our pooled intraday scheme (Appendix B.2). The 15:30 origin is that of the closing interval $i = c$; the pooled models also forecast every other interval (t, i) of their estimation sample, each at its own origin, the start of the interval, from the information available then (Table 1).

- **Notation.** We index trading days by $t = 1, \ldots, T$ and the 30-minute intraday intervals of day t (including the overnight session) by $i = 1, \ldots, M$, where $i = c$ denotes the closing half-hour interval (15:30–16:00 ET).
  - Realized variance $\mathrm{RV}_{t,i}$ and the lag operator $(t, i - j)$ are defined in Section 3. Our primary forecast target is the realized variance of the closing interval, $\mathrm{RV}_t \equiv \mathrm{RV}_{t,c}$.
  - We denote by $\mathcal{F}_t$ the information set containing all data observed through 15:30 ET on day t, and write $\mathbb{E}_t[\cdot] \equiv \mathbb{E}[\cdot \mid \mathcal{F}_t]$ and $\mathrm{Var}_t(\cdot) \equiv \mathrm{Var}(\cdot \mid \mathcal{F}_t)$ for the corresponding conditional expectation and variance.
- **Diurnal adjustment.** Following Andersen and Bollerslev (1997), we remove the intraday diurnal pattern using the trailing sample mean of realized variance in the same time-of-day interval strictly prior to day t:

$$s^2_{t,i} = \frac{1}{|\mathcal{D}_{t,i}|} \sum_{t' \in \mathcal{D}_{t,i}} \mathrm{RV}_{t',i},$$

$$\mathcal{D}_{t,i} = \{\text{last } n_{\mathrm{d}} \text{ trading days } t' < t \text{ with interval } i\}, \qquad |\mathcal{D}_{t,i}| \geq \underline{n}_{\mathrm{d}}.$$

  - For intervals with $|\mathcal{D}_{t,i}| < \underline{n}_{\mathrm{d}}$, we set $s^2_{t,i} = 1$ and exclude the interval from evaluation. The window parameters n_{d} and $\underline{n}_{\mathrm{d}}$ are given in Appendix B.1.
- **Regression target.** To stabilize the variance, we take the square root of the diurnally adjusted realized variance and winsorize the resulting series at rolling lower and upper empirical quantiles:

$$y_{t,i} = \sqrt{\mathrm{RV}_{t,i}/s^2_{t,i}}, \qquad \mathcal{Y}_{t,i} = \{y_{t,i-j} : j = 1, \ldots, n_{\mathrm{w}}\},$$
$$\tilde{y}_{t,i} = \mathrm{clip}(y_{t,i},\ \underline{Q}_{t,i},\ \overline{Q}_{t,i}).$$

  - Here $\underline{Q}_{t,i}$ and $\overline{Q}_{t,i}$ denote the lower and upper empirical quantiles computed over $\mathcal{Y}_{t,i}$, which pools the preceding n_{w} intervals across all times of day, and clip truncates its first argument to that range. The quantile levels and window length n_{w} are reported in Appendix B.1.
  - All models are estimated on the winsorized target $\tilde{y}_{t,i}$, whereas out-of-sample evaluation is conducted against the unwinsorized realized variance $\mathrm{RV}_{t,c}$.
  - For the scored interval $i = c$ (the closing half hour that every model forecasts), we write $y_t \equiv y_{t,c}$ and $\tilde{y}_t \equiv \tilde{y}_{t,c}$.
- **Back-transformation.** To obtain a variance forecast in raw units, we square the fitted volatility and multiply by the diurnal factor without post-estimation recalibration:

$$\hat{\sigma}^2_t = \hat{y}_t^{\,2}\, s^2_{t,c},$$

where $\hat{y}_t \equiv \hat{y}_{t,c}$ is the forecast of the winsorized target $\tilde{y}_t$, not of the unclipped y_t, formed using $\mathcal{F}_t$.

  - Because $\mathbb{E}_t[y_t^2] - (\mathbb{E}_t[y_t])^2 = \mathrm{Var}_t(y_t) \geq 0$, squaring the level forecast introduces a downward Jensen inequality bias that we leave uncorrected (note E33).

- **Rolling estimation window.** We let $\mathcal{I}$ denote the subset of intraday intervals included in estimation. Under the intraday scheme, the estimation window at origin (t, i) is $\mathcal{W}_{t,i}$ (Appendix B.2), consisting of the W most recent intervals in $\mathcal{I}$ prior to (t, i) (the intervals in $\mathcal{I}$ and the value of W are given in Section 6.1).
- **Hyperparameter-selection splits.** In line with Gu et al. (2020), every split runs forward in time: an estimation subsample, an embargo gap of n_{gap} observations, and a validation subsample of n_{val} observations. We use two forms of split.
  - *Pre-sample split.* The pooled linear models of Table 1 choose their penalties once, on a split that ends before the first forecast, and hold them fixed for every forecast; the closing-interval linear models choose theirs on pre-sample folds of the same form (Appendix G.1). The estimation subsample is a full window of W observations, so that each candidate penalty is fitted on as many observations as a forecast. For a split whose validation subsample ends just before (t, i),

$$\begin{aligned}\mathcal{W}_{t,i}^{\text{est}} &= \{(t,\, i - n_{\text{val}} - n_{\text{gap}} - W), \ldots, (t,\, i - n_{\text{val}} - n_{\text{gap}} - 1)\},\\ \mathcal{W}_{t,i}^{\text{gap}} &= \{(t,\, i - n_{\text{val}} - n_{\text{gap}}), \ldots, (t,\, i - n_{\text{val}} - 1)\},\\ \mathcal{W}_{t,i}^{\text{val}} &= \{(t,\, i - n_{\text{val}}), \ldots, (t,\, i - 1)\}.\end{aligned}$$

    For the pooled linear models this split is the last pre-sample fold (validation year 2013), with the first scored forecast at the interval that follows it. The pooled tree ensembles keep one configuration per method, selected on the pre-sample and held fixed (Appendix B.4).
  - *Split of the rolling window.* A model that re-selects its hyperparameters during the evaluation partitions its own rolling window $\mathcal{W}_{t,i}$, so that the three subsamples together hold W observations:

$$\mathcal{W}_{t,i}^{\text{est}} = \{(t,\, i - W), \ldots, (t,\, i - n_{\text{val}} - n_{\text{gap}} - 1)\},$$

    with $\mathcal{W}_{t,i}^{\text{gap}}$ and $\mathcal{W}_{t,i}^{\text{val}}$ as above.

  Target interval (t, i) never enters any of these sets. The lengths n_{gap} and n_{val} of each split, and the models that use each form, are given in Appendix B.1.
- **Estimation objective.** Each model is estimated by minimizing the sum of squared errors on the scale of $\tilde{y}$. At forecast origin (t, i), the parameter vector $\boldsymbol{\theta}$ is fitted over the full rolling window $\mathcal{W}_{t,i}$:

$$\hat{\boldsymbol{\theta}}_{t,i} = \arg\min_{\boldsymbol{\theta}} \sum_{(t',i') \in \mathcal{W}_{t,i}} \big(\tilde{y}_{t',i'} - \hat{y}_{t',i'}(\boldsymbol{\theta})\big)^2.$$

  - Here (t', i') indexes the historical intervals in the window $\mathcal{W}_{t,i}$, and $\boldsymbol{\theta}$ collects the model parameters.
  - During hyperparameter tuning only, candidate configurations are estimated on $\mathcal{W}_{t,i}^{\text{est}}$ rather than $\mathcal{W}_{t,i}$ and evaluated on $\mathcal{W}_{t,i}^{\text{val}}$.

5.2 Models

- **HAR and the HAR benchmark via OLS.** Following Corsi (2009), we specify the linear heterogeneous autoregressive model for each intraday interval (t, i) as

$$\hat{y}_{t,i} = \beta_0 + \sum_{k \in \mathcal{K}} \beta^{(k)}\, \bar{y}_{t,i}^{(k)} + \boldsymbol{\delta}^\top \boldsymbol{D}_{t,i}.$$

– The horizon set $\mathcal{K}$ and the lagged volatility averages $\bar{y}_{t,i}^{(k)}$ are defined in Section 4, where $\bar{y}_{t,i}^{(k)}$ is the mean of the winsorized target over the k half-hour intervals preceding (t,i), including overnight intervals. The scored forecast is the closing-interval prediction $\hat{y}_t \equiv \hat{y}_{t,c}$.
– The vector $\boldsymbol{D}_{t,i}$ collects the calendar dummy variables with coefficient vector $\boldsymbol{\delta}$; a pure HAR specification restricts $\boldsymbol{\delta} = \mathbf{0}$. Our HAR benchmark ($\mathcal{X} = \text{HAR}$) includes both the HAR components and the calendar dummies.
– The full predictor vector $\boldsymbol{x}_{t,i} \in \mathbb{R}^p$ ($\mathcal{X} = \text{Full}$) augments the HAR benchmark with the multi-horizon averages $\bar{u}_{t,i}^{(k)}$ ($k \in \mathcal{K}$) and the corresponding availability averages $\bar{a}_{t,i}^{(k)}$ for every exogenous series u (Section 4); the closing-interval model also carries the averages $\bar{\chi}_{t,i}^{(k)}$ of the activity indicators (Appendix G.2).

- **Penalized linear regressions: ridge (Hoerl and Kennard, 1970), lasso (Tibshirani, 1996), and elastic net (Zou and Hastie, 2005).** To accommodate the high-dimensional predictor vector $\boldsymbol{x}_{t,i}$, we augment the residual sum of squares with an ℓ_1, ℓ_2, or combined penalty, following the penalized HAR formulation of Zhang et al. (2024), and update the coefficient estimates as the estimation window rolls forward:

$$\hat{\boldsymbol{\beta}}_{t,i}^{\text{ridge}} = \underset{\beta_0,\boldsymbol{\beta}}{\arg\min} \sum_{(t',i')\in\mathcal{W}_{t,i}} (\tilde{y}_{t',i'} - \beta_0 - \boldsymbol{x}_{t',i'}^\top \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{p} \omega_j \, \beta_j^2,$$

$$\hat{\boldsymbol{\beta}}_{t,i}^{\text{enet}} = \underset{\beta_0,\boldsymbol{\beta}}{\arg\min} \sum_{(t',i')\in\mathcal{W}_{t,i}} (\tilde{y}_{t',i'} - \beta_0 - \boldsymbol{x}_{t',i'}^\top \boldsymbol{\beta})^2 + \lambda\alpha \sum_{j=1}^{p} \omega_j |\beta_j| + \frac{\lambda(1-\alpha)}{2} \sum_{j=1}^{p} \omega_j \, \beta_j^2,$$

where $\lambda > 0$ governs overall penalty strength, $\alpha \in [0,1]$ is the elastic-net mixing parameter, and $\omega_j \geq 0$ is the predictor-specific penalty weight (with the intercept β_0 unpenalized, $\omega_0 = 0$). For ridge, λ coincides directly with the grid value reported in Table 12. For lasso and elastic net, Table 12 reports the `scikit-learn` penalty parameter, so the penalty multiplier λ in the unnormalized objective above equals $2n$ times the tabulated value, where n is the number of observations in the window.

– **Differential block penalty.** When estimating ridge, lasso, and elastic net on the full predictor set, we penalize the HAR benchmark predictors 100 times more lightly than the exogenous predictors:

$$\omega_j = \begin{cases} \underline{\omega} = 0.01 & j \in \text{HAR benchmark}, \\ 1 & j \text{ an additional predictor}, \\ 0 & j = 0 \text{ (intercept)}. \end{cases}$$

– When a penalized regression is fitted on the HAR benchmark alone, we apply a uniform penalty weight $\omega_j = 1$ across all predictors.
– **Standardization within the estimation window.** The penalties apply to standardized coefficients (Tibshirani, 1996; Friedman et al., 2010). Each predictor column is centered and scaled by its mean m_j and standard deviation s_j over $\mathcal{W}_{t,i}$; the objectives above are solved for the coefficients γ_j of the standardized columns $(x_j - m_j)/s_j$, with the weights ω_j, and transformed back, $\beta_j = \gamma_j/s_j$, the intercept absorbing the centering. Without this step the penalty on a column's contribution to the forecast would scale with $1/s_j^2$ for ridge and $1/s_j$ for the lasso, so columns with little spread, such as long-horizon averages, would be penalized far more heavily for no economic reason.

The robust standardization of Section 4 does not achieve this: it equalizes interquartile ranges rather than variances, and its scale factors come from the rolling grid rather than from the estimation window. For ridge, m_j and s_j are updated at every step; for the lasso and elastic net, at every cold restart of the homotopy described below, and held fixed in between.

- A column of an exogenous series enters a fit only if the series' own one-interval column differs from its modal value on at least 250 intervals of $\mathcal{W}_{t,i}$, the evidence requirement of the activity indicator (Section 4). A k-interval average repeats each observation k times, so its own count would overstate the evidence. Columns that fail are screened out like constant columns (below).
- An availability average over $k > 1$ intervals is a proportion, with sampling resolution $\sqrt{p(1-p)/k}$ at its window mean p. It is screened out of a fit when its standard deviation over $\mathcal{W}_{t,i}$ does not exceed that resolution, or when the standard deviation of its difference from an earlier retained availability average of the same horizon does not exceed $\sqrt{2}$ times the resolution at the pair's mean: its movement is then no larger than independent missing observations would produce, so it carries no detectable information. These are mostly the long-horizon availability averages of regular-hours series, which record the trading calendar rather than the series.
- The solvers work directly on the standardized columns, which gives the same estimates as penalizing the raw columns with weights $\omega_j s_j$ and $\omega_j s_j^2$ but keeps the systems they solve well conditioned.
- The lasso corresponds to $\alpha = 1$, and our elastic net specification fixes $\alpha = 0.5$.
- For ridge regression, we update $(\boldsymbol{X}_{t,i}^\top \boldsymbol{X}_{t,i} + \lambda \operatorname{diag}(\boldsymbol{\omega}))^{-1}$ via exact rank-one downdates and updates as observations exit and enter $\mathcal{W}_{t,i}$, where $\boldsymbol{X}_{t,i}$ stacks the rows $(1, \boldsymbol{x}_{t',i'}^\top)$ for $(t', i') \in \mathcal{W}_{t,i}$ and $\boldsymbol{\omega} = (\omega_0, \ldots, \omega_p)^\top$.
- For the lasso and elastic net, we update the active set and coefficient vector via exact online homotopy (Garrigues and El Ghaoui, 2008), adding the incoming observation and dropping the outgoing observation at each step.
- The homotopy restarts from a cold solve on the current window at fixed intervals. Between restarts, after each update we check the Karush–Kuhn–Tucker conditions of the current window's problem; where they fail, the updated solution is replaced by a cold solve on the current window, from which the homotopy continues.
- At every estimation step, only the screened predictor subset $\mathcal{J}_{t,i}$ enters the solver: columns that are constant within $\mathcal{W}_{t,i}$ or exact duplicates of an earlier retained column are omitted. In models estimated solely on the closing interval, this screen automatically removes the hour-of-day and overnight indicators (Section 4).

- **Tree ensembles.** To capture potential nonlinearities and high-order interactions among predictors, we consider two gradient-boosted tree implementations (LightGBM and XGBoost) and random forests.
  - **Gradient boosting** (Friedman, 2001), implemented via LightGBM (Ke et al., 2017) and XGBoost (Chen and Guestrin, 2016), minimizes the squared-error loss $\frac{1}{2}(\tilde{y}-F)^2$ over boosting rounds $b = 1, \ldots, B$ with learning rate η. Each regression tree f_b partitions the predictor space into J leaves with leaf weights $\boldsymbol{w}$, where leaf j contains the observation subset $\mathcal{R}_j$ of size $n_j = |\mathcal{R}_j|$. When the ℓ_1 leaf penalty $\lambda_1^{\text{tree}} = 0$, the additive update and

optimal leaf weight take the form

$$F_b = F_{b-1} + \eta f_b,$$
$$w_j^* = -\frac{\sum_{(t',i')\in\mathcal{R}_j}\big(F_{b-1}(\boldsymbol{x}_{t',i'}) - \tilde{y}_{t',i'}\big)}{n_j + \lambda_2^{\text{tree}}}.$$

At round b, tree f_b is grown to minimize the regularized second-order objective

$$\sum_{(t',i')\in\mathcal{W}_{t,i}}\Big[\big(F_{b-1}(\boldsymbol{x}_{t',i'}) - \tilde{y}_{t',i'}\big)f_b(\boldsymbol{x}_{t',i'}) + \tfrac{1}{2}f_b(\boldsymbol{x}_{t',i'})^2\Big] + \gamma^{\text{tree}}J + \lambda_1^{\text{tree}}\|\boldsymbol{w}\|_1 + \tfrac{1}{2}\lambda_2^{\text{tree}}\|\boldsymbol{w}\|_2^2,$$

where γ^{tree} penalizes tree complexity per leaf and $\lambda_1^{\text{tree}}, \lambda_2^{\text{tree}}$ apply ℓ_1 and ℓ_2 shrinkage to the leaf weights $\boldsymbol{w}$.

– Every leaf is required to contain at least $n_{\min}$ observations ($n_j \geq n_{\min}$), which corresponds to the minimum leaf size in LightGBM and, under unit Hessian for squared-error loss, the minimum child weight in XGBoost.

– **Random forest** (Breiman, 2001) forms the equally weighted average of B regression trees f_b, each grown independently on a bootstrap sample of the estimation window with a random subset of candidate predictors evaluated at each split:

$$\hat{y}_{t,i} = \frac{1}{B}\sum_{b=1}^{B} f_b(\boldsymbol{x}_{t,i}).$$

5.3 Forecast evaluation

- **Forecast and realization.** We evaluate the unadjusted back-transformed variance forecast $\hat{\sigma}_t^2$ defined in Section 5.1 against the actual, unwinsorized realized variance of the closing interval, $\text{RV}_t \equiv \text{RV}_{t,c}$ (winsorization is applied to $\tilde{y}_{t,c}$ during model estimation only).
- **Every-interval scoring of the pooled models.** The definitions below are written for the closing interval. Table 1 applies them interval by interval to the pooled models, on the 33,527 regular-hours intervals of the days in $\mathcal{T}_{\text{oos}}$ with a positive realized variance: each forecast $\hat{\sigma}_{t,i}^2 = \hat{y}_{t,i}^2\, s_{t,i}^2$ is scored against $\text{RV}_{t,i}$; $\overline{\text{QLIKE}}$ and the MSE average over the scored intervals; the reference forecast of R^2_{OOS} is the historical mean realized variance of the same interval of the day, over every earlier trading day from 5 January 1998; and the Diebold–Mariano test is applied to each day's average loss.
- **Evaluation set.** Because every forecast is generated by a model estimated on a strictly trailing window, all predictions are out of sample. We let $\mathcal{T}_{\text{oos}}$ denote the set of common forecast-origin trading days following the initial estimation window and hyperparameter presample on which all competing forecasts are available, excluding early-close days, and write $T_{\text{oos}} = |\mathcal{T}_{\text{oos}}|$ (the exact sample dates and T_{oos} are given in Section 6.1).
- **Primary loss criterion.** Our primary statistical loss is the daily quasi-likelihood (QLIKE) loss of Patton (2011), which is robust to measurement noise in the realized-variance proxy. Because the diurnal factor $s_{t,c}^2$ cancels between $\text{RV}_t = y_t^2 s_{t,c}^2$ (with unwinsorized y_t) and $\hat{\sigma}_t^2 = \hat{y}_t^2 s_{t,c}^2$, the daily loss takes the form

$$L_t = \frac{y_t^2}{\hat{y}_t^2} - \log\frac{y_t^2}{\hat{y}_t^2} - 1 = \text{QLIKE}(\text{RV}_t,\, \hat{\sigma}_t^2),$$

with sample average over $\mathcal{T}_{\text{oos}}$ given by

$$\overline{\text{QLIKE}} = \bar{L} = \frac{1}{T_{\text{oos}}}\sum_{t\in\mathcal{T}_{\text{oos}}} L_t.$$

- **Benchmark forecast.** Our baseline is the closing-interval HAR benchmark: a HAR-type model in the spirit of Corsi (2009), estimated by OLS on the trailing averages $\bar{y}^{(k)}$ of the target over $k \in \{1, 5, 25, 125, 625, 3125\}$ samples and the 16 calendar columns $\boldsymbol{D}$ (Section 5.2), using closing-interval observations only (Appendix G.3).
- **Relative loss.** When comparing a candidate forecast A against a reference forecast B (the closing-interval HAR benchmark unless noted otherwise), we report the percentage difference in mean QLIKE:

$$\Delta\% = 100\Big(\frac{\bar{L}^A}{\bar{L}^B} - 1\Big),$$

  and assess its statistical significance via the Diebold–Mariano test detailed in Appendix I.
- **Squared-error loss.** In variance units, we also compute the mean squared error over $\mathcal{T}_{\text{oos}}$:

$$\text{MSE} = \frac{1}{T_{\text{oos}}} \sum_{t \in \mathcal{T}_{\text{oos}}} \big(\text{RV}_t - \hat{\sigma}_t^2\big)^2.$$

- **Out-of-sample R^2.** Following Campbell and Thompson (2008), we compute R^2_{OOS} relative to the expanding historical mean of closing-interval realized variance known at the forecast origin:

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{t \in \mathcal{T}_{\text{oos}}} \big(\text{RV}_t - \hat{\sigma}_t^2\big)^2}{\sum_{t \in \mathcal{T}_{\text{oos}}} \big(\text{RV}_t - \overline{\text{RV}}_{t-1}\big)^2}, \qquad \overline{\text{RV}}_{t-1} = \frac{1}{|\{t' < t\}|} \sum_{t' < t} \text{RV}_{t'},$$

  where the expanding average $\overline{\text{RV}}_{t-1}$ is taken over all prior trading days from the sample start specified in Section 6.1.

E33. Decision: to decide: compute MAE and the calibration diagnostics (the ratio $\overline{\text{RV}}/\overline{\hat{\sigma}^2}$ and the under-prediction frequency) for the forecasts of Table 1, in make_rehaul_oos_rerun.py, or drop them.
Problem: MAE and calibration diagnostics promised here are not yet computed/reported.
No generator of the paper (writeup/make_rehaul_*.py) computes MAE, the ratio or the under-prediction frequency. The calibration files under results/atm_straddle_0dte_1530/ (recal_grid/, recal_2026-09-24/) come from the earlier recalibration studies, not from the current forecasts, and generated/campaign_numbers.tex holds one MAE of an earlier campaign, which this paper does not load.

6 Experiments and results

6.1 Experimental setup

- **Estimation samples** (see Appendix Table 11). The pooled intraday sample comprises 300,317 half-hour intervals beginning in June 1998,

$$300{,}317 = 303{,}442 \text{ intervals on the grid} - 3{,}125 \text{ burn-in samples};$$

  the pooled linear models and tree ensembles of Table 1 and of this section and Section 7 are trained on its 13 regular-hours bars with rolling windows of $W = 39{,}000$ bars (3,000 trading days; training scheme below), and the pooled linear runs of the model set on all its intervals with $W = 96{,}000$ intervals (Appendix B.2).
- **Evaluation span $\mathcal{T}_{\text{oos}}$** (see Section 5.3). We evaluate the rolling out-of-sample forecasts over the period from 3 January 2020 to 30 April 2024, which comprises $T_{\text{oos}} = |\mathcal{T}_{\text{oos}}| = 1{,}082$ trading days on which all competing forecasts are available (after excluding 6 early-close sessions). The pooled linear models and tree ensembles of Table 1 are scored on a longer span, from 2 January 2014 to 30 April 2024 (2,579 trading days, after excluding 20 early-close

sessions), with a rolling window of 3,000 trading days; the linear penalties are chosen once on the last pre-sample fold (validation year 2013), before the first forecast, and then frozen, and each tree ensemble keeps one configuration from an Optuna search on the pre-sample.

- **Training scheme.** Under the pooled intraday estimation scheme, the linear specifications and the tree ensembles are fit across the 13 regular-hours bars of each trading day using a rolling window of $W = 39{,}000$ bars (3,000 trading days), re-estimated at every bar; the pooled linear runs of the model set use all $M = 48$ half-hour intervals with $W = 96{,}000$ intervals. The models of Table 1, the linear models and the tree ensembles, are fit on the 13 regular-hours bars with a rolling window of 39,000 bars (3,000 trading days).
- **Benchmark forecast.** Our baseline benchmark is the closing-interval HAR benchmark, a HAR-type model in the spirit of Corsi (2009) on the target's trailing averages over $k \in \{1, 5, 25, 125, 625, 3125\}$ samples and the 16 calendar columns, estimated by OLS on closing-interval observations (Appendix G.3); Appendix G.5 examines all model classes trained analogously on the closing 30-minute interval alone.
- **Rolling re-estimation with frozen hyperparameters.** The pooled intraday models are re-estimated at every 30-minute interval (see Appendix B.2). Their hyperparameters are chosen once, before the first forecast, and then held fixed for the whole evaluation span, with no re-tuning (Appendix D): the linear penalties on the last pre-sample fold (validation year 2013), and for each tree ensemble the one configuration that an Optuna search selected on the pre-sample with a window of 24,000 intervals (Appendix B.3). On the HAR benchmark the frozen penalties at this window are 10^{-3} (ridge), 10^{-8} (lasso) and 10^{-7} (elastic net), numerically zero: the three penalized models reproduce the OLS forecast, and their rows in Table 1 coincide with the OLS row.
- **Evaluation metric and inference.** We evaluate predictive accuracy via the QLIKE loss against actual realized variance (see Section 5.3) and conduct pairwise inference using Diebold–Mariano tests with the Harvey–Leybourne–Newbold small-sample adjustment and a Newey–West long-run variance estimator ($q_{\mathrm{NW}} = 6$ on $\mathcal{T}_{\mathrm{oos}}$).[1]

We refer the reader to Appendix B for comprehensive implementation details.

6.2 Out-of-sample accuracy

- **Overview of forecasting performance.** Table 1 reports the out-of-sample MSE, average QLIKE loss, and out-of-sample R^2_{OOS} for each pooled intraday model class under both the HAR benchmark and the full predictor set across the 2,579 trading days from 2 January 2014 to 30 April 2024. The corresponding metrics for specifications estimated exclusively on the closing half-hour interval—including the OLS regression on the HAR benchmark predictor set, our closing-interval HAR benchmark forecast—are presented in Table 10 (see Appendix G.5). For the linear models, split at the start of 2020, the full predictor set has the lower QLIKE in both parts (2014–2019: 0.1709–0.1731 against 0.1864 for the HAR benchmark; 2020–2024: 0.1851–0.1875 against 0.1933), and a higher R^2_{OOS} on 2014–2019 (64.8–65.5% against 61.4%) but a lower one on 2020–2024 (17.5–24.5% against 66.1%), so its lower R^2_{OOS} over the whole span comes from 2020–2024.
- **Computation of R^2_{OOS}.** Following Campbell and Thompson (2008), the reference forecast in R^2_{OOS} is an expanding historical mean of realized variance computed from 5 January 1998 through day $t-1$. For the pooled models of Table 1, linear and tree ensembles, it is the mean

[1] Appendix I provides the complete specification of these statistical tests.

realized variance of the same half-hour bar over every earlier trading day; for the closing-interval models of Table 10 it is the mean realized variance of the closing half hour. Both rely solely on information available prior to the forecast origin.

6.3 The full predictor set versus the HAR benchmark

- **Paired comparison within each model class.** To isolate the predictive contribution of the exogenous features within each intraday model class m ($\mathcal{I} = \{1, \ldots, M\}$), we form the daily loss differential $d_t = L_t^{\text{Full}} - L_t^{\text{HAR}}$, where $L_t^{\mathcal{X}}$ denotes the QLIKE loss of the forecast $f_{m,\mathcal{X},\{1,\ldots,M\}}$. We report the percentage change in average loss $\Delta\%$ defined in Section 5.3, along with the Diebold–Mariano statistic DM computed on d_t over $\mathcal{T}_{\text{oos}}$ using lag truncation $q_{\text{NW}} = 6$.
- **QLIKE: full predictor set against the HAR benchmark.** Scored on every bar of Table 1, from 2 January 2014 to 30 April 2024, augmenting the HAR benchmark with the full predictor set lowers the average out-of-sample QLIKE loss in 6 of the 6 intraday model classes, 6 of them significantly at the 5% level. For the forecasts of the closing half hour over $\mathcal{T}_{\text{oos}}$, it lowers the loss in 3 of the 6 classes, the three tree ensembles, with 0 of these reductions statistically significant at the 5% level, and raises it for the three penalized linear models:
  - Regarding the tree ensembles, the reductions are $\Delta\% = -1.1$ for LightGBM (DM $= -0.39$, $p = 0.70$), -2.1 for XGBoost (DM $= -0.76$, $p = 0.44$), and -0.7 for the random forest (DM $= -0.26$, $p = 0.79$); the reduction is statistically significant at the 5% level in 0 of the three classes.
  - Regarding the penalized linear models, the full predictor set raises the QLIKE loss by $\Delta\% = 10.9$ for ridge regression (DM $= 2.31$, $p = 0.021$), 14.4 for the lasso (DM $= 2.51$, $p = 0.012$), and 8.5 for the elastic net (DM $= 1.91$, $p = 0.056$); at the 5% level the increase is statistically significant for ridge regression and the lasso and not for the elastic net.
- **Robustness to the window length.** With a rolling window of 3,500 trading days in place of 3,000 (Table 7), the full predictor set remains significantly more accurate than the HAR benchmark on every bar in 6 of the 6 model classes, and no pooled forecast's average QLIKE moves by more than 1.3% on every bar or 4.4% on the closing half hour. The changes that are statistically significant at the 5% level (5 of the 12 pooled forecasts on every bar, 3 on the closing half hour) are all in the penalized linear models; for example, the closing-half-hour QLIKE of the lasso on the full predictor set changes by -4.4% ($p = 0.004$).

6.4 Model classes

- **Exogenous features and the predictability of realized variance, by model class.** Comparing across model classes, the average QLIKE losses of the pooled penalized linear models and of the pooled tree ensembles on each predictor set are:
  - Under the full predictor set, the linear models attain average QLIKE losses between 0.1234 and 0.1301 (0.1234 for the elastic net, 0.1260 for ridge, and 0.1301 for the lasso), and the tree ensembles 0.1079–0.1123.
  - Under the HAR benchmark, the linear specifications achieve an average QLIKE loss of 0.1137, identical for the three of them because their HAR-benchmark penalties are numerically zero at this window, and the tree ensembles 0.1101–0.1131.
- **Comparison against the closing-interval HAR benchmark.** Every pooled intraday

specification incurs a higher average QLIKE loss than the closing-interval HAR benchmark ($\overline{\text{QLIKE}} = 0.1033$):

- Under the full predictor set, the average QLIKE loss of the pooled linear models exceeds that of the closing-interval HAR benchmark by +19.5 to +26.0%, and the difference is statistically significant at the 5% level for 3 of the three; for the pooled tree ensembles, the excess loss ranges from +4.5 to +8.8%, and it is significant at the 5% level for 1 of the three.
- Under the HAR benchmark, the excess QLIKE loss of the pooled intraday models ranges from +6.6 to +10.1%; the difference is statistically significant at the 5% level for 6 of the 6 models and at the 1% level for 4.
- Incorporating the exogenous covariates narrows the accuracy gap relative to the closing-interval HAR benchmark in 3 of the 6 intraday model classes, the tree ensembles, and widens it for the three penalized linear models.
- Because the closing-interval HAR benchmark is estimated solely on the closing 30-minute interval of each day, we extend this closing-interval training scheme to all model classes in Appendix G.5.

Table 1: Forecast accuracy of the intraday models, 2 January 2014 to 30 April 2024 (3,000-day window, hyperparameters frozen)

	HAR benchmark			All features			
Model	MSE (10^{-10})	QLIKE	R^2_{OOS} (%)	MSE (10^{-10})	QLIKE	R^2_{OOS} (%)	DM
OLS	1.11	0.1893	65.4	–	–	–	–
Ridge	1.11	0.1893	65.4	2.24	0.1791	30.4	−5.63 (<0.001)
Lasso	1.11	0.1893	65.4	2.37	0.1775	26.5	−4.72 (<0.001)
Elastic net	1.11	0.1893	65.4	2.43	0.1778	24.4	−6.98 (<0.001)
Tree ensembles							
LightGBM	1.18	0.1875	63.3	1.41	0.1613	56.1	−10.32 (<0.001)
XGBoost	1.16	0.1888	63.9	1.34	0.1627	58.2	−9.94 (<0.001)
Random forest	1.19	0.1895	63.0	1.32	0.1685	58.9	−9.85 (<0.001)

Notes: Pooled intraday linear models and tree ensembles, one design for every row: the HAR components at horizons $k \in \{1, 5, 25, 125, 625, 3125\}$, the full predictor set with the FOMC series, a rolling window of 3,000 trading days, re-estimated at every bar (Section 5), trained and scored on the 13 regular-hours half-hour bars ending 10:00 to 16:00 ET. Hyperparameters are chosen once and then frozen: the linear penalties on the last pre-sample fold (validation year 2013), before the first forecast; the tree ensembles' configurations, one per method, by an Optuna search on the pre-sample (Appendix B.3). Every bar is scored on the $T_{\text{oos}} = 2{,}579$ trading days from 2 January 2014 to 30 April 2024 (early closes dropped), 33,527 bars (bars with a non-positive realized variance dropped); each forecast $\hat{\sigma}^2_{t,i} = \hat{y}^2_{t,i}\, s^2_{t,i}$ is scored against the actual realized variance $\text{RV}_{t,i}$ of its interval. MSE (in units of 10^{-10}) and QLIKE are averages over the scored intervals. $R^2_{\text{OOS}} = 1 - \sum(\text{RV}_{t,i} - \hat{\sigma}^2_{t,i})^2 / \sum(\text{RV}_{t,i} - \overline{\text{RV}}_{t-1,i})^2$, in percent, where $\overline{\text{RV}}_{t-1,i}$ is the historical mean realized variance of the same half-hour interval over every earlier trading day from 5 January 1998 (Campbell and Thompson, 2008). DM: the Diebold–Mariano statistic, with the correction of Harvey et al. (1997), for the daily average QLIKE of the full predictor set minus that of the HAR benchmark of the same model (negative: the full set has the lower loss), Newey–West lag 8, p-value in parentheses. Models trained on the last 30 minutes are in Table 10.

6.5 Variable importance

RR. Decision (user, 2026-10-22/23): variable importance on the final pooled runs at the main window, the linear models' standardized coefficients and the tree ensembles' TreeSHAP values at 126 monthly origins; the preliminary pooled run (results/rehaul_fi_pooled13_2026-10-22) is superseded.
Still to come: the ablations (leave one category out, then one series out within the categories that matter; not run); the MinGRU network (parked, not covered); the trees' SHAP values at the 3,500-day robustness window (only the linear models' coefficients are computed there). The trees' accuracy across windows is in: 24 of the 24 walks at 1,000–3,500 days (Table 5). The forest's walk gate is complete (126 of 126 origins).

- **Questions.** Which predictors carry the pooled intraday forecasts, and does that change over time? Is the exogenous signal dense and weak, spread thinly over many series, or concentrated in a few? Do the series that matter do so only on some days, at some bars of the day, or through interactions with other predictors? And how do the nonlinear models, the tree ensembles, respond to the non-standard series, the ones outside the HAR benchmark, the futures' own realized moments and trading activity, and the VIX?
- **What is computed.** The pooled intraday models of the final design (rolling window of 3,000 trading days, 39,000 bars; full predictor set of 564 predictors in 48 groups: each of the 46 exogenous series with its 12 predictors, the HAR components and the calendar dummies), explained at monthly origins.
  - Origins $k = 1, \ldots, K$, $K = 126$: one every 21 trading days, at the 10:00 bar, from 2 January 2014 to 5 April 2024. Origin k's model forecasts its block $\mathcal{O}_k$, the 21 trading days up to the next origin (the last block: 18 days); the blocks partition all $|\mathcal{O}| = 33{,}970$ bars from 2 January 2014 to 30 April 2024. The penalties and the tree configurations were chosen on the last pre-sample fold, which ends on 31 December 2013, so no explained bar lies in a tuning fold.
  - Tree ensembles (LightGBM, XGBoost, random forest): at each origin the model is refit on the 39,000 bars before it, as its every-bar walk refits at that bar, and held for the block. Path-dependent TreeSHAP values (Lundberg et al., 2020) of every bar $(t, i) \in \mathcal{O}_k$, on the scale of the forecast, add up to it, and a group's value sums its predictors' values:

$$\hat{y}_{t,i} = \phi_{k,0} + \sum_{j=1}^{p} \phi_{t,i,j}, \qquad \phi_{t,i,A} = \sum_{j \in A} \phi_{t,i,j}, \qquad s_A = \frac{I_A}{\sum_B I_B}, \quad I_A = \frac{1}{|\mathcal{O}|} \sum_{(t,i) \in \mathcal{O}} |\phi_{t,i,A}|,$$

    with $\phi_{k,0}$ the model's expected forecast over its own training sample (the trees' node covers), so no background sample is chosen.
  - Linear models (ridge, lasso, elastic net): the walk's own coefficients at each origin on the window-standardized predictors, which the penalty acts on, $\gamma_{k,j} = \hat{\beta}_{k,j}\, \mathrm{sd}_{k,j}$, and a group's share $S_A = \sum_k \sum_{j \in A} |\gamma_{k,j}| / \sum_k \sum_j |\gamma_{k,j}|$; penalties as frozen for this window (ridge 10^3, lasso 10^{-4}, elastic net 10^{-3}), with the penalty factor 0.01 on the HAR components and the calendar dummies.
  - Two classes, fixed before any SHAP value was computed. Standard (12 groups): the HAR components, the calendar dummies, the futures' seven realized moments, the VIX, and the futures' volume and trade count. Non-standard: the other 36 series (constituent liquidity, the equal- and value-weighted market moments, StockTwits sentiment, volatility demand, the FOMC calendar, VVIX and VIX3M).
  - Intervals: 95% moving-block bootstrap over the origins (blocks of 12 origins, about a year; 2,000 resamples), paired across models. They measure the variation over the span's origins, not the sampling error of one fit.

- **Scope.**
  - The trees and the linear models are measured differently (mean absolute SHAP value against $|\gamma|$), so the comparison is of rankings and of shares of each model's own measure, not of magnitudes. With the penalty factor on the HAR block, the linear models' HAR shares are not comparable with the trees'; comparisons across model types use the exogenous total.
  - A SHAP share measures how much a series moves the forecast, not how much it improves its accuracy.
  - Between correlated predictors (the equal- and value-weighted versions of a series, a series' value and availability averages) the credit follows the trees' split choices; summing over a group removes the split within the group, not the split between groups.
  - SHAP explains each origin's refit held for its block, whereas the walks refit at every bar; the two coincide at the origin bars. There the refits equal the walks' forecasts bit for bit at 126, 126 and 126 of the 126 origins (LightGBM, XGBoost, random forest). The linear coefficients are the walks' own (their replayed forecasts match to 8.9×10^{-16}).
  - XGBoost computes SHAP values in single precision: they add up to the forecast within 2.9×10^{-6} (tolerance 10^{-5}); LightGBM's and the forest's within 4.8×10^{-14}.
  - Not covered yet: the MinGRU network (parked), and the ablations (leave one category out, then one series out within the categories that matter), which are not run.
- **Which series matter** (Table 2, Figure 1).
  - In the trees the HAR components carry 51.6–57.6% of the total attribution and the futures' realized moments 20.8–23.6%. The two leading exogenous series are the same in all three, `sumabsret` and `sumret`; `sumabsret` alone holds 20.3, 24.8 and 29.2% of the exogenous total.
  - The non-standard series hold 21.8% [20.4, 23.4], 20.6% [19.3, 22.2] and 18.6% [17.2, 20.3] of the total (LightGBM, XGBoost, random forest), 43.7–45.9% of the exogenous total. Their weight sits in a few series; the five largest by the mean over the three trees are `sumabsret_vwstock` (2.4–3.9% of the total), `sumabsret_ewstock` (1.9–3.9%), `fomc_until_inv` (1.1–2.0%), `ofi_vwstock` (1.4–1.8%) and `effspread_vwstock` (1.2–1.3%). The rest carry little: the StockTwits series together 0.6–0.8%, the volatility-demand series together 0.2–0.3%, VVIX 0.2–0.3%, VIX3M 0.3–0.7%.
  - The ridge ranks the series differently. Its largest exogenous series is `sumret` (6.2% of its exogenous total), and it puts the FOMC calendar near the top: `fomc_until_inv` at rank 2 and `fomc_day` at rank 4 of 46, where the trees rank them 6–11 and 38–43. Spearman rank correlation of the 46 shares with the ridge's: 0.59, 0.59, 0.55 (the two top-10 lists share 6, 6 and 4 series); among the trees: 0.99 (LightGBM, XGBoost), 0.90 and 0.89 (each against the forest).
- **Dense and weak, or concentrated?**
  - Linear models across windows (Table 4). If the exogenous signal were dense and weak, more observations per coefficient would bring more series in: a longer window would select more exogenous series (P1) and spread the weight over more of them (P2). The test refits nothing new: the walks at W from 1,000 to 3,500 trading days, at the same 126 origins, with the predictions and the rule fixed before computing. At a penalty held fixed across windows, none of the 105 statistics rises with W (102 fall, 3 stay flat); for example the lasso at 10^{-3} selects 38.7 exogenous series on average at $W = 1{,}000$ and 31.9 at 3,500. At the penalties tuned for each window, P1 is "supported" for the elastic

net and "partly" for the lasso, but the counts rise only where the tuned penalty falls tenfold (elastic net: from 10^{-2} at $W = 1{,}000$ to 10^{-3} at 1,500); the ridge's weight does not spread (N_{eff} 29.2 at 1,000 and 28.2 at 3,500 days, slope −0.10 per 1,000 days [−0.25, 0.12]). The dense-and-weak reading is not supported for the linear models. A falling count is also what a signal that changes over time would produce, since a longer window averages over more regimes; the study does not test that, so it does not show a sparse and strong signal either.

  - Tree ensembles across windows (Table 5; descriptive: no rule fixed this comparison before the walks). The trees' SHAP values are computed at the main window only, so for them the test is on accuracy: if the exogenous signal were dense and weak, the full predictor set would gain more from a longer window than the HAR benchmark does. The walks share the design of Table 1, at $W = 1{,}000$, 2,000, 3,000 and 3,500 trading days (refit at every bar, frozen configurations, the same 33,527 scored bars). The extra gain from W_1 to W_2 is the change of the full set's daily QLIKE minus the HAR benchmark's (negative: the full set gains more), with a Newey–West t over the 2,579 days. On every bar the full set is more accurate than the benchmark at every window (t from −10.64 to −9.41), its lead widens up to 3,000 days in all three trees, and the extra gain is negative and significant at the 5% level for all three trees from 1,000 to 2,000 days ($t = -6.71$, −2.58 and −4.65 for LightGBM, XGBoost and the forest), for LightGBM and the random forest from 2,000 to 3,000 ($t = -2.15$ and −3.87; XGBoost −1.12) and for none of the trees from 3,000 to 3,500 days. The full set's own loss falls significantly from 1,000 to 2,000 days in all three trees (DM −7.59, −3.67, −5.97) and from 2,000 to 3,000 in the random forest (DM −3.37); LightGBM's extra gain at that step comes mostly from the benchmark's loss rising (DM 1.77). At the 16:00 bar the full set's lead is significant at the 5% level in 3 of the 12 model–window cells, and the extra gain has no consistent sign: it is significant toward the full set in 2 of the 9 tree–window steps (LightGBM and XGBoost from 2,000 to 3,000 days, $t = -3.72$ and −2.64), toward the benchmark in 1 (LightGBM from 1,000 to 2,000 days, $t = 2.51$), and never for the forest. On every bar, then, the full set's advantage grows with the window until 3,000 days and then stops, as a dense and weak signal would predict; a signal that needs more data to be fitted nonlinearly would do the same, and accuracy alone does not say which series the longer window brings in.
  - Concentration at the main window (Table 2, Panel D; descriptive, computed from the pooled shares). The effective number of exogenous series, $N_{\text{eff}} = 1/\sum_A s_A^2$ over the exogenous shares, is 11.0, 9.1 and 8.9 for the trees against 31.0 for the ridge (lasso 12.2, elastic net 17.2); half of the trees' exogenous attribution sits in 4, 4 and 4 series, half of the ridge's in 11. The trees lean on a handful of series; the ridge spreads its weight more widely, as an L_2 penalty tends to across correlated predictors.

- **Trees against linear models on the non-standard series** (Table 2, Panels B and C). Of the exogenous total, the non-standard series hold 45.9, 43.7 and 44.3% in the trees, against 67.3% in the ridge, 73.9% in the lasso and 56.0% in the elastic net. The difference, tree minus ridge, is $D = -21.4$ [−23.4, −19.6], −23.6 [−25.5, −21.7] and −23.0 [−24.8, −21.4] percentage points, and the interval lies below zero in all 27 of 27 comparisons (three trees, three linear models, three denominators: the exogenous total, the total, and the exogenous total with the futures' volume and trade count counted as non-standard). The trees lean on the non-standard series less than the linear models do, not more.
- **Over time** (Figure 2). Every tree's non-standard share falls below its pooled value from 2018

to 2022 and is lowest in 2021 (18.4, 17.5 and 15.2% of the total); the HAR components gain in the same years (LightGBM: 57.2% in 2021, against 44.9% in 2017). The ridge's non-standard share of its exogenous total also dips, to its lowest in 2021 (63.8%, against 70.9% in 2017; linear models dated by the origin). Panels (b) and (c) show the leading series year by year.

- **Only sometimes, or conditionally?** (Figure 3). The trees split on most series at every origin (42, 42 and 46 of the 46); what changes with the conditions is how much the series move the forecast.
  - FOMC statement days (78 days in the span): the non-standard share is 41.7, 40.1 and 28.1% of the total, against 20.9, 19.8 and 18.3% on other days (differences +20.8 [+19.0, +22.5], +20.3 [+18.3, +22.1] and +9.8 [+7.7, +12.0] percentage points). Almost all of it is `fomc_until_inv`: 25.1, 22.9 and 11.4% of the total on statement days, against 1.05 and 1.15% on other days (LightGBM, XGBoost).
  - Bar of the day: the non-standard share is highest at the first bar, 10:00 (24.0, 22.7, 20.2%), and at 14:30 and 15:00 (LightGBM 24.3–25.2%), the bars where the FOMC calendar's share peaks (LightGBM 5.5%, against a median of 1.5% over the other bars). It is lower at the last bar: 16:00 minus 10:00 −3.5 [−4.4, −2.7], −3.1 [−4.0, −2.3] and −2.3 [−3.2, −1.6] percentage points.
  - Volatility regime (tercile of the VIX close before the origin): the non-standard share is lower in the high-VIX tercile than in the low one, −2.9 [−5.1, −0.7], −2.8 [−4.8, −0.7] and −3.3 [−5.1, −1.4] percentage points of the total, and the HAR components gain instead (+4.9, +3.7, +4.2). As a share of the exogenous total the drop is clear for XGBoost and the forest and not for LightGBM (−1.9 [−4.1, +0.2]).
- **Shape of the response** (Table 3, Figure 4). For each non-standard series with a share of at least 0.5% in a tree (11, 10 and 13 series), the mean SHAP value by decile of its dominant predictor is classified by a rule fixed in advance: monotone 7, 6 and 10; irregular 3, 3 and 3; a step 1, 0 and 0; flat 0, 1 and 0; none U-shaped, inverted-U or a threshold. Most responses are one-sided (asymmetric: 9, 7 and 13).
  - The market's cross-sectional absolute returns and the spreads rise monotonically and are asymmetric toward the top tail: `sumabsret_vwstock` moves the forecast from −1.4% of the mean forecast in its bottom decile to 3.9% in its top decile (LightGBM; XGBoost 4.3%, forest 2.3%).
  - `ofi_vwstock` falls monotonically: a larger order-flow imbalance lowers the forecast, from 0.8% to −1.1% (LightGBM).
  - `fomc_until_inv` (dominant predictor: its average over $k = 25$ samples) is labeled irregular by the rule; read from the figure (post hoc), it is flat through the lower deciles and jumps in the top decile, the approach of a statement, to 2.9, 2.8 and 1.4% of the mean forecast.
- **Interactions** (Table 6; boosted trees, 1,008 bars, descriptive). Interactions carry 33.1 and 31.0% of the attribution at the group level (45.4 and 42.6% at the predictor level). They carry 49.7 and 45.7% of a non-standard series' attribution against 26.8 and 26.0% for the standard groups; more than half of that gap comes from the HAR components, which act mostly on their own (17.0% through interactions in LightGBM), and the calendar dummies: the standard exogenous series alone reach 40.1 and 37.8% (descriptive). The largest pairs with a non-standard series pair it with the HAR components, HAR components × `sumabsret_vwstock` (1.26 and 1.23% of the pair total) and HAR components × `sumabsret_ewstock` (0.86 and 0.83%), and the HAR components are in 7 and 8 of the ten largest. The weakest series

act mostly through interactions: 55–66% of the attribution of the StockTwits series, VVIX, VIX3M and `fomc_day` (LightGBM).

- **In short.** In the trees, the HAR components and the futures' realized moments carry most of the attribution. For the linear models the exogenous signal is not dense and weak in the sense tested: a longer window does not bring more series in. The trees concentrate their exogenous attribution on a few series, yet on every bar their full predictor set's lead over the HAR benchmark widens with the window until 3,000 trading days and then stops (at the 16:00 bar the change has no consistent sign), as a dense and weak signal would predict. The non-standard series matter conditionally, on FOMC statement days, at the first bar and at the bars where the FOMC calendar peaks, and in calm markets, and about half of their attribution runs through interactions, the largest of them with the HAR components. The tree ensembles use them less than the linear models do, through monotone, one-sided responses.

7 Economic value of the forecasts

7.1 The straddle strategy

- **Option-trading sample.** We evaluate the strategy on the option-trading subsample $\mathcal{T}_{\text{opt}} \subset \mathcal{T}_{\text{oos}}$ (Section 6.1), the trading days with a same-day SPXW expiration:

$$T_{\text{opt}} = 866 \text{ trading days} = 872 \text{ SPXW expirations} - 6 \text{ early-close days}.$$

  The 419 subsequent SPXW expiration dates (413 full trading days) are held out from the present analysis.
- **Position.** We trade same-day-expiring (0DTE) SPXW options. With S_t denoting the underlying print on the option chain at the 15:30 ET forecast origin, we select the call and put at the nearest adjacent strikes with live quotes:

$$K_C = \min\{K \geq S_t\}, \qquad K_P = \max\{K \leq S_t\}.$$

  When $K_C = K_P = S_t$, the package forms an at-the-money straddle; otherwise, it forms a narrow strangle on the two strikes bracketing S_t. We refer to both positions as the straddle: an approximately delta-neutral package entered at 15:30 ET and held unhedged until expiration at 16:00 ET. Every 15:30 chain in $\mathcal{T}_{\text{opt}}$ has a call at or above and a put at or below S_t with a positive mid-quote. We also screen each chain for a quote outage, that is, fewer than 10 contracts with a positive mid-quote or a wider strike more than 10 index points from S_t (more than one strike missing near the money); no chain in the sample fails this screen, so no day is dropped for want of quotes; early-close days are already outside $\mathcal{T}_{\text{opt}}$.
- **Entry prices.** From the legs' bid quotes b and ask quotes a at 15:30 ET, we define the package mid-quote P_t, bid P_t^b, and ask P_t^a as

$$P_t = \tfrac{1}{2}(b_t^C + a_t^C) + \tfrac{1}{2}(b_t^P + a_t^P), \qquad P_t^b = b_t^C + b_t^P, \qquad P_t^a = a_t^C + a_t^P.$$

- **Settlement.** The package settles in cash at the official S&P 500 close S_t^{close} with no intraday delta hedging; the simple return on the mid-quote premium paid at entry is

$$X_t = \left(S_t^{\text{close}} - K_C\right)^+ + \left(K_P - S_t^{\text{close}}\right)^+, \qquad R_t = \frac{X_t}{P_t} - 1 \;\geq\; -1.$$

Table 2: Attribution by category and class: tree ensembles and linear models

	Tree ensembles: mean $\lvert\phi\rvert$			Linear models: $\lvert\gamma\rvert$		
	LightGBM	XGBoost	Random forest	Ridge	Lasso	Elastic net
Panel A: Share of the total attribution, by category (%)						
HAR components	51.6 [49.5, 53.6]	51.9 [50.0, 53.7]	57.6 [55.4, 59.6]	18.4	9.9	28.6
Calendar dummies	0.9 [0.8, 1.2]	0.9 [0.8, 1.2]	0.3 [0.2, 0.4]	1.8	1.1	2.4
Realized moments	22.3 [21.3, 23.4]	23.6 [22.6, 24.6]	20.8 [19.8, 21.7]	19.6	16.3	23.0
Liquidity	8.8 [8.3, 9.5]	8.0 [7.4, 8.6]	9.4 [8.8, 10.0]	20.0	14.0	11.5
Market moments, EW	5.6 [5.0, 6.1]	5.1 [4.7, 5.6]	3.8 [3.5, 4.2]	10.4	7.0	6.8
Market moments, VW	5.1 [4.5, 5.8]	5.3 [4.7, 6.0]	4.5 [4.0, 5.0]	10.6	7.9	8.8
Sentiment	0.8 [0.6, 0.9]	0.7 [0.6, 0.8]	0.6 [0.5, 0.7]	1.8	0.8	0.8
Implied volatility	2.3 [2.0, 2.6]	2.0 [1.7, 2.3]	1.4 [1.3, 1.6]	6.4	6.8	7.1
Volatility demand	0.3 [0.2, 0.4]	0.2 [0.1, 0.3]	0.2 [0.2, 0.4]	1.2	0.7	0.5
FOMC announcements	2.3 [2.1, 2.6]	2.2 [2.0, 2.5]	1.3 [1.2, 1.5]	9.9	35.3	10.6
Panel B: Share by class (%)						
Standard	78.2 [76.6, 79.6]	79.4 [77.8, 80.7]	81.4 [79.7, 82.8]	46.3	34.3	61.3
Non-standard	21.8 [20.4, 23.4]	20.6 [19.3, 22.2]	18.6 [17.2, 20.3]	53.7	65.7	38.7
of the exogenous total	45.9 [44.3, 47.7]	43.7 [41.8, 45.6]	44.3 [42.2, 46.5]	67.3	73.9	56.0
Panel C: D, tree minus linear model, non-standard share of the exogenous total (pp)						
vs ridge	−21.4 [−23.4, −19.6]	−23.6 [−25.5, −21.7]	−23.0 [−24.8, −21.4]			
vs lasso	−28.0 [−30.4, −25.9]	−30.2 [−32.4, −28.2]	−29.6 [−31.6, −27.8]			
vs elastic net	−10.1 [−11.7, −8.6]	−12.3 [−13.8, −10.8]	−11.8 [−13.4, −10.0]			
Panel D: Concentration and rank agreement over the 46 exogenous series						
N_{eff}	11.0	9.1	8.9	31.0	12.2	17.2
Series holding half	4	4	4	11	5	6
Series holding 80%	13	11	17	27	18	16
Spearman, ridge	0.59 (6)	0.59 (6)	0.55 (4)			
Spearman, lasso	0.50 (5)	0.47 (5)	0.43 (3)			
Spearman, elastic net	0.69 (7)	0.70 (7)	0.64 (5)			

Notes: Pooled intraday models of the final design, rolling window of 3,000 trading days, full predictor set (564 predictors in 48 groups: 46 exogenous series, the HAR components, the calendar dummies), re-estimated at 126 monthly origins (2 January 2014 to 5 April 2024); together the origins' models forecast all 33,970 bars from 2 January 2014 to 30 April 2024. Tree ensembles: the share s_A of group A in the total mean absolute TreeSHAP value over all bars, ϕ_A summed over the group's predictors, with 95% moving-block bootstrap intervals over the origins (blocks of 12 origins, 2,000 resamples) in brackets. Linear models: the share of the absolute coefficients on the window-standardized predictors, $\lvert\gamma_{k,j}\rvert = \lvert\hat{\beta}_{k,j}\rvert\,\mathrm{sd}_{k,j}$, summed over the origins (penalties: ridge 10^3, lasso 10^{-4}, elastic net 10^{-3}); the HAR components carry the penalty factor 0.01, so their linear shares and the linear "of the total" rows are not comparable with the trees'. The two measures differ, so the table compares shares of each model's own measure. Standard: the HAR components, the calendar dummies, the seven futures realized moments, the VIX, the futures volume and trade count (12 groups); non-standard: the other 36 series. *D*: paired bootstrap interval. Panel D, computed here from the pooled shares (descriptive): N_{eff}, the effective number of series, and the fewest series whose shares sum to half and to 80% of the exogenous total; Spearman rank correlation of the 46 shares with the linear model's, and in parentheses the overlap of the two top-10 lists.

Table 3: The leading non-standard series in the tree ensembles

	Share (%)			Rank		Response shape			Interaction (%)	
Series	LGBM	XGB	RF	Trees	Ridge	LGBM	XGB	RF	LGBM	XGB
Panel A: Non-standard series with a share of at least 0.5% in some tree ensemble										
`sumabsret_vwstock`	3.48	3.91	2.36	3	7	$M\uparrow^{t}$	$M\uparrow^{t}$	$M\uparrow^{t}$	38	35
`sumabsret_ewstock`	3.93	3.83	1.92	4	11	$M\uparrow^{t}$	$M\uparrow^{t}$	$M\uparrow^{t}$	34	33
`fomc_until_inv`	2.01	2.01	1.05	6	2	I^{t}	I^{t}	I^{t}	32	33
`ofi_vwstock`	1.80	1.69	1.40	7	12	$M\downarrow$	$M\downarrow$	$M\downarrow^{b}$	40	38
`effspread_vwstock`	1.26	1.25	1.27	10	30	$M\uparrow^{t}$	$M\uparrow^{t}$	$M\uparrow^{t}$	41	38
`sumret3_vwstock`	0.90	0.86	1.06	12	21	$M\downarrow^{b}$	$M\downarrow^{b}$	$M\downarrow^{b}$	50	47
`sumret3_ewstock`	0.71	0.63	0.83	14	40	I^{b}	I^{b}	$M\downarrow^{b}$	58	55
`sellturnover_vwstock`	0.72	0.72	0.69	13	20	I^{t}	I^{t}	I^{t}	56	53
`effspread_ewstock`	0.68	0.57	0.70	15	33	$M\uparrow^{t}$	$M\uparrow$	$M\uparrow^{t}$	53	48
`spread_vwstock`	0.52	0.50	0.63	17	10	$M\uparrow^{t}$	–	$M\uparrow^{t}$	53	48
`vix3m`	0.67	0.55	0.27	16	18	$S\uparrow$	F	–	58	54
`sellturnover_ewstock`	0.38	0.36	0.55	21	39	–	–	I^{t}	68	67
`spread_ewstock`	0.42	0.30	0.56	21	15	–	–	$M\uparrow^{t}$	58	56
`ofi_ewstock`	0.24	0.24	0.63	25	28	–	–	$M\downarrow^{b}$	69	62
Panel B: Non-standard series in the ridge's top ten, below 0.5% in every tree ensemble										
`sumret2_ewstock`	0.39	0.29	0.26	23	8	–	–	–	65	66
`fomc_day`	0.12	0.13	0.10	42	4	–	–	–	61	64

Notes: Pooled intraday models of the final design, rolling window of 3,000 trading days, full predictor set (564 predictors in 48 groups: 46 exogenous series, the HAR components, the calendar dummies), re-estimated at 126 monthly origins (2 January 2014 to 5 April 2024). Share: of the total mean absolute SHAP value (LGBM: LightGBM, XGB: XGBoost, RF: random forest). Rank: among the 46 exogenous series, the median over the three tree ensembles and the ridge's (by its share of $|\gamma|$, Table 2). Response shape: the mean SHAP value of the series in each decile of its dominant predictor (the value average with the largest mean absolute SHAP), classified by the rule fixed before computing (Figure 4): M↑/M↓ monotone up/down (absolute Spearman correlation of the decile means with the decile index at least 0.9), S step (three bins or fewer), F flat (the decile means move less than half the mean absolute SHAP value), I irregular; t/b: asymmetric, the top/bottom arm at least twice the other; – below 0.5% in that model, not classified. No profile was classified as a threshold, U-shaped or inverted-U. Interaction: the share of the series' SHAP interaction attribution that sits off the diagonal (with other groups), on 1,008 bars (8 drawn per origin); not computed for the random forest.

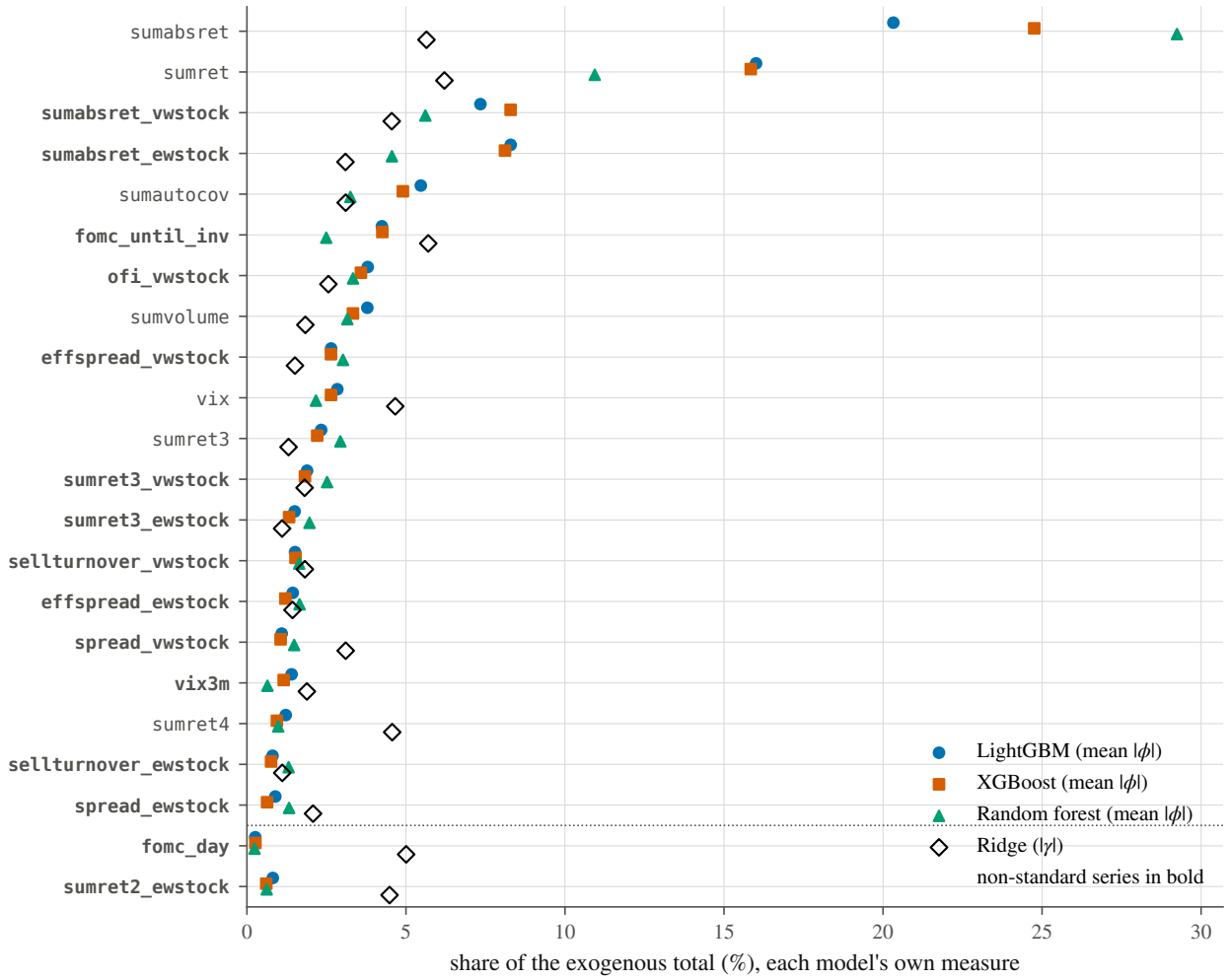


Figure 1: Which exogenous series the models use. Each series' share of the exogenous attribution, pooled over the 126 monthly origins: the tree ensembles' share of the mean absolute SHAP value (filled markers) and the ridge's share of the absolute standardized coefficients $|\gamma|$ (open diamonds). Rows: the 20 series with the largest mean share over the three tree ensembles, in that order, then (below the dotted line) the two series in the ridge's top ten that are not among them. Non-standard series are labeled in bold. The two measures differ; the figure compares rankings and shares, not magnitudes (Table 2).

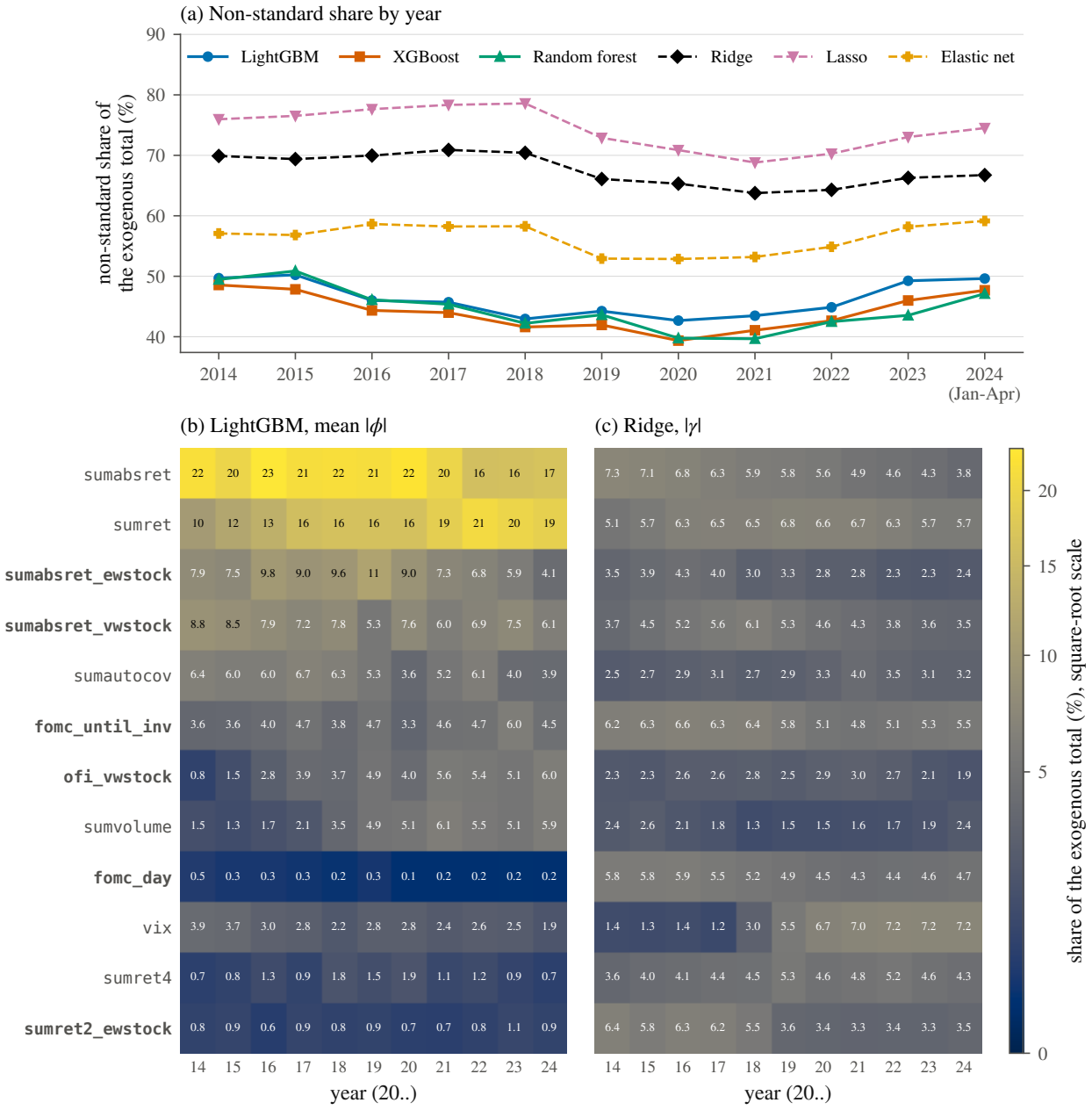


Figure 2: Variable importance over time. (a) The 36 non-standard series' share of each model's exogenous attribution by calendar year: tree ensembles (solid) by the year of the forecast bar, linear models (dashed) by the year of the origin whose coefficients enter. (b), (c) The share of the exogenous total by year for the leading series of LightGBM and of the ridge (the union of each model's eight largest pooled shares; non-standard series in bold), on one square-root color scale. Shares of each model's own measure: mean absolute SHAP value for the trees, $|\gamma|$ for the linear models. The linear shares by year are computed here from the coefficients at the origins (descriptive). 2024 covers January to April.

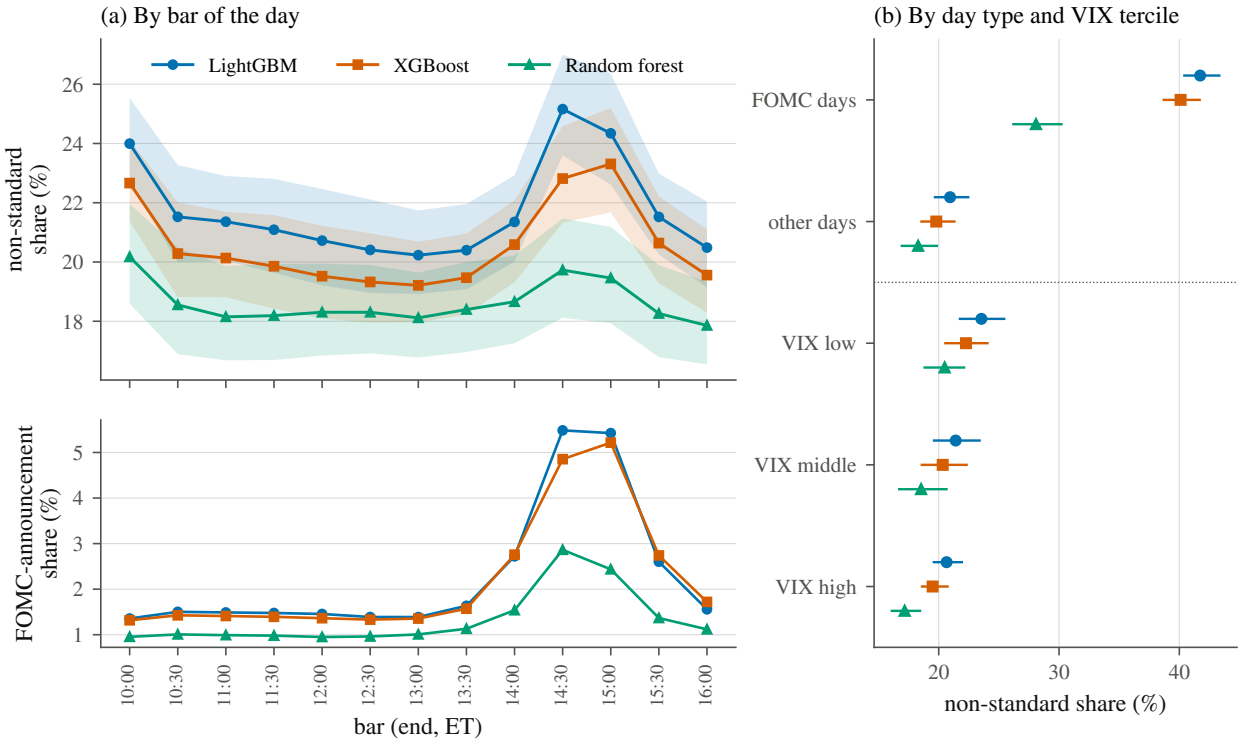


Figure 3: When the tree ensembles use the non-standard series. Share of the non-standard series in the total mean absolute SHAP value on subsets of the 33,970 bars, with 95% block-bootstrap intervals over the origins. (a) By bar of the day (top; bands: intervals) and the FOMC-announcement category's share (bottom). (b) On the 78 scheduled FOMC statement days and on all other days, and by the tercile of the VIX close before the origin (cut points 13.96 and 19.02; every bar of an origin's block takes that origin's tercile).

Table 4: Does a longer window bring in more exogenous series? The linear models across training windows

		Series selected		N_{eff}, mean over origins			Prediction supported	
Model	Penalty	W = 1,000	3,500	W = 1,000	3,500	Slope[a]	P1	P2
Ridge	tuned: $10^4, 10^3$	–	–	29.2	28.2	−0.10 [−0.25, 0.12]	–	no (0/3/6)
	fixed: 10^3	–	–	33.9	28.2	−2.33 [−2.56, −2.11]	–	no (0/0/9)
	fixed: 10^4	–	–	29.2	23.5	−2.37 [−2.68, −2.03]	–	no (0/0/9)
	fixed: $10^4\,W$/1,000	–	–	29.2	22.4	−2.79 [−3.08, −2.43]	–	no (0/0/9)
Lasso	tuned: $10^{-3}, 10^{-4}$	38.8	41.9	16.7	10.9	−2.28 [−2.52, −1.89]	partly (5/1/0)	no (0/0/9)
	fixed: 10^{-4}	45.6	41.9	18.3	10.9	−2.89 [−3.46, −2.49]	no (0/0/6)	no (0/0/9)
	fixed: 10^{-3}	38.7	31.9	16.7	11.2	−2.15 [−2.38, −1.85]	no (0/1/5)	no (0/0/9)
Elastic net	tuned: $10^{-2}, 10^{-3}$	19.1	37.1	6.3	14.7	1.89 [1.51, 2.22]	yes (6/0/0)	yes (9/0/0)
	fixed: 10^{-3}	43.2	37.1	20.5	14.7	−2.17 [−2.74, −1.74]	no (0/0/6)	no (0/0/9)
	fixed: 10^{-2}	19.2	11.7	6.3	4.6	−0.55 [−0.73, −0.37]	no (0/1/5)	no (0/1/8)

Notes: The pooled ridge, lasso and elastic net on the full predictor set, rolling windows W = 1,000, 1,500, 2,000, 2,500, 3,000, 3,500 trading days, each walk's coefficients at the same 126 monthly origins; tuned: each W's walk at the penalty its pre-sample tuning chose (listed from the shortest to the longest window where it changes); fixed: the fit at the origin at one penalty for every W (ridge also at a penalty proportional to W). Series selected: mean number of exogenous series with a nonzero coefficient (the ridge keeps every series that has data). N_{eff}: effective number of exogenous series, $1/\sum_A w_A^2$ with w_A the series' share of $\sum_j |\gamma_j|$ over the exogenous predictors at an origin; [a]slope of the per-W value on W, per 1,000 days, with its 95% paired moving-block bootstrap interval. If the exogenous signal were dense and weak, a longer window would select more series (P1, lasso and elastic net: series and predictors selected, mean, median and in half the origins; six measures) and spread the weight over more series (P2: N_{eff} and the fewest series holding half and 80% of the weight, per origin and pooled; nine measures). Supported: every measure rises; partly: some rise; no: none rises; in parentheses the numbers of measures that rise, stay flat and fall. Predictions, measures and rule were fixed before computing.

Table 5: Does the full predictor set gain more from a longer window? The tree ensembles across training windows

	Gap, full minus HAR, by window W				Extra gain, from W_1 to W_2		
Model	1,000	2,000	3,000	3,500	1,000 → 2,000	2,000 → 3,000	3,000 → 3,500
Panel A: every bar							
LightGBM	-19.16^{***}	-25.32^{***}	-26.23^{***}	-26.22^{***}	-6.16^{***}	-0.91^{**}	0.01
	(−10.64)	(−10.47)	(−10.32)	(−10.03)	(−6.71)	(−2.15)	(0.04)
XGBoost	-23.94^{***}	-25.52^{***}	-26.05^{***}	-25.43^{***}	-1.58^{***}	−0.53	0.63
	(−9.72)	(−10.08)	(−9.94)	(−9.97)	(−2.58)	(−1.12)	(1.32)
Random forest	-14.36^{***}	-18.65^{***}	-20.97^{***}	-21.23^{***}	-4.28^{***}	-2.33^{***}	−0.26
	(−10.61)	(−10.26)	(−9.85)	(−9.41)	(−4.65)	(−3.87)	(−0.60)
Panel B: the 16:00 bar							
LightGBM	-3.57^{*}	−0.71	-3.38^{*}	−2.93	2.86^{**}	-2.67^{***}	0.44
	(−1.79)	(−0.38)	(−1.81)	(−1.62)	(2.51)	(−3.72)	(0.92)
XGBoost	−1.93	−2.28	-4.18^{**}	-4.00^{**}	−0.34	-1.90^{***}	0.18
	(−1.03)	(−1.22)	(−2.18)	(−2.19)	(−0.34)	(−2.64)	(0.37)
Random forest	-3.64^{**}	−1.24	−1.80	−2.29	2.39^{*}	−0.56	−0.49
	(−2.05)	(−0.62)	(−0.92)	(−1.15)	(1.68)	(−0.54)	(−0.71)

Notes: The pooled LightGBM, XGBoost and random forest on the HAR benchmark and on the full predictor set, refit at every bar on rolling windows of W trading days, each on its frozen configuration, first evaluation day 2 January 2014; QLIKE on the 33,527 bars of the 2,579 evaluation days (the scored set of Table 1, the same at every window; Panel B: their 16:00 bars), in units of 10^{-3}. Gap: the mean over days of the full set's daily QLIKE minus the HAR benchmark's (negative: the full set is more accurate). Extra gain: the mean over days of the change of the full set's daily QLIKE from W_1 to W_2 minus the HAR benchmark's change (negative: the full set gains more from the longer window). In parentheses: Newey–West t-statistics (Bartlett kernel, lag 8); stars: two-sided normal p below 10%, 5% and 1%. Descriptive: no rule fixed this comparison before the walks.

Table 6: SHAP interactions in the boosted trees

	LightGBM	XGBoost
Panel A: Interaction share of the attribution (%)		
All groups, group level	33.1	31.0
All predictors, predictor level	45.4	42.6
Non-standard series	49.7	45.7
Standard groups	26.8	26.0
standard exogenous series only[a]	40.1	37.8
HAR components	17.0	17.7
Panel B: Largest pairs with a non-standard series, share of the pair total (%)		
HAR components × `sumabsret_vwstock`	1.26	1.23
HAR components × `sumabsret_ewstock`	0.86	0.83
HAR components × `sellturnover_vwstock`	0.31	0.36
HAR components × `effspread_vwstock`	0.30	0.31
HAR components × `stocktwits_sentiment`	0.28	0.31
`sumret` × `sumabsret_ewstock`	0.26	0.22
HAR components × `fomc_until_inv`	0.25	0.40
`sumabsret` × `sumabsret_vwstock`	0.25	0.29
HAR components × `ofi_vwstock`	0.23	0.30
`sumret` × `sumabsret_vwstock`	0.20	0.21

Notes: LightGBM and XGBoost of Table 2; SHAP interaction values ϕ_{jl} on 1,008 bars, 8 drawn at random from each origin's block and explained by that origin's model; summed over the predictors of each group, $G_{AB} = \sum_{j \in A} \sum_{l \in B} \phi_{jl}$. Panel A: the share of $\sum |G_{AB}|$ off the diagonal ($A \neq B$), for all groups and at the predictor level, and pooled over the members of each class. [a]Computed here from the per-series file (descriptive): the standard class without the HAR components and the calendar dummies. Panel B: the ten largest pairs by LightGBM, mean $|2G_{AB}|$ as a share of the total over all pairs $A \leq B$ (diagonal included). No intervals: the interaction values are descriptive.

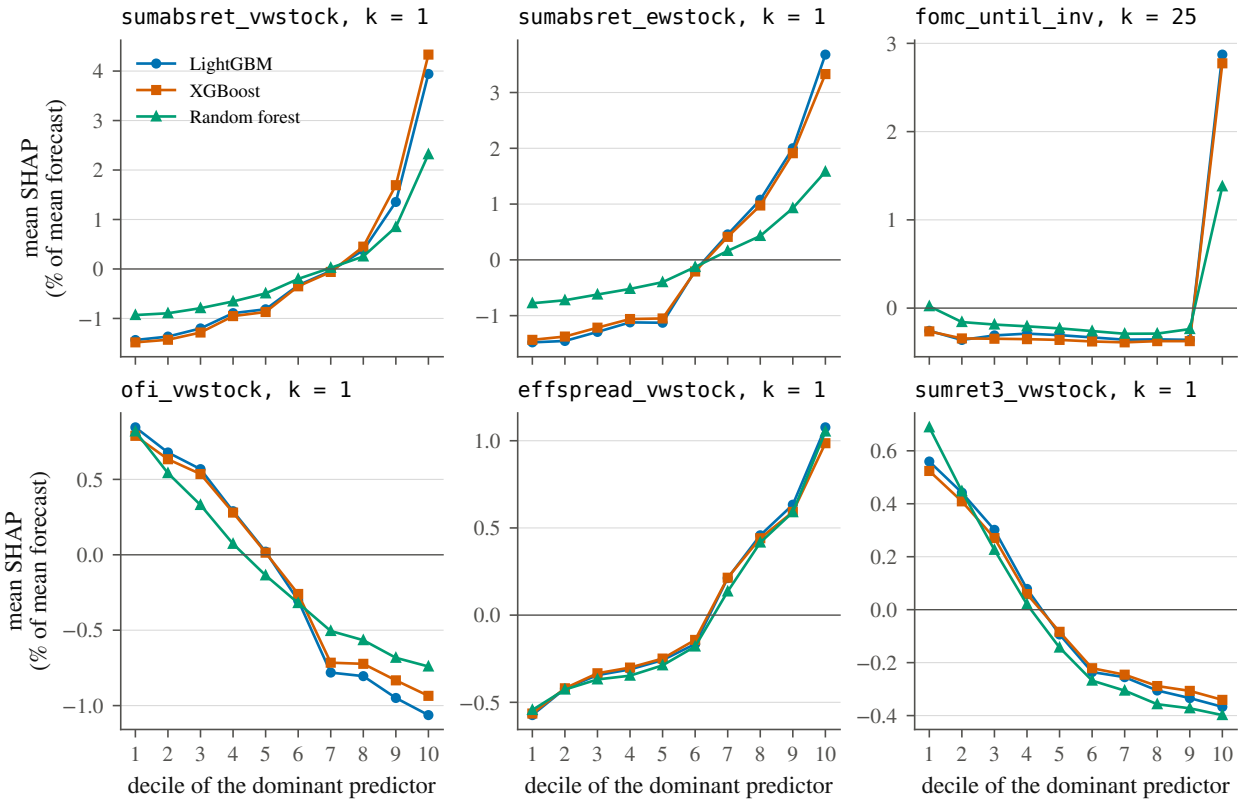


Figure 4: Response shapes of the leading non-standard series. For each series, the mean SHAP value of the series (all its predictors) in each decile of its dominant predictor, the value average at horizon k with the largest mean absolute SHAP value, as a percentage of the mean forecast; over all 33,970 bars. The predictor values are the robust-scaled inputs the trees see. A model's line is drawn where the series' share is at least 0.5% (the classification in Table 3); the five series with the largest mean share over the three tree ensembles and the next row of that table.

- **Implied variance.** Let IV_t^C and IV_t^P denote the call and put implied volatilities at 15:30 ET, quoted on a per-hour basis. The implied volatility over the closing half hour is

$$\mathrm{IV}_t = \frac{\mathrm{IV}_t^C + \mathrm{IV}_t^P}{2\sqrt{2}},$$

  and IV_t^2 is the corresponding implied variance to expiration. The vendor solves each leg's implied volatility by bisection and, when the bisection does not converge, reports a node of its bracket instead of a volatility. On the 5 days of $\mathcal{T}_{\mathrm{opt}}$ on which a leg carries such a node, we recover IV_t from the package instead: $\mathrm{IV}_t = \sigma$, the volatility over the remaining half hour at which the Black–Scholes value of the package, with a zero interest rate and no dividends, equals its mid-quote (Black and Scholes, 1973),

$$P_t = S_t\,\Phi(d_C) - K_C\,\Phi(d_C - \sigma) + K_P\,\Phi(\sigma - d_P) - S_t\,\Phi(-d_P),$$

  where $d_j = \big(\ln(S_t/K_j) + \sigma^2/2\big)/\sigma$ for $j \in \{C, P\}$ and Φ is the standard normal distribution function. All of these days lie within the first 252 days of $\mathcal{T}_{\mathrm{opt}}$, which only initialize the sizing rules below, so none of them is a scored day.
- **Forecast variance risk premium.** Following Carr and Wu (2009) and Bollerslev et al. (2009), the variance risk premium over the closing half hour is $\mathrm{IV}_t^2 - \mathbb{E}_t[\mathrm{RV}_t]$, which we forecast by $\mathrm{IV}_t^2 - \hat{\sigma}_t^2$, where $\hat{\sigma}_t^2$ is the model variance forecast from Section 5.1. Every trading rule below conditions on the forecast exclusively through the variance gap $\hat{\sigma}_t^2 - \mathrm{IV}_t^2$, taking a long straddle position when forecast variance exceeds implied variance.
- **Trading strategies.** A strategy chooses a daily straddle position q_t ($q_t > 0$ long, $q_t < 0$ short) and earns the daily return $r_t = q_t R_t$. Because the variance forecast enters solely through q_t, comparing performance across rules isolates the economic value of the forecast (Fleming et al., 2001). We consider four rules:
  - *Unconditional short-straddle benchmark:* a static short-variance position setting $q_t \equiv -1$;
  - *Sign strategy:* $q_t = \mathrm{sign}(\hat{\sigma}_t^2 - \mathrm{IV}_t^2)$, with ties mapped to $q_t = -1$;
  - *Robust clip:* $q_t = \mathrm{clip}\big((\hat{\sigma}_t^2 - \mathrm{IV}_t^2)/\mathrm{MAD}_t, -1, 1\big)$, where $\mathrm{MAD}_t = \mathrm{median}_{t'<t}|\hat{\sigma}_{t'}^2 - \mathrm{IV}_{t'}^2|$ is the expanding median absolute variance gap over earlier days $t' < t$ in $\mathcal{T}_{\mathrm{opt}}$;
  - *Percentile rank:* the sign of the gap scaled by the empirical percentile rank of its magnitude across the $t-1$ preceding days $t' < t$ in $\mathcal{T}_{\mathrm{opt}}$:

$$q_t = \mathrm{sign}(\hat{\sigma}_t^2 - \mathrm{IV}_t^2) \cdot \frac{1}{t-1} \sum_{t'<t} \mathbf{1}\{|\hat{\sigma}_{t'}^2 - \mathrm{IV}_{t'}^2| \le |\hat{\sigma}_t^2 - \mathrm{IV}_t^2|\}.$$

- **Sizing sample.** Every position q_t is constructed causally using days $t' < t$ only. We reserve the first 252 days of $\mathcal{T}_{\mathrm{opt}}$ to initialize the scaling factor MAD_t, and evaluate all trading rules—including the sign strategy—on the remaining 614 post-warmup trading days ($t > 252$).
- **Execution.** At mid-quotes, the strategy return is $r_t = q_t R_t$. At quoted bid and ask prices, buying the straddle pays the ask P_t^a and selling receives the bid P_t^b; because settlement occurs in cash at expiration, the bid–ask spread is incurred at entry only:

$$r_t^{\mathrm{ba}} = \begin{cases} |q_t|\,(X_t/P_t^a - 1), & q_t > 0, \\ |q_t|\,(1 - X_t/P_t^b), & q_t < 0. \end{cases}$$

- **Transaction costs.** Executing at the quoted bid or ask rather than the mid-quote imposes a median half-spread cost of 2.9% of the straddle premium. No additional commissions or exchange fees are deducted.

7.2 Performance measures

- **Annualized Sharpe ratio.** We compute the annualized Sharpe ratio from daily strategy returns, both at mid-quotes and at quoted bid/ask prices, over the 614 post-warmup days of $\mathcal{T}_{\text{opt}}$ ($t > 252$):

$$\text{SR} = \frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} \text{SR}_y, \qquad \text{SR}_y = \sqrt{A_y}\, \frac{\bar{r}_y}{\text{sd}_y(r)}, \qquad A_y = \frac{N_y}{c_y},$$

  where $\mathcal{Y}$ is the set of the 4 calendar years of those 614 days, N_y is the number of them that fall in year y (option days, i.e. days with a same-day SPXW expiration), $\bar{r}_y$ and $\text{sd}_y(r)$ are the sample mean and standard deviation (divisor $N_y - 1$) of r_t over those N_y days, and c_y is the fraction of calendar year y that the sample covers, from the later of 1 January and the sample's first day to the earlier of 31 December and its last day. A_y is thus the number of option days per calendar year at year y's expiration schedule: 163.4, 219.0, 248.0 and 251.1 for 2021 to 2024 (N_y = 64, 219, 248, 83; c_y = 0.392, 1.000, 1.000, 0.331). Each calendar year enters with equal weight and is annualized at its own trading frequency. The frequency changes within the sample: SPXW expirations were listed on Mondays, Wednesdays and Fridays until spring 2022 and on every weekday from 16 May 2022. Equal weights also keep a year with many option days from dominating the average if the strategy's performance changes over the sample. Table 8 (Appendix F) reports the main Sharpe ratio comparison under two other definitions: the per-year ratio with $\sqrt{N_y}$ in place of $\sqrt{A_y}$, the years weighted by N_y, and the ratio of the mean to the standard deviation of all 614 daily returns, scaled by $\sqrt{252}$.

  D6. Decision (user, 2026-10-22): the Sharpe ratio is the definition above (D6): for each calendar year y of the post-warmup sample, $\text{SR}_y = \sqrt{A_y}\,\bar{r}_y/\text{sd}_y(r)$ with $A_y = N_y/c_y$ the year's option days per calendar year, and SR the equal-weight mean of the SR_y; every statement about a Sharpe ratio difference uses the paired block-bootstrap test of Appendix I, which resamples D6 itself.
  Disclosure: this definition was adopted after the results under the previous one (the per-year ratio with $\sqrt{N_y}$, N_y-weighted, of 2026-10-18) had been seen. That ratio (D1) and the ratio of the pooled daily returns scaled by $\sqrt{252}$ (D2), with the Ledoit and Wolf (2008) test of the pooled ratios, are reported in Table 8, together with D6 before and from 16 May 2022.

  The 216 days of $\mathcal{T}_{\text{oos}}$ without a same-day SPXW expiration enter the statistical accuracy evaluation only. Because the position is held for 30 minutes intraday, the risk-free return over the holding period is negligible and omitted.
- **Mean return and its t-statistic.** We test the statistical significance of mean daily returns over the 614 post-warmup days $\mathcal{T}^{+}_{\text{opt}} = \{t \in \mathcal{T}_{\text{opt}} : t > 252\}$, the sample of the Sharpe ratio, with the test that Lazarus et al. (2018) recommend: the t-statistic with the Newey–West (Bartlett kernel) long-run variance at bandwidth $S = \lceil 1.3\,(614)^{1/2} \rceil = 33$, whose autocovariances divide by the number of days with no small-sample factor, and fixed-b critical values (Kiefer and Vogelsang, 2005); the two-sided 5% critical value is 2.11 (Appendix I).
- **Hit rate and position counts.** We also record the empirical hit rate $\frac{1}{614} \sum_{t \in \mathcal{T}^{+}_{\text{opt}}} \mathbf{1}\{r_t > 0\}$ and the number of long-straddle days $\#\{t \in \mathcal{T}^{+}_{\text{opt}} : q_t > 0\}$.
- **Gain over the short-straddle benchmark.** For any strategy A compared against a benchmark strategy B (the unconditional short straddle unless noted otherwise), we define the daily return differential $\Delta_t = r^A_t - r^B_t$ and the Sharpe ratio gain

$$\Delta\text{SR} = \text{SR}(r^A) - \text{SR}(r^B).$$

  The t-statistic of the previous item, applied to Δ_t, tests whether the mean return differential differs from zero. Because it does not test the equality of the two Sharpe ratios, every

statement about ΔSR uses a paired circular block bootstrap of ΔSR itself on the same days (Appendix I); the studentized bootstrap test of Ledoit and Wolf (2008), which tests the equality of the Sharpe ratios of the pooled daily returns, is reported as a robustness check.

- **Elementary score.** To measure forecast accuracy specifically at the decision threshold IV_t^2, we evaluate the elementary score of Ehm et al. (2016) normalized by implied variance:

$$\mathrm{ES}_{1,t} = \left|\frac{\mathrm{RV}_t}{\mathrm{IV}_t^2} - 1\right| \mathbf{1}\{\min(\hat{\sigma}_t^2, \mathrm{RV}_t) \le \mathrm{IV}_t^2 < \max(\hat{\sigma}_t^2, \mathrm{RV}_t)\},$$

  and denote its sample mean over the post-warmup days $\mathcal{T}_{\mathrm{opt}}^+$ by $\overline{\mathrm{ES}}_1$.
- **Decision-threshold property.** Note that $\mathrm{ES}_{1,t}$ is strictly positive if and only if the forecast $\hat{\sigma}_t^2$ and the realization RV_t fall on opposite sides of the implied variance IV_t^2—that is, precisely when the sign strategy takes the wrong directional position.
- **Compounding.** When evaluating compounded wealth trajectories from initial capital $V_0 = 1$, we allocate a fixed fraction $f = 3\%$ of current wealth to the straddle premium each day, yielding terminal portfolio value and annualized log growth rate

$$V = \prod_{t \in \mathcal{T}_{\mathrm{opt}}^+} (1 + f\, r_t), \qquad G = \frac{252}{614} \sum_{t \in \mathcal{T}_{\mathrm{opt}}^+} \log(1 + f\, r_t).$$

- Under bid/ask execution, r_t^{ba} replaces $r_t = q_t R_t$ in the compounding recursion. We define maximum drawdown as $\max_t(1 - V_t / \max_{t' \le t} V_{t'})$, where V_t is the cumulative portfolio value at the end of day t.

7.3 Results

- **Unconditional short-straddle benchmark.** Selling the closing straddle unconditionally ($q_t \equiv -1$; labeled "Short straddle" in the tables) yields SR $= -0.12$ at mid-quotes and -0.46 at quoted bid/ask prices. It has a positive return on 62.2% of the post-warmup days, and its mean daily return is statistically indistinguishable from zero ($t = 0.67$, fixed-b $p = 0.52$), and the strategy loses money in 2 of the 4 calendar years: passively selling the closing 0DTE straddle earns no statistically detectable variance risk premium in our option-data subsample.
- **Sign strategies of the intraday models versus the short-straddle benchmark.** In point estimates, 12 of the 12 pooled intraday forecasts achieve a positive Sharpe ratio gain ($\Delta\mathrm{SR} > 0$) over the unconditional short straddle.
  - The mean return gain over the short-straddle benchmark is statistically significant at the 5% level in 1 of the 12 specifications (fixed-b test of the mean daily return differential on the 614 post-warmup days, bandwidth $S = 33$); the t-statistics range from -0.01 to 2.17.
- **Sharpe ratios of the pooled intraday models.** On the HAR benchmark, the Sharpe ratios of the 6 pooled intraday specifications range from 0.01 to 0.98 at mid-quotes and from -0.32 to 0.62 at quoted bid/ask prices; on the full predictor set, they range from 1.67 to 1.96 at mid-quotes and from 1.31 to 1.60 at quoted bid/ask prices.
- **Full predictor set versus the HAR benchmark.** Expanding the information set from the HAR benchmark to the full predictor set increases the mid-quote Sharpe ratio in all 6 intraday model classes; across classes, the mid-quote Sharpe ratio ranges from 0.01 to 0.98 on the HAR benchmark and from 1.67 to 1.96 on the full predictor set.

  - The resulting gain ΔSR spans +0.74 to +1.95 across classes, with a median of +0.91 (elastic net +0.96, ridge +0.86, lasso +0.74, LightGBM +1.14, XGBoost +0.76, random forest +1.95).
  - The mean daily return differential is statistically significant at the 5% level in 1 of the 6 classes (fixed-b test, bandwidth $S = 33$).
  - The Sharpe ratio gain is statistically significant at the 5% level in 1 of the 6 classes in which it is positive (block-bootstrap test of equal Sharpe ratios, Appendix I); the Ledoit and Wolf (2008) test of the Sharpe ratios of the pooled daily returns rejects their equality at the 5% level in 2 of the 6 classes (Table 8).
- **Position sizing.** Across the 12 intraday forecasts, the median mid-quote Sharpe ratio is 1.32 under the sign strategy, 1.09 under the robust clip rule, and 1.16 under the percentile-rank rule.
  - In point estimates, continuous sizing via the robust clip outperforms the binary sign strategy ($\mathrm{SR}^{\mathrm{clip}} > \mathrm{SR}^{\mathrm{sign}}$) for 3 of the 12 intraday forecasts.
- **Concentration of economic gains.** As is typical of short-dated option strategies, cumulative performance is concentrated in a small subset of trading days:
  - removing each forecast's 10 best days renders the mean return differential $\bar{\Delta}$ over the short-straddle benchmark negative for 11 of the 12 intraday forecasts;
  - trading days on which the straddle expires worthless ($R_t = -1$, corresponding to settlement pinning near the strike; 137 of the 614 post-warmup days) account for 37% of the positive returns of the median pooled intraday forecast (34–39% across the 12 forecasts).
- **Statistical accuracy versus economic value.** Across the 12 pooled intraday forecasts evaluated on $\mathcal{T}_{\mathrm{opt}}$, the Spearman rank correlations between average statistical loss and the post-warmup Sharpe ratio (negative: the forecasts with the lower loss have the higher SR; positive: the lower SR) are

$$\rho_S(\overline{\mathrm{QLIKE}}, \mathrm{SR}) = 0.26, \qquad \rho_S(\overline{\mathrm{ES}}_1, \mathrm{SR}) = -0.45.$$

  Among the pooled intraday forecasts, the rank correlation of the Sharpe ratio is positive with average QLIKE loss, 0.26, and negative with $\overline{\mathrm{ES}}_1$, the loss at the implied-variance threshold, -0.45. With the per-year ratio scaled by $\sqrt{N_y}$ and weighted by N_y in place of SR, the two correlations are 0.37 and -0.35; with the ratio of the pooled daily returns scaled by $\sqrt{252}$, they are 0.23 and -0.34 (note D6). Their signs are the same under all three definitions.
- **Robustness checks** (detailed in Appendix F):
  - *Interval-shift test:* when we replace $\hat{y}_{t,c}$ in the sign rule by the forecast $\hat{y}_{t,c+j}$ from j half-hour bars away, $q_t = \mathrm{sign}(\hat{y}^2_{t,c+j}\, s^2_{t,c} - \mathrm{IV}^2_t)$, for $j = -12, \ldots, 6$ ($j < 0$ earlier on day t; $j > 0$ on day $t+1$, not known at the 15:30 origin, a placebo), the median Sharpe ratio across the 12 pooled intraday forecasts (613 post-warmup days on which every shifted forecast exists; 1.51 at $j = 0$) ranges from -0.48 to 2.01 across j (a half-range of ± 1.25), against the ± 1.35 95% sampling band of the Sharpe ratio under i.i.d. returns (Table 9). Every earlier bar ($j < 0$) lowers the median, by -1.91 to -0.70 against $j = 0$, and at each earlier bar between 8 and 12 of the 12 forecasts have a lower Sharpe ratio than at $j = 0$;

E26b. Decision: to decide (James).
Open choices for the interval-shift test: (i) the days: every j on the 613 days on which every shifted forecast exists (above; the original test's rule, which drops the last option day, 30 April 2024, as it has no next session; median Sharpe ratio at $j = 0$: 1.51), or only $j \leq 0$ on all 614 post-warmup days of the printed Sharpe ratios (median at $j = 0$: 1.32, range -0.62 to 1.32, half-range ± 0.97); (ii) the sampling band: $1.96\sqrt{\sum_y A_y/N_y}/K$ over the K calendar years, the i.i.d. band of the Sharpe ratio of note D6 at SR $= 0$, in place of the original $1.96\sqrt{252/n}$ (writeup/make_rehaul_pooled_extras.py); (iii) how to read the pattern for $j < 0$. The earlier reading ("well within" the band, "consistent with sampling variation") is parked.

- *Subsample stability:* we verify performance across subsamples split around May 2022 (when CBOE expanded SPXW expirations to all five weekdays; Table 8);
- *Compounded wealth:* across the 12 pooled intraday forecasts, compounding at $f = 3\%$ of wealth per day over the 614 post-warmup days yields a median terminal portfolio value V of 4.1 at mid-quotes (1.1 to 9.4) and 2.5 at quoted bid/ask prices (0.6 to 5.8), against 1.1 and 0.6 for the unconditional short straddle, with maximum drawdowns reaching up to 45% and 50%, respectively (52% and 60% for the unconditional short straddle; Table 9);
- *Window length:* with a rolling window of 3,500 trading days in place of 3,000 (Table 7), the full predictor set raises the mid-quote Sharpe ratio in 6 of the 6 model classes, 2 of them significantly at the 5% level; no pooled forecast's Sharpe ratio differs significantly between the two windows (0 of 12), and the random forest on the HAR benchmark moves most, from 0.01 to 0.74 ($p = 0.14$).

Appendix G.5 reports each model class trained on the closing half hour alone.

8 Conclusion

- **Exogenous predictors and QLIKE loss in pooled intraday models.** Relative to the HAR benchmark, augmenting the model with the full exogenous predictor set lowers mean QLIKE on every regular-hours bar in all 6 pooled intraday model classes, each reduction statistically significant at the 5% level. For the forecast of the closing half hour it lowers mean QLIKE in 3 of the 6 classes, the three tree ensembles, none of these reductions statistically significant at the 5% level, and raises it for the three penalized linear models (Section 6). With a window of 3,500 trading days the every-bar result holds in 6 of the 6 classes (Table 7).
- **Comparison with the closing-interval HAR benchmark.** Every pooled intraday specification incurs a higher mean QLIKE than the closing-interval HAR benchmark, the OLS HAR regression estimated solely on closing-interval data. Incorporating the full exogenous predictor set narrows this loss gap in 3 of the 6 model classes, the tree ensembles, and widens it for the penalized linear models; on the full predictor set the gap is statistically significant at the 5% level for 3 of the three penalized linear models and for 1 of the three tree ensembles (Section 6.4).
- **Models trained on the closing half hour.** Results for each model class estimated on the closing interval alone are reported in Appendix G.5.
- **Economic value in 0DTE straddles.** When the variance forecasts are used to trade same-day-expiring SPXW straddles over the closing half hour on the option-data subsample, adding the exogenous predictors increases the sign strategy's post-warmup Sharpe ratio in every one of the 6 intraday model classes, in point estimates, and significantly at the 5% level in 1 of them. Across the 12 pooled intraday forecasts, the rank correlation of the straddle Sharpe ratio is positive with average QLIKE loss, 0.26, and negative with $\overline{\mathrm{ES}}_1$, the loss at the implied-variance threshold, -0.45 (Section 7.3).
- **Which series the pooled models use.** In the tree ensembles the non-standard series, those outside the HAR components, the futures' own realized moments and trading activity, and the VIX, hold 43.7–45.9% of the exogenous attribution, less than in the penalized linear models (56.0–73.9%); the trees' attribution is concentrated in a few series and their non-standard share is higher on FOMC statement days (Section 6.5).
- **Scope and limitations.**
  - Our out-of-sample evaluation spans $T_{\mathrm{oos}} = 1{,}082$ trading days from 3 January 2020 to 30 April 2024; the forecast accuracy of the pooled linear models and tree ensembles (Table 1) is scored on 2,579 trading days from 2 January 2014 to 30 April 2024.
  - The empirical analysis focuses on a single equity index (the S&P 500) and a single target horizon (the 15:30–16:00 ET closing half hour).
  - In our economic evaluation, the straddle is held unhedged to cash settlement, and executing at quoted bid and ask prices rather than mid-quotes materially reduces the strategy's Sharpe ratio (Section 7).
- **Directions for future work.** Natural extensions include out-of-sample evaluation on post-30 April 2024 option expirations, cross-market validation on additional index option chains, and intraday variance forecasting at earlier decision origins prior to 15:30 ET.

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A Appendix structure

- **Implementation details** (Appendix B): samples and windows, the training schemes in practice, the tuned hyperparameters and the tuning procedures.
- **Data and predictors** (Appendix C):
  - sources and spans;
  - predictor dictionary: 46 exogenous series and 564 intraday predictors in the full predictor set (47 series and 628 predictors in the closing-interval model as fitted, which still carries the FOMC statement-interval dummy dropped from the dictionary; note E35) (Appendix Table 13);
  - missing and stale data;
  - removal of duplicate predictors; the window filter (constant and duplicate predictors dropped in each estimation window $\mathcal{W}_t$);
  - SPXW option chain; implied-volatility inversion;
  - FOMC calendar.
- **Hyperparameter selection by model** (Appendix D):
  - linear models: penalty strength λ, elastic-net mixing α; block-penalized ridge, penalty weights ω_j;
  - tree ensembles: fixed hyperparameters, Optuna budgets, selected trials; boosting rounds $b = 1, \ldots, B$, learning rate η;
  - re-selection frequency, on the estimation sample $\mathcal{W}_t^{\text{est}}$, gap $\mathcal{W}_t^{\text{gap}}$ and validation sample $\mathcal{W}_t^{\text{val}}$; re-estimation frequency;
- **Robustness, statistical** (Appendix E):
  - full model table, tree ensembles included (not reported);
  - Diebold–Mariano and HAC conventions (DM with the Harvey–Leybourne–Newbold correction and Newey–West lag truncation q_{NW}; fixed-b tests of mean returns; the block-bootstrap test of equal Sharpe ratios and the Ledoit–Wolf test of the pooled ratios);
  - model confidence set, superior predictive ability, StepM;
  - regime and break tests (not reported);
  - closing-interval vs intraday models;
  - remaining variance through the trading day (not reported).
- **Robustness, trading** (Appendix F):
  - date-shift test;
  - Murphy diagrams (elementary scores); subsample splits;
  - P&L decomposition (not reported);

  - compounding;
  - position sizing;
  - strategy variations (not reported); other forecast origins (not reported); month-ends (not reported).
- **Closing-interval models** (Appendix G): the closing-interval sample (Appendix G.1), predictors and column counts (Appendix G.2), the closing-interval HAR benchmark and its estimation (Appendix G.3), closing-interval standardization (Appendix G.4), and every model class trained on the last 30 minutes (Appendix G.5).
- **Reproducibility** (Appendix H):
  - software lock files (not reported);
  - hashes of the design matrices (not reported);
  - computing environment;
  - CPU classes (not reported);
  - compute time (not reported);
  - the generator script behind every reported number.
- **Statistical tests** (Appendix I): the Diebold–Mariano statistic, the Newey–West variance, the fixed-b test of a mean return, the block-bootstrap test of equal Sharpe ratios, the Ledoit–Wolf test of the pooled ratios, and the robustness of the strategy-return tests to the long-run variance estimator.

S13. Decision: future work; keep the ten items in the overview, each marked (not reported), and report them properly in a later draft.
Problem: the overview lists ten items that the appendices do not report. Robustness, statistical (Appendix E): the full model table with the tree ensembles; regime and break tests; remaining variance through the trading day. Robustness, trading (Appendix F): the P&L decomposition; strategy variations; month-ends. Reproducibility (Appendix H): software lock files; hashes of the design matrices; CPU classes; compute time.

B Implementation details

B.1 Samples and windows

- **Estimation and evaluation samples:** Table 11.
- **Burn-in:** the first 3,125 samples of each series as built, the longest HAR horizon, are dropped (Section 3; Table 11): 3,125 trading days of the closing-interval series, 1,560 trading days of the series of the two intervals of the last hour (the linear pre-sample), and 3,125 30-minute intervals, 104 trading days, of the intraday grid.
- **Diurnal adjustment** $s^2_{t,i}$ (Section 5.1): $n_{\mathrm{d}} = 20$ trading days, at least $\underline{n}_{\mathrm{d}} = 5$ of them; intervals with fewer have $s^2_{t,i} = 1$ and are not scored.
- **Winsorization** of the target: $n_{\mathrm{w}} = 240$ intervals, all intervals of the day pooled; the bounds are the empirical 1st and 99th percentiles, $\underline{Q}_{t,i} = Q_{0.01}(\mathcal{Y}_{t,i})$ and $\overline{Q}_{t,i} = Q_{0.99}(\mathcal{Y}_{t,i})$.
- **Hyperparameter-selection splits:** each split leaves a gap of n_{gap} observations between an estimation sample and the validation sample that follows it.
  - Pre-sample folds of the linear penalties (Section B.4): for the closing-interval linear models, $n_{\mathrm{gap}} = 25$ trading days before a validation sample of 250 trading days; for the pooled linear models of Table 1, $n_{\mathrm{gap}} = 25$ trading days (325 bars) before a validation sample of 250 trading days (3,250 bars).
  - Splits of the window at a re-selection: the validation sample is the last $n_{\mathrm{val}} = 125$

observations of the window, after a gap of $n_{\text{gap}} = 25$ observations. The observations are trading days in the random-search and Optuna rungs of the closing-interval trees, and 30-minute intervals in the margin re-tune of the pooled linear models on all 48 intervals of the model set (note E23).

E21. Decision: label each length with its unit and which step uses it.
Problem: done (2026-10-23), for the author to confirm: the item now gives each length's unit and step as the code counts them, which reverses the reading of 17 October 2026 (both lengths in 30-minute intervals). Each length counts rows of the design its split is applied to (src/models/reclasso_har.py, forward_window_split; specs/causal_tune_linear.py, VAL_TAIL and EMBARGO). The gap $n_{\text{gap}} = 25$ of every pre-sample fold is in trading days (experiments/close_linear_pretune.py, pre_folds; experiments/pooled_linear_rerun_2026_10_19_13.py, PRE_EMBARGO_SESS, 25 trading days, 325 bars), so the "25-day gap" of Table 11 is right. $n_{\text{val}} = 125$ is in trading days in the closing-interval tree rungs and in 30-minute intervals only in the margin re-tune of the 48-interval pooled linear models.

B.2 Training schemes in practice

- **Closing-interval scheme** ($\mathcal{I} = \{c\}$): Appendix G.3.
- **Intraday scheme** ($\mathcal{I} = \{1, \ldots, M\}$), the paper's training scheme: one observation per interval, so a period is an intraday interval and $\mathcal{W}_{t,i}$ holds the W observations before (t, i) (Section 5.1). The pooled models of Table 1 and Sections 6–7 are estimated on the 13 regular-hours bars of each day, bar ends 10:00 to 16:00 ET, with the HAR components and trailing averages computed on the full grid of $M = 48$ intervals, overnight included (Section 4). The pooled linear models of the model set, which are not in Table 1, are estimated on all M intervals of each day, overnight included. In the design of the pooled models of Table 1, the full predictor set has 564 predictors (Section 4).
  - For the closing-interval models and the straddle strategy, only the forecasts of the closing interval $i = c$, made at the 15:30 ET forecast origin, are scored and traded; the pooled linear models and tree ensembles of Table 1 are scored on every one of the 13 regular-hours bars.
  - Linear models: $W = 39{,}000$ bars (3,000 trading days of the 13 regular-hours bars), re-estimated every interval.
  - Trees: $W = 39{,}000$ bars (3,000 trading days of the 13 regular-hours bars), the design of the linear models of Table 1; one fixed hyperparameter configuration per tree method, never re-selected, re-estimated every bar.

B.3 Tuned hyperparameters

- **Linear penalties,** chosen by hyperparameter selection on a pre-sample (Section B.4), one grid value per decade (Table 12):

$$\lambda^{\text{ridge}} \in \{10^{-2}, \ldots, 10^{3}\}, \qquad \lambda^{\text{lasso}}, \lambda^{\text{enet}} \in \{10^{-6}, \ldots, 10^{-2}\}.$$

  For lasso and elastic net these are the scikit-learn parameters (Section 5.2).
  - closing-interval linear models (Table 12, Panel A): on the full predictor set ridge 10^{3}, lasso 10^{-2} and elastic net 10^{-2} (top edge); on the HAR benchmark ridge 10^{-2}, lasso 10^{-6} and elastic net 10^{-6} (the lower edge of each grid);
  - pooled linear models of Table 1 and Sections 6–7: one penalty per estimator and predictor set, chosen by the same criterion on the last pre-sample fold (validation year 2013) (Section B.4); on the full predictor set ridge 10^{3}, lasso 10^{-4} and elastic net 10^{-3}, on

the HAR benchmark ridge 10^{-3}, lasso 10^{-8} and elastic net 10^{-7} (not in Table 12). These penalties act on the coefficients of the columns standardized within the estimation window (Section 5.2). A pick at either end of its grid adds the next decade in that direction, one decade at a time, until the pick is interior, the validation MSE at the new end differs from that at the previous end by less than 10^{-6} relative, or 6 decades have been added, so a pick can lie outside the grids above;

- each of these penalties is held fixed at every forecast origin of the evaluation span, with no re-tuning; the pooled linear models on all 48 intervals of the model set, which are not in Table 1, keep the margin re-tune of their own study (note E23).

E23. Decision: one pre-sample pick per estimator and predictor set, held fixed for the whole evaluation; no causal adaptive re-tuning during evaluation.
Problem (2026-10-23): the rule holds for every forecast of Table 1 and Sections 6–7. The closing-interval linear forecasts are the frozen arms (since 2026-10-18), and the pooled linear models hold the penalties frozen on the last pre-sample fold (validation year 2013) (results/window_sweep_2014start_2026-10-22/w3000/frozen_penalties.json; the same six in w3500). The margin re-tune still enters the six 48-interval pooled linear forecasts of the master table's model set (poolpt_*: arm f of results/pool_linear_pretune_2026-10-06, experiments/stack_pool_pretune_yhat.py, f_series), which count in the aggregates over the 112 forecasts of the model set (Appendices F and G.5) but are not in Table 1.
Pending: the sensitivity check of the forecasts at neighboring grid values, on which the robustness reasoning below rests, has not been run.
Each estimator and predictor set gets one penalty, chosen once by hyperparameter selection on the fixed pre-sample window and held constant at every forecast origin. Pre-tuning on a fixed window and holding the penalty constant is a simplification, and it is justified by robustness: the forecasts are not very sensitive to the penalty, so a penalty re-selected during the evaluation would change them little. The text does not claim that the exact value matters. The item above reports the pre-sample pick so that the forecasts can be reproduced, and the decision rests on that insensitivity, which a sensitivity check of the forecasts at neighboring grid values would have to confirm.

- **Trees,** fixed hyperparameters from a search on a 1998–2002 pre-sample (Table 12):
  - LightGBM and XGBoost: the best trial of an Optuna search (Akiba et al., 2019);
  - random forest: trial 255 of 400 ($B = 485$ trees, maximum depth 20, share of predictors tried at each split 0.90, bootstrap sample share 0.41);
  - minimum leaf size $n_{\min}$ (observations per leaf): the number outside the parentheses in Table 12 is the minimum leaf size selected on a window of 24,000 intervals; the number in parentheses is that minimum leaf size rescaled to the 2,000-day window of the closing-interval trees, $n_{\min} \leftarrow n_{\min} \cdot 2{,}000/24{,}000$, and it is used by the closing-interval trees only. The pooled trees of Table 1 use the number outside the parentheses as selected, without rescaling, on their window of 39,000 bars (Section B.2).

B.4 Tuning procedures

- **Linear:**
  - closing-interval linear models: hyperparameter selection on validation MSE over a pre-sample of four folds of 250 trading days (6 January 2016 to 2 January 2020), before the first forecast:

$$\hat{\lambda} = \arg\min_{\lambda} \frac{1}{4} \sum_{v=1}^{4} \mathrm{MSE}_v^{\mathrm{val}}(\lambda),$$

    with $\mathrm{MSE}_v^{\mathrm{val}}(\lambda)$ the mean squared error of $\tilde{y}$ on the validation sample of fold v;
  - pooled linear models of Table 1: hyperparameter selection on the validation MSE of the last pre-sample fold (validation year 2013), fitted on the $W = 39{,}000$ bars before its gap

of n_{gap} trading days (Section B.1), before the first forecast:

$$\hat{\lambda} = \arg\min_{\lambda} \text{MSE}^{\text{val}}(\lambda),$$

with $\text{MSE}^{\text{val}}(\lambda)$ the mean squared error of $\tilde{y}$ over every regular-hours bar of that validation sample, and the grid extended at its ends (Section B.3);

  - $\hat{\lambda}$ is then held fixed at every forecast origin of the evaluation span, with no re-tuning (note E23); the closing-interval forecasts are the frozen arms of the pre-tune study (note N24).

- **Pooled trees,** the tree cells of Table 1 and the trees of Sections 6–7: one frozen configuration per method (Section B.3), never re-selected, refit at every bar on the $W = 39{,}000$ bars before it (Section B.2). None of the closing-interval rungs below enters Table 1.
- **Closing-interval trees,** in increasing order of tuning effort:
  - fixed hyperparameters, re-estimated every 10 trading days or daily;
  - random search every 250 trading days;
  - Optuna (Akiba et al., 2019): 50 trials, re-tuned every 1, 5, 25 or 250 trading days, best of the first 10, 25 or 50 trials.
- **Every selection uses only data available at the forecast origin;** no forecast is scored on a day that entered the selection of its own hyperparameters, and the evaluation span $\mathcal{T}_{\text{oos}}$ starts after the pre-sample.

C Data and predictors

- **Sources and spans.** Eight groups, in Section 3. Closing-interval models use all 47 exogenous series as fitted (the 46 of the dictionary and the FOMC statement-interval dummy, dropped on 22 October 2026; note E35). Intraday models use the 46 series in every group, FOMC included.
- **Predictor dictionary.** Appendix Table 13 defines each series. Trailing averages, activity indicators, and calendar columns are added in Section 4.
- **Missing and stale values.** For an exogenous series u, the availability indicator $a_{t,i} = \mathbf{1}\{u \text{ observed at } (t,i)\}$ is recorded before any filling (Section 3). The series is then forward-filled with no staleness limit. On every interval with $a_{t,i} = 0$ the stored regressor is the imputation of Section 4, step 4: the median of the series' observed adjusted values on earlier retained intervals, or the adjusted value on its first observed interval when no earlier observation exists.
- **Duplicate and constant predictors.** At each estimation the window screen of Section 4 drops a column that is constant on $\mathcal{W}_{t,i}$ or an exact bitwise copy of an earlier kept column. The first copy is kept. On the closing-interval model with the full predictor set the screen keeps a median of 369 of the 628 columns.
- **SPXW chain.** Quotes on S&P 500 Weeklys that expire the same day, every 30 minutes: bid, ask, mid-quote, underlying level, implied volatility, and delta. The file covers 1,291 expirations, 2020–2025. The straddle in Section 7.1 uses the 15:30 quotes: K_C and K_P are picked among the 15:30 contracts with a positive mid-quote, and every 15:30 chain in $\mathcal{T}_{\text{opt}}$ has both legs. The chains are also screened for a quote outage, fewer than 10 contracts with a positive mid-quote or a wider strike more than 10 index points from S_t; no 15:30 chain fails the screen, so it drops no day. S_t is the 15:30 underlying print on the chain, and the official close is the S&P 500 close.

- **Implied-volatility inversion.** The vendor quote on each leg is a volatility per hour, from a geometric bisection. A solve that does not converge returns a node of the bisection's bracket instead of a volatility. On the 5 days of $\mathcal{T}_{\text{opt}}$ on which a leg carries such a node, the implied volatility of the package is recovered by inverting its Black–Scholes value at the mid-quote, as in Section 7.1; no day is left without an implied volatility. The 16:00 stamp is not used. The conversion from the hourly quote to the closing half hour is the display in Section 7.1.
- **FOMC calendar.** The four scheduled-announcement series of Section 3, on closing-interval models only. The schedule is known before the interval.

D Hyperparameter selection by model

- **Penalty parameters.** Ridge, lasso, and elastic net minimize the raw sum of squares plus the penalty in Section 5.2. λ is the coefficient on that penalty for ridge. For lasso and elastic net the printed grid is the scikit-learn parameter, and the coefficient in the sum is $2n$ times the printed value. Lasso is $\alpha = 1$. Elastic net is $\alpha = 0.5$. The block weights ω_j are 0.01 on the HAR benchmark, 1 on an additional predictor, and 0 on the intercept. The HAR benchmark fitted alone uses $\omega_j = 1$ on every predictor.
- **Where λ is chosen.** One grid value per decade, Table 12. The pre-sample is four folds of 250 trading days, 6 January 2016 to 2 January 2020, for the closing-interval linear models and the last pre-sample fold (validation year 2013) for the pooled linear models of Table 1, whose grid is extended at its ends (Section B.3); the criterion is validation MSE (Section B.4). The value is chosen once on this pre-sample and held fixed for the whole evaluation span, with no re-tuning during evaluation (note E23). Panel A of the table gives the closing-interval picks, the penalty of every closing-interval linear forecast at every forecast origin; on the full predictor set each is at the top of its grid. The pooled linear picks are in Section B.3. The pooled linear models on all 48 intervals of the model set, which are not in Table 1, keep the margin re-tune of their own study (note E23).
- **Trees.** LightGBM and XGBoost take the best trial of an Optuna search on a 1998–2002 pre-sample. The random forest takes trial 255 of 400. The selected configuration and the leaf minimum are in Section B.3. The tree cells of Table 1 are the pooled trees, which hold this configuration fixed and are refit at every bar; the three tuning rungs in Section B.4 are the closing-interval trees'.
- **What may enter a choice.** A selection uses only data available at the forecast origin. The evaluation span starts after the pre-sample.

E Robustness of the statistical results

- **Reported comparisons.** Forecast accuracy of the pooled linear models and tree ensembles is Table 1, from 2 January 2014 to 30 April 2024. The Diebold–Mariano statistic, with the Harvey–Leybourne–Newbold correction, the Newey–West variance and the tests of strategy returns are Appendix I.
- **Estimation window.** Table 7 refits the models of Table 1 on a rolling window of 3,500 trading days ($W = 45{,}500$ bars) instead of 3,000, with the same first evaluation day, the same frozen hyperparameters and a refit at every bar. With the longer window the full predictor set is significantly more accurate than the HAR benchmark on every bar for 6 of the 6 model classes (6 of 6 at 3,000 days), and mean QLIKE moves by at most 1.3% on every bar and 4.4% on the 16:00 bar; the 5 forecasts whose every-bar QLIKE differs significantly between

the windows, and the 3 at 16:00, are all linear models. The Sharpe ratio changes significantly between the windows for 0 of the 12 forecasts at mid-quotes and 0 at quoted bid and ask prices; the largest change at mid-quotes is that of the random forest on the HAR benchmark, 0.74 ($p = 0.14$).

- **Model confidence set, superior predictive ability, and StepM.** These three procedures are named in that appendix for the comparison of predictor sets. No table reports an MCS membership, an SPA p-value, or a StepM rejection (note E14).
- **Regime and break tests.** Regime tests and break tests have no reported output.
- **Closing-interval against intraday.** Both schemes forecast the same closing half hour. The comparison of their average QLIKE is in Section 6. A forecast of the variance remaining from an earlier origin to the close is not reported.

F Robustness of the trade

- **Definition of the Sharpe ratio.** Table 8 repeats the comparison of the full predictor set with the HAR benchmark for the 6 pooled intraday model classes under the Sharpe ratio SR of Section 7.2 and under two other definitions (note D6). Under SR the change is positive in 6 of the 6 classes (from +0.74 to +1.95), and the bootstrap test finds it statistically significant at the 5% level in 1. With each year scaled by $\sqrt{N_y}$ and the years weighted by N_y, the change is positive in 6 classes (from +0.22 to +1.00). For the ratio of the pooled daily returns scaled by $\sqrt{252}$, it is positive in 6 classes (from +0.52 to +1.65), and the Ledoit and Wolf (2008) test finds it statistically significant at the 5% level in 2. Computing SR separately on the 125 post-warmup days before 16 May 2022 and on the 489 days from that date, the change is positive in 6 and 6 classes, respectively (from +0.17 to +2.27 and from +0.63 to +1.24).
- **Date-shift test.** The sign strategy is re-scored with the forecast of the interval j half hours away from the closing interval, placed on the closing interval's diurnal adjustment:

$$q_t = \operatorname{sign}(\hat{y}^2_{t,c+j}\, s^2_{t,c} - \mathrm{IV}^2_t).$$

  Here $j = -12, \ldots, 6$: $j < 0$ reads an earlier interval of day t; $j > 0$ reads day $t + 1$, not available at the 15:30 origin (a placebo). The test is scored on the 12 pooled intraday forecasts, on the 613 post-warmup days on which every shifted forecast exists (the last post-warmup day has no next session), with the Sharpe ratio of Section 7.2 (Table 9). Across j, the median Sharpe ratio runs from −0.48 to 2.01 (a swing of ±1.25; over $j \leq 0$, from −0.48 to 1.51), against ±1.35, the 95% band of that ratio under i.i.d. returns at a true ratio of zero, $1.96\sqrt{\sum_y A_y/N_y}/|\mathcal{Y}|$.
- **Murphy diagram.** The elementary score at the implied variance is $\mathrm{ES}_{1,t}$ in Section 7.2. Its rank correlation with the Sharpe ratio, across the 12 pooled intraday forecasts, is in Section 7.3. The subsample split is around May 2022.
- **Profit and loss.** The reported return is R_t in Section 7.1. A decomposition of that return into separate variance and direction components is not reported.
- **Compounding.** Wealth starts at 1 and the premium each day is the fraction $f = 3\%$ in Section 7.2. Across the 12 pooled intraday forecasts, on the 614 post-warmup days, the median terminal value is 4.1 at mid-quotes and 2.5 at quoted bid and ask prices (12 and 11 end above 1), the median G is +0.55 and +0.35, and the maximum drawdowns reach 45% and 50% (always short: $V = 1.1$ and 0.6, drawdowns 52% and 60%; Table 9).
- **Position sizing.** The sign strategy and the two sized rules, the robust clip and the percentile rank, are in Section 7.1. All three are scored on the 614 days after the warmup of 252 days.

Table 7: Robustness: a rolling window of 3,500 trading days

		Every bar			16:00 bar		
Model	Window	HAR	Full	DM	HAR	Full	DM
Panel A: QLIKE							
Ridge	3,000	0.1893	0.1791	-5.63^{***}	0.1269	0.1279	0.31
	3,500	0.1897	0.1777	-7.06^{***}	0.1272	0.1233	-1.55
	Change	$(2.48)^{**}$	$(-2.26)^{**}$		(1.31)	$(-3.10)^{***}$	
Lasso	3,000	0.1893	0.1775	-4.72^{***}	0.1269	0.1310	1.04
	3,500	0.1897	0.1752	-6.39^{***}	0.1272	0.1253	-0.69
	Change	$(2.48)^{**}$	$(-2.04)^{**}$		(1.31)	$(-2.89)^{***}$	
Elastic net	3,000	0.1893	0.1778	-6.98^{***}	0.1269	0.1262	-0.21
	3,500	0.1897	0.1770	-8.45^{***}	0.1272	0.1223	-2.15^{**}
	Change	$(2.48)^{**}$	(-1.47)		(1.31)	$(-2.93)^{***}$	
LightGBM	3,000	0.1875	0.1613	-10.32^{***}	0.1235	0.1201	-1.81^{*}
	3,500	0.1876	0.1613	-10.02^{***}	0.1233	0.1204	-1.62
	Change	(0.20)	(0.20)		(-0.45)	(0.55)	
XGBoost	3,000	0.1888	0.1627	-9.94^{***}	0.1234	0.1192	-2.18^{**}
	3,500	0.1890	0.1636	-9.97^{***}	0.1232	0.1192	-2.19^{**}
	Change	(1.13)	$(1.87)^{*}$		(-0.55)	(-0.15)	
Random forest	3,000	0.1895	0.1685	-9.85^{***}	0.1266	0.1248	-0.92
	3,500	0.1898	0.1685	-9.40^{***}	0.1264	0.1241	-1.15
	Change	(1.15)	(0.02)		(-0.36)	(-1.43)	
Panel B: Sharpe ratio of the sign strategy							
		Mid-quotes			Bid/ask prices		
		HAR	Full	ΔSR	HAR	Full	ΔSR
Ridge	3,000	0.98	1.85	0.86	0.62	1.49	0.87
	3,500	1.00	1.74	0.74	0.64	1.38	0.74
	Change	0.02	-0.10		0.02	-0.11	
Lasso	3,000	0.98	1.72	0.74	0.62	1.36	0.74
	3,500	1.00	1.73	0.74	0.64	1.37	0.73
	Change	0.02	0.01		0.02	0.01	
Elastic net	3,000	0.98	1.94	0.96	0.62	1.58	0.96
	3,500	1.00	1.97	0.98	0.64	1.61	0.98^{*}
	Change	0.02	0.03		0.02	0.03	
LightGBM	3,000	0.61	1.75	1.14^{*}	0.25	1.40	1.15^{**}
	3,500	0.47	1.62	1.16^{**}	0.11	1.27	1.16^{**}
	Change	-0.14	-0.12		-0.14	-0.13	
XGBoost	3,000	0.91	1.67	0.76	0.55	1.31	0.76
	3,500	0.66	1.54	0.88^{*}	0.31	1.19	0.88^{*}
	Change	-0.25	-0.13		-0.24	-0.13	
Random forest	3,000	0.01	1.96	1.95^{***}	-0.32	1.60	1.93^{***}
	3,500	0.74	1.92	1.18^{**}	0.39	1.57	1.18^{**}
	Change	0.74	-0.03		0.71	-0.03	

Notes: The models of Table 1 on a rolling window of 3,500 instead of 3,000 trading days, all else equal; HAR: HAR benchmark; Full: full predictor set. Panel A: mean QLIKE on every bar (33,527 bars) and on the 16:00 bar of the $T_{\text{oos}} = 2{,}579$ trading days, 2 January 2014 to 30 April 2024. DM: Diebold–Mariano statistic (Harvey–Leybourne–Newbold, $q_{\text{NW}} = 8$) of the daily loss, full minus HAR (negative: full more accurate); Change, in parentheses: 3,500-day minus 3,000-day loss (positive: the longer window is less accurate). Panel B: Sharpe ratio of the sign strategy (Section 7.2) on the 614 post-warmup option days; ΔSR: full minus HAR; Change: 3,500-day minus 3,000-day ratio; stars from the paired block-bootstrap test of equal Sharpe ratios (Section I). *, **, ***: significance at the 10%, 5% and 1% levels.

Table 8: Sharpe ratio change under alternative definitions of the Sharpe ratio

	All 614 post-warmup days			SR by subsample	
	SR (paper)	Per-year, N_y-weighted	Pooled, $\sqrt{252}$	Before 16 May 2022	From 16 May 2022
Ridge	0.86	0.61	0.86	0.23	0.81
	[0.161]		[0.106]		
Lasso	0.74	0.64	0.81	0.17	0.74
	[0.227]		[0.119]		
Elastic net	0.96	0.66	0.95*	0.24	0.93
	[0.135]		[0.077]		
LightGBM	1.14*	0.63	0.99**	0.83	0.97
	[0.051]		[0.040]		
XGBoost	0.76	0.22	0.52	0.50	0.63
	[0.141]		[0.269]		
Random forest	1.95***	1.00	1.65**	2.27	1.24
	[0.007]		[0.020]		

Notes: Each row is a pooled intraday model class; each entry is the Sharpe ratio of the sign strategy at mid-quotes on the full predictor set minus that on the HAR benchmark. SR is the paper's Sharpe ratio (Section 7.2): the equal-weight mean over calendar years of SR_y, each year annualized by the square root of its option days per calendar year; in brackets, the p-value of the paired block-bootstrap test of equal Sharpe ratios (Section I). Per-year, N_y-weighted: each year's ratio scaled by the square root of its number of option days N_y, the years weighted by N_y. Pooled, $\sqrt{252}$: the ratio of the mean to the standard deviation of all 614 daily returns, times $\sqrt{252}$; in brackets, the p-value of the Ledoit and Wolf (2008) test, which tests the equality of these pooled ratios. The last two columns give SR on the 125 post-warmup days before 16 May 2022, the first week with an expiration on every weekday, and on the 489 days from that date, each subsample annualized as a sample of its own. *, ** and *** denote significance at the 10%, 5% and 1% levels.

Table 9: Interval-shift test and compounded wealth of the pooled intraday forecasts

		Interval-shift test: SR			Compounding			
			Over j		V		Max. drawdown (%)	
Model	Predictors	$j = 0$	Min	Max	Mid	Bid/ask	Mid	Bid/ask
Ridge	HAR	1.15	−0.54	2.23	2.64	1.62	43	47
	Full	2.03	−0.55	2.03	8.13	4.99	24	26
Lasso	HAR	1.15	−0.54	2.23	2.64	1.62	43	47
	Full	1.90	−0.78	1.90	7.63	4.68	24	26
Elastic net	HAR	1.15	−0.54	2.23	2.64	1.62	43	47
	Full	2.13	−0.40	2.13	9.12	5.60	25	30
LightGBM	HAR	0.76	−0.34	2.13	1.74	1.07	44	49
	Full	1.94	−0.32	1.94	6.32	3.89	26	28
XGBoost	HAR	1.07	−0.35	2.00	2.75	1.69	32	36
	Full	1.87	−0.41	2.15	5.40	3.32	23	28
Random forest	HAR	0.15	−0.83	2.07	1.07	0.65	45	50
	Full	2.14	−0.42	2.14	9.45	5.80	21	28

Notes: Interval-shift test: the sign strategy with the forecast of bar $c + j$ in place of that of the closing interval c, placed on the closing interval's diurnal adjustment, $q_t = \text{sign}\big(\hat{y}^2_{t,c+j}\, s^2_{t,c} - \text{IV}^2_t\big)$, for $j = -12, \ldots, 6$ ($j < 0$: earlier intervals of day t; $j > 0$: the first six intervals of day $t + 1$). SR is the paper's Sharpe ratio at mid-quotes (Section 7.2) on the 613 post-warmup days on which every shifted forecast exists (the last post-warmup day has no next session), so the $j = 0$ column differs from the Sharpe ratios on all 614 post-warmup days; Min and Max run over j, $j = 0$ included. Compounding: a fixed $f = 3\%$ of wealth per day over the 614 post-warmup days, terminal value V and maximum drawdown at mid-quotes and at quoted bid/ask prices.

- **Other origins.** No other forecast origin is reported. The interval-shift test above keeps the 15:30 close, implied volatility and straddle, and takes only the model's forecast from another interval. A separate month-end strategy is not reported.

G Closing-interval models

G.1 Closing-interval sample

- The closing-interval specifications incorporate all 47 exogenous series as fitted, the FOMC group included (the dictionary lists 46 series and 3 FOMC series since the statement-interval dummy was dropped on 22 October 2026; note E35); so do the pooled intraday specifications, with 46 series (Section 3).
- The closing-interval sample ($i = c$) consists of 3,469 trading days from July 2010 through April 2024, estimated on a rolling window of $W = 2{,}000$ trading days.
- This rolling procedure yields 1,469 out-of-sample closing-interval forecasts starting in June 2018:

$$1{,}469 = 3{,}469 - 2{,}000.$$

- Hyperparameter selection for the linear specifications is conducted on an initial pre-sample using closing-interval observations starting on 13 April 2004, with validation windows spanning 6 January 2016 to 2 January 2020.

G.2 Closing-interval predictors

- The horizon set of the closing-interval model is base 5,

$$\mathcal{K} = \{1, 5, 25, 125, 625, 3125\},$$

  the same set as the intraday model. Here, k counts retained intervals across the full intraday grid $\mathcal{G}$ (including overnight sessions), while the closing-interval specification restricts estimation to the rows $i = c$.
- In the closing-interval panel, the k-interval trailing averages of the activity indicator, $\bar{\chi}_{t,i}^{(k)}$, are computed for the 6 mode-inflated series of the dictionary and, in the fitted forecasts, for the FOMC statement-interval dummy dropped on 22 October 2026 (notes E6, E35).
- Under the closing-interval specification with the full predictor set, the window filter retains a median of 369 out of 628 candidate columns across rolling windows.
- **Predictor set dimensions.** In the closing-interval specification, the HAR benchmark contains 22 candidate columns, whereas the full predictor set contains 628 columns:

$$628 = \underbrace{22}_{\bar{y}^{(k)},\ \boldsymbol{D}} + \underbrace{47 \times 6}_{\bar{u}^{(k)}} + \underbrace{47 \times 6}_{\bar{a}^{(k)}} + \underbrace{(6+1) \times 6}_{\bar{\chi}^{(k)}}, \qquad 22 = 6 + 16.$$

G.3 Closing-interval HAR benchmark and estimation

- **Closing-interval HAR benchmark.** The benchmark forecast of the main text is a HAR-type model in the spirit of Corsi (2009), estimated by OLS on the HAR components and calendar dummies (Section 5.2) using closing-interval observations only: one observation per trading day, re-estimated daily on a rolling window of $W = 2{,}000$ trading days.
- Under the closing-interval scheme, a period is a trading day and the target is $y_t \equiv y_{t,c}$; the refit cadence of each model class is in the next item.
- **Closing-interval scheme:** $W = 2{,}000$ trading days for every model class; re-estimated every trading day, except the tree specifications re-estimated every 10 trading days (fixed hyperparameters or random search).

E32. Decision: user decision 2026-10-18: the tree specifications refit every session (every trading session), not every 10 trading days; "daily" applies only to the closing-interval linear models. The rerun is still pending.
Problem: the closing-interval tree specifications refit every 10 trading days (specs/causal_tune_trees.py:35–40, REFIT_EVERY = 10 sessions; jobs/sge/unification_tree.sge:17, TREE_REFIT_EVERY=10), so their printed numbers are not every-session refits.
Requires a rerun of the closing-interval tree specifications that refit every 10 trading days, with an every-session refit (the fixed-hyperparameter rung, subtree_daily_*, already refits every session: experiments/close_studies_export.py:75 and 80, refit_every = 1). The pooled intraday models already refit at every interval (experiments/pool_linear_pretune.py:8–9 and specs/causal_tune_linear.py:159, REFIT_FREQUENCY = 1; experiments/pooled_trees_score.py:139–140), and the closing-interval linear models and the HAR benchmark every trading day (experiments/close_linear_pretune.py:11).

G.4 Closing-interval standardization

The closing-interval model of Section G.5 standardizes the averages of series values on its own window $\mathcal{W}_t$ (Section B.2). Rows t' are trading days ($i = c$ suppressed):

$$z_t^{(k)} = \frac{\bar{u}_t^{(k)} - \operatorname{med}_{t' \in \mathcal{W}_t} \bar{u}_{t'}^{(k)}}{\max\{\mathrm{IQR}_t^{(k)},\ 0.01 \max_{t' \le t} \mathrm{IQR}_{t'}^{(k)},\ \mathrm{IQR}_t^{*(k)}\}}, \qquad \mathrm{IQR}_t^{(k)} = \mathrm{IQR}\{\bar{u}_{t'}^{(k)} : t' \in \mathcal{W}_t\}.$$

- $\mathrm{IQR}_t^{*(k)}$: IQR of $\bar{u}_{t'}^{(k)}$, $t' < t$, over rows whose k-interval averaging window is fully observed.
- Column whose mode accounts for at least 20% of rows: $\mathrm{IQR}_t^{*(k)}$ also excludes rows at the mode.
- $\mathrm{IQR}_t^{*(k)} = 1$ until 8 such rows exist.
- $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$: not standardized. HAR components, calendar dummies: $\mathrm{IQR}_t^{*(k)} = 0$.
- $t < W$: the first window's statistics.

G.5 Training on the last 30 minutes

- **Closing-interval estimation scheme ($\mathcal{I} = \{c\}$).** As a final refinement, we restrict estimation of each model class in Section 5.2 exclusively to observations from the closing half hour $i = c$ (15:30–16:00 ET), matching the estimation domain of the daily HAR benchmark.
  - Under this scheme, each trading day contributes a single observation, so a period corresponds to one trading day and the rolling window $\mathcal{W}_t = \{t - W, \ldots, t - 1\}$ spans $W = 2{,}000$ trading days (Appendix B.2). Rolling standardization of the exogenous series averages on $\mathcal{W}_t$ follows Appendix G.4.
  - We hold the predictor sets, evaluation sample $\mathcal{T}_{\text{oos}}$, and straddle trading rules unchanged; statistical accuracy is reported in Table 10 for the 15:30 forecasts of the closing half hour over $\mathcal{T}_{\text{oos}}$.
- **Improvements in QLIKE accuracy.** Training solely on the closing interval reduces mean QLIKE relative to pooled intraday training in every model class and under both predictor sets. Specifically, the percentage excess loss of the pooled intraday forecast $f_{m,\mathcal{X},\{1,\ldots,M\}}$ relative to the closing-interval forecast $f_{m,\mathcal{X},\{c\}}$, $\Delta\% = 100(\bar{L}^{\text{pool}}/\bar{L}^{\text{close}} - 1)$, is strictly positive throughout:
  - for penalized linear models, pooling increases mean QLIKE by +10.1% on the HAR benchmark for each of the three (DM = 2.61, significant at the 5% level; the pooled HAR-set penalized models coincide with OLS at this window, Section 6), and by +21.6% to +31.2% on the full predictor set (significant at the 5% level for 3 of the three linear classes);
  - for tree ensembles, pooling increases mean QLIKE by +0.7% to +5.9% on the HAR benchmark and by +3.9% to +6.0% on the full predictor set; the difference is significant at the 5% level for 0 of the three tree classes on the HAR benchmark and for 0 on the full predictor set.
- **Preservation of the exogenous-predictor gains.** Within the closing-interval scheme, expanding from the HAR benchmark to the full predictor set lowers mean QLIKE in all 6 model classes, with relative reductions $\Delta\%$ between -4.1% and -0.1%, although none of the single-class reductions is individually significant at the 5% level (for lasso, $\Delta\% = -4.0\%$, DM $= -1.91$, $p = 0.056$).
  - Across all 51 matched predictor-set pairs in the full model set (45 tree and 3 linear specifications trained on the closing interval, plus 3 pooled intraday specifications), adding the exogenous predictors reduces mean QLIKE in 35 pairs, 4 of which are significant at the 5% level.
- Across the 12 pooled intraday forecasts, 0 achieve a lower mean QLIKE than the daily HAR benchmark (0 significantly), while 10 perform significantly worse; the closing-interval forecasts are left out of this count.
- Comparing tree ensembles directly against the closing-interval ridge regression on the same

predictor set, the tree specification incurs a higher mean QLIKE in 48 of 48 pairs (23 significantly).

- **Sensitivity of squared-error loss to extreme days.** In contrast to QLIKE, evaluated under MSE the full predictor set increases squared-error loss in 49 of the 51 matched pairs across the model set (0 significantly).
  - A single date outside the option-data subsample, 19 March 2020, accounts for 28–39% of the total cumulative squared error of the closing-interval tree models over $\mathcal{T}_{\text{oos}}$, compared with at most 0.9% of their total QLIKE loss.
  - Similarly, five trading days in March 2020 account for 73–87% of the cumulative squared error of the daily HAR benchmark and of the closing-interval lasso and XGBoost on the full predictor set; when February–April 2020 are excluded, the closing-interval lasso achieves a lower MSE than the benchmark forecast.
- **Economic value under the closing-interval scheme** (sign strategy evaluated on the 614 post-warmup days of $\mathcal{T}_{\text{opt}}$):
  - The closing-interval elastic net on the full predictor set attains SR $= 1.68$ at mid-quotes and 1.33 at quoted bid/ask prices; its mean return gain over the short-straddle benchmark has $t = 1.92$ (fixed-b $p = 0.072$). The closing-interval lasso on the full predictor set attains SR $= 1.44$ at mid-quotes and 1.09 at quoted bid/ask prices.
  - Closing-interval LightGBM on the full predictor set, with fixed hyperparameters re-estimated every trading day (the specification of the closing-interval tree rows of the main results table), attains SR $= 1.19$ at mid-quotes; the tree rows of the straddle table are the pooled intraday trees.
  - Replacing the HAR benchmark with the full predictor set changes the closing-interval sign-strategy Sharpe ratio by $\Delta\text{SR} = +0.21$ for the elastic net and -0.03 for the lasso (neither statistically significant at the 5% level by the block-bootstrap test of equal Sharpe ratios); across all 51 matched predictor-set pairs in the model set, $\Delta\text{SR} > 0$ in 47 pairs (median $+0.71$), and the same test finds a statistically significant gain at the 5% level in 13 pairs and a statistically significant loss in 0.
  - Relative to the unconditional short-straddle benchmark, $\Delta\text{SR} > 0$ for 106 of the 112 forecasts in the model set; the mean return gain is statistically significant at the 5% level (fixed-b test) for 0 of them, 0 of which use the full predictor set.
  - Finally, of the 12 mean return differentials between a pooled intraday model and the closing-interval model of the same class and predictor set, 1 is statistically significant at the 5% level (fixed-b test of the mean daily return differential, bandwidth $S = 33$, on the 614 post-warmup days).

H Reproducibility

- **Generator of each reported number.**
  - `writeup/make_rehaul_data_methods.py`: samples, windows, hyperparameter tables, and the predictor dictionary.
  - `writeup/make_rehaul_oos_rerun.py`: Table 1.
  - `writeup/make_rehaul_oos_table.py`: Table 10.
  - `writeup/make_rehaul_results.py`: the accuracy macros in Section 6.
  - `writeup/make_rehaul_economics.py`: the straddle macros in Section 7.

Table 10: Forecast accuracy when training on the last 30 minutes

	HAR benchmark			All features		
Model	MSE (10^{-10})	QLIKE	R^2_{OOS} (%)	MSE (10^{-10})	QLIKE	R^2_{OOS} (%)
OLS (benchmark forecast)	4.15	0.1033	78.2	–	–	–
Panel A: Linear models						
Ridge	4.15	0.1033	78.2	7.30	0.1031	61.7
Lasso	4.15	0.1033	78.2	5.94	0.0992	68.8
Elastic net	4.15	0.1033	78.2	6.89	0.1014	63.9
Panel B: Tree ensembles						
LightGBM	5.75	0.1094	69.8	17.19	0.1049	9.8
XGBoost	5.27	0.1076	72.4	20.94	0.1031	−9.9
Random forest	4.62	0.1068	75.8	14.15	0.1060	25.8

Notes: Rolling forecasts from 3 January 2020 to 30 April 2024, every forecast made with a model estimated on its trailing window only, so every forecast is out of sample; each is scored against the actual realized variance of the closing half hour, RV_t, on the $T_{\text{oos}} = 1{,}082$ trading days on which every forecast of both training schemes exists, early closes dropped (Section 5.3). The span starts with the intraday linear forecasts, on the first trading day after the pre-sample on which the penalties were selected (6 January 2016 to 2 January 2020). The MSE entries are in units of 10^{-10}: the printed number times 10^{-10} is the mean squared error $T_{\text{oos}}^{-1}\sum_t(\text{RV}_t - \hat{\sigma}_t^2)^2$. QLIKE is the mean QLIKE loss. $R^2_{\text{OOS}} = 1 - \sum_t(\text{RV}_t - \hat{\sigma}_t^2)^2 / \sum_t(\text{RV}_t - \overline{\text{RV}}_{t-1})^2$, in percent, where the benchmark $\overline{\text{RV}}_{t-1}$ is the historical mean of the closing-half-hour realized variance over all trading days from 5 January 1998 to day $t-1$ (Campbell and Thompson, 2008). All features is the full predictor set of Section 4; the OLS is estimated on the HAR benchmark only (–: no forecast). Each model is estimated on the last 30 minutes of each trading day only (one observation per day); the OLS on the HAR benchmark is the benchmark forecast. Trees use fixed hyperparameters and are re-estimated daily (Section B.2). The intraday models on the same days are in Table 1.

- `writeup/make_rehaul_fi.py`: the penalty weight on the HAR block in Appendix D and the closing-interval forecast count in Appendix G.1.
- `writeup/make_rehaul_appendix.py`: the appendix-table macros.
- `writeup/make_table_close_main.py`: the closing-interval master-table macros, including the closing-interval predictor counts.
- `writeup/make_fig_qlike_sharpe.py`: the matched predictor-set pair counts.
- `writeup/hac_inference.py`: the fixed-b test of a mean return, the Sharpe ratio of Section 7.2 and the block-bootstrap test of its differences, and the Ledoit–Wolf test of equal Sharpe ratios, used by the three generators above; `writeup/make_rehaul_results.py` writes Table 14 and `writeup/make_rehaul_economics.py` Table 8.
- `writeup/final_run.py`: the switch (`REHAUL_FINAL_RUN`) that points the two loaders below and the generators of the pooled numbers at the final pooled runs, and the final design (first evaluation day, pre-sample fold, window): the linear models in results/window_sweep_2014start_2026-10-22/w3000, LightGBM and XGBoost in results/pooled_trees_final_2014start_w3000, and the random forest in results/pooled_rf_skx_2014start_w3000; under the robustness window, the w3500 directories of the same three runs (results/pooled_rf_skx_2014start_w3500 for the random forest).
- `writeup/pooled_trees_13.py`: the forecasts of the pooled tree ensembles, fitted by `experiments/pooled_trees_rerun_13.py` (LightGBM and XGBoost in results/pooled_trees_final_2014start_w3000, the random forest, on Intel's oneDAL engine, in results/pooled_rf_skx_2014start_w3000). They are read

by `writeup/make_rehaul_results.py`, `writeup/make_rehaul_economics.py`, `writeup/make_rehaul_oos_table.py`, `writeup/make_rehaul_oos_rerun.py` and `writeup/make_rehaul_data_methods.py`.

  - `writeup/pooled_linear_13.py`: the forecasts and frozen penalties of the pooled linear models of Table 1 (results/window_sweep_2014start_2026-10-22/w3000, fitted by `experiments/pooled_linear_rerun_2026_10_19_13.py`), read by `writeup/make_rehaul_results.py`, `writeup/make_rehaul_economics.py` and `writeup/make_rehaul_data_methods.py`; `writeup/make_rehaul_oos_rerun.py` reads the same run directly.
  - `writeup/make_buckets_tex.py`: the group sizes and the exogenous series count.
  - `experiments/close_murphy_splits_headline.py`: the Murphy-diagram macros.
  - `writeup/make_rehaul_pooled_extras.py`: the interval-shift test and the compounding of the pooled intraday forecasts in Appendix F and Table 9.
  - `writeup/make_rehaul_robust_window.py`: the comparison of the windows of 3,000 and 3,500 trading days in Appendix E and Table 7.
  - `writeup/make_rehaul_fi_pooled.py`: the variable-importance numbers of the pooled models in Section 6.5.
  - `writeup/make_rehaul_disclosures.py`: the disclosure numbers of Section 3, the ends of the feeds and the data fixes of the final design.
  - `writeup/tests/check_pending.py`: checks that no generated file carries a number for a forecast whose final run has no forecasts yet.

- **Software list.** `requirements.txt` at the repository root names the packages and a version range for each. It is not a pinned lock file, and the draft does not record a second lock file.
- **Design matrices, CPU class, and compute time.** The generators read the forecasts and tables already on disk. The draft does not record a hash of the design matrix behind Table 1, the CPU class of those fits, or their compute time.

Table 11: Estimation and evaluation samples.

Sample	Start	End	Obs.	Window W
Intraday grid: 30-minute intervals with an observed $RV_{t,i}$, outside the weekend break	1998-01-05	2024-04-30	303,442	–
Panel A: Closing-interval models (trading days)				
Closing-interval observations, after the burn-in	2010-07-12	2024-04-30	3,469	2,000
Rolling-window forecasts	2018-06-25	2024-04-30	1,469	–
Hyperparameter-selection pre-sample (linear models)	2004-04-13	2020-01-02	3,947	2,000
Validation samples (4×250 days)	2016-01-06	2020-01-02	1,000	2,000
Panel B: Intraday models (30-minute intervals)				
Estimation sample, after the 3,125-interval burn-in	1998-06-04	2024-04-30	300,317	96,000
Regular-hours bars, bar ends 10:00–16:00 ET: the trees and the pooled linear models of Table 1 and Sections 6–7	1998-06-04	2024-04-30	85,314	39,000
Rolling-window forecasts from the first evaluation day, less early-close days (20)	2014-01-02	2024-04-30	33,970	–
Penalty selection (pooled linear models): validation sample of pre-sample fold 3	2013-01-02	2013-12-31	3,250	39,000
Tree hyperparameter pre-sample (before 2003)	1998-06-04	2002-12-31	36,285	24,000
Panel C: Evaluation samples (trading days)				
Evaluation span (T_{oos}): trading days on which every forecast exists, less early-close days (6)	2020-01-03	2024-04-30	1,082	–
Option-data subsample (T_{opt}): SPXW expiration days (872) less early-close days (6)	2020-01-03	2024-04-30	866	–
Later SPXW expiration days, held out (413 full trading days)	2024-05-01	2025-12-31	419	–

Notes: Dates are interval-end dates (ET); Obs. and the rolling estimation window W are in the units of each panel. The burn-in is the longest HAR horizon, 3,125 observations of each series as built: on the closing-interval series, one observation a trading day, 3,125 trading days, so the closing-interval sample starts in 2010; on the series of the two intervals of the last hour (the linear pre-sample) 1,560 trading days, so the pre-sample starts in 2004; on the intraday grid (Panel B) 3,125 30-minute intervals, 104 trading days. The trees and the pooled linear models of Table 1 and Sections 6–7 are re-estimated at every bar on the previous 39,000 regular-hours bars (3,000 trading days of 13 bars); the pooled linear models on all 48 intervals (in the model set) use $W = 96{,}000$ intervals (2,000 trading days). The tree hyperparameters are chosen once, by an Optuna search on the 1998–2002 development sample that fits each candidate on the previous 24,000 intervals and scores it on the next 250, stepping forward 250 intervals at a time. Each validation sample follows a 25-day gap and a 2,000-day estimation sample (Section B.1). The penalties of the pooled linear models on the regular-hours bars are chosen once, on pre-sample fold 3: each candidate is fitted on the W bars before a gap of 25 sessions (325 bars) and scored on the next 250 sessions (3,250 bars, the validation row above); a penalty selected at an end of its grid adds the next decade in that direction until the selection is interior, stopping when the criterion changes by less than 10^{-6} relative or after 6 decades. Full trading days: expiration days that are not early-close days. Panel C: the evaluation span holds every trading day on which all rolling forecasts exist (Section 5.3); the straddle economics use its option-data subsample (Section 7).

Table 12: Hyperparameters: penalized regressions and tree ensembles.

Model	Hyperparameter	Search space	Value: HAR benchmark	Value: Full predictor set
Panel A: Penalized linear models, selected on the pre-sample				
OLS	–	–	–	–
Ridge	Penalty, λ	$\{10^{-2}, 10^{-1}, \ldots, 10^{3}\}$ (6)	10^{-2}	10^{3}
Lasso	Penalty, λ	$\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5)	10^{-6}	10^{-2}
Elastic net	Penalty, λ	$\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5)	10^{-6}	10^{-2}
	Mixing weight, α	fixed	0.5	0.5
All three	Penalty weight on the HAR benchmark, $\underline{\omega}$	fixed	1	0.01
Panel B: Tree ensembles, fixed hyperparameters (selected on the 1998–2002 pre-sample)				
LightGBM	Leaves per tree, J	[15, 255] (log)	20	
	Learning rate, η	[0.01, 0.3] (log)	0.013	
	Boosting rounds, B	[50, 500]	392	
	Minimum leaf size, $n_{\min}$	[20, 500] (log)	98 (8)	
	Share of predictors per tree	[0.5, 1]	0.70	
	Share of observations per tree	[0.5, 1]	0.56	
	L_1 penalty on leaf weights, λ_1^{tree}	$[10^{-8}, 10]$ (log)	10^{-8}	
	L_2 penalty on leaf weights, λ_2^{tree}	$[10^{-8}, 10]$ (log)	0.016	
XGBoost	Maximum depth	[3, 10]	5	
	Learning rate, η	[0.01, 0.3] (log)	0.0123	
	Boosting rounds, B	[50, 500]	367	
	Minimum leaf size, $n_{\min}$	[1, 100] (log)	50.2 (4.18)	
	Share of observations per tree	[0.5, 1]	0.66	
	Share of predictors per tree	[0.5, 1]	0.65	
	L_1 penalty on leaf weights, λ_1^{tree}	$[10^{-8}, 10]$ (log)	0.0215	
	L_2 penalty on leaf weights, λ_2^{tree}	$[10^{-8}, 10]$ (log)	1.2×10^{-5}	
	Penalty per leaf, γ	$[10^{-8}, 1]$ (log)	1.5×10^{-6}	
Random forest	Number of trees, N	[50, 500]	485	
	Maximum depth	[2, 32] (log)	20	
	Share of predictors per split	[0.02, 1] (log)	0.90	
	Bootstrap sample share	[0.05, 1] (log)	0.41	
	Minimum leaf size, $n_{\min}$	[1, 1000] (log)	12 (1)	

Notes: Panel A: Section 5.2 states the scale of λ. One grid value per decade (grid size in parentheses), selected by mean validation MSE over the 4 folds of the pre-sample (Table 11). The selected penalty is then held fixed at every forecast origin, with no re-tuning. Panel B: the best trial of an Optuna search on the intraday pre-sample (24,000-interval window; random forest: trial 255 of 400), ranked by MSE and then held fixed for both predictor sets; (log): searched on a logarithmic scale. $n_{\min}$ was selected on 24,000 intervals and is rescaled to the 2,000-day window (rescaled value in parentheses). Implementations: LightGBM (Ke et al., 2017) with bagging at every round, XGBoost (Chen and Guestrin, 2016) with histogram-based split finding and the random forest of Intel's oneDAL library (through the Intel Extension for Scikit-learn) with bootstrap samples and histogram-based split finding (256 bins); random seeds are fixed.

Table 13: Predictor definitions.

Variable	Description	Source	Diurnal adjustment	Transform $g(v)$	Predictors, $k \in \mathcal{K}$
Realized moments (7; front-month E-mini, one-minute log-price changes)					
`sumret`	Sum of one-minute changes in the log price of the front-month contract, $\texttt{sumret}_{t,i} = \sum_k r_{t,i,k}$	E-mini S&P 500 futures	s.d.	v	$u^{(k)}$, $\bar{a}^{(k)}$
`sumabsret`	Realized absolute variation, $\sum_k \lvert r_{t,i,k}\rvert$	E-mini S&P 500 futures	mean	v	$u^{(k)}$, $\bar{a}^{(k)}$
`sumret3`	Realized skewness, unnormalized, $\sum_k r_{t,i,k}^3$	E-mini S&P 500 futures	s.d.	$v^{1/3}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sumret4`	Realized quarticity, unscaled, $\sum_k r_{t,i,k}^4$	E-mini S&P 500 futures	mean	$\text{sign}(v)\,\lvert v\rvert^{1/4}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sumpret2`	Positive realized semivariance, $\sum_k r_{t,i,k}^2\,\mathbf{1}\{r_{t,i,k} > 0\}$	E-mini S&P 500 futures	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sumbipow`	Bipower variation, $\texttt{sumbipow}_{t,i} = \sum_{k\geq 2} \lvert r_{t,i,k}\rvert\,\lvert r_{t,i,k-1}\rvert$	E-mini S&P 500 futures	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sumautocov`	First-order realized autocovariance, $\texttt{sumautocov}_{t,i} = \sum_{k\geq 2} r_{t,i,k}\,r_{t,i,k-1}$	E-mini S&P 500 futures	s.d.	$\text{sign}(v)\,\lvert v\rvert^{1/2}$	$u^{(k)}$, $\bar{a}^{(k)}$
Liquidity (14)					
`sumvolume`	Number of front-month contracts traded in the half hour	E-mini S&P 500 futures	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
`numobs`	Number of non-missing one-minute intervals in the half hour, $\texttt{numobs}_{t,i} \leq 30$; full-sample mode 30 on 72% of intervals	E-mini S&P 500 futures	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)\dagger}$
`turnover_ewstock`	Equal-weighted turnover, shares traded in the half hour divided by the previous day's shares outstanding	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`buyturnover_ewstock`	Equal-weighted buyer-initiated turnover, the same ratio using estimated buy orders	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sellturnover_ewstock`	Equal-weighted seller-initiated turnover, the same ratio using estimated sell orders	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`effspread_ewstock`	Equal-weighted effective spread, the mean percent difference between the trade price and the preceding midquote	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`spread_ewstock`	Equal-weighted quoted spread, the mean percent difference between the bid and the ask	Constituents	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$

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Table 13 (continued)

Variable	Description	Source	Diurnal adjustment	Transform $g(v)$	Predictors, $k \in \mathcal{K}$
`turnover_vwstock`	Value-weighted turnover, shares traded in the half hour divided by the previous day's shares outstanding	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`buyturnover_vwstock`	Value-weighted buyer-initiated turnover, the same ratio using estimated buy orders	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sellturnover_vwstock`	Value-weighted seller-initiated turnover, the same ratio using estimated sell orders	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`effspread_vwstock`	Value-weighted effective spread, the mean percent difference between the trade price and the preceding midquote	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`spread_vwstock`	Value-weighted quoted spread, the mean percent difference between the bid and the ask	Constituents	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
`ofi_ewstock`	Equal-weighted order-flow imbalance, (Buy − Sell)/(Buy + Sell) of turnover	Constituents	s.d.	v	$u^{(k)}$, $\bar{a}^{(k)}$
`ofi_vwstock`	Value-weighted order-flow imbalance, (Buy − Sell)/(Buy + Sell) of turnover	Constituents	s.d.	v	$u^{(k)}$, $\bar{a}^{(k)}$
Equal-weighted market (6; simple averages of S&P 500 constituents)					
`sumret2_ewstock`	Equal-weighted realized variance of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sumret3_ewstock`	Equal-weighted realized skewness of 5-minute log returns, unnormalized	Constituents	s.d.	$v^{1/3}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sumret4_ewstock`	Equal-weighted realized quarticity of 5-minute log returns, unscaled	Constituents	mean	$\text{sign}(v)\,\lvert v\rvert^{1/4}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sumabsret_ewstock`	Equal-weighted realized absolute variation of 5-minute log returns	Constituents	mean	v	$u^{(k)}$, $\bar{a}^{(k)}$
`sumbipow_ewstock`	Equal-weighted bipower variation of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sumpret2_ewstock`	Equal-weighted positive realized semivariance of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
Value-weighted market (6; previous day's market capitalization)					
`sumret2_vwstock`	Value-weighted realized variance of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sumret3_vwstock`	Value-weighted realized skewness of 5-minute log returns, unnormalized	Constituents	s.d.	$v^{1/3}$	$u^{(k)}$, $\bar{a}^{(k)}$

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Table 13 (continued)

Variable	Description	Source	Diurnal adjustment	Transform $g(v)$	Predictors, $k \in \mathcal{K}$
`sumret4_vwstock`	Value-weighted realized quarticity of 5-minute log returns, unscaled	Constituents	mean	$\text{sign}(v)\,\lvert v\rvert^{1/4}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sumabsret_vwstock`	Value-weighted realized absolute variation of 5-minute log returns	Constituents	mean	v	$u^{(k)}$, $\bar{a}^{(k)}$
`sumbipow_vwstock`	Value-weighted bipower variation of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
`sumpret2_vwstock`	Value-weighted positive realized semivariance of 5-minute log returns	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
Sentiment (3)					
`stocktwits_attention`	Number of StockTwits posts in the half hour	StockTwits	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
`stocktwits_sentiment`	Score of tagged posts, 1 if all are bullish and −1 if all are bearish	StockTwits	none	v	$u^{(k)}$, $\bar{a}^{(k)}$
`stocktwits_sentcount`	Number of posts in the half hour that carry a bullish or bearish tag	StockTwits	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
Implied volatility (3)					
`vix`	VIX index (30-day S&P 500 implied volatility)	Cboe	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
`vvix`	VVIX index (implied volatility of VIX)	Cboe	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
`vix3m`	VIX3M index (3-month S&P 500 implied volatility)	Cboe	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
Volatility demand (4; net vega bought by non-market-makers)					
`voldemand_spx_open_and_close`	Net vega bought by non-market-makers, S&P 500 index and top S&P 500 ETF options, trades that open or close a position; full-sample mode 0 on 59% of intervals	Cboe	none	$\text{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)\dagger}$
`voldemand_spx_open_only`	Net vega bought by non-market-makers, S&P 500 index and top S&P 500 ETF options, trades that open a position; full-sample mode 0 on 60% of intervals	Cboe	none	$\text{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)\dagger}$
`voldemand_all_open_and_close`	Net vega bought by non-market-makers, all traded options, trades that open or close a position; full-sample mode 0 on 59% of intervals	Cboe	none	$\text{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)\dagger}$
`voldemand_all_open_only`	Net vega bought by non-market-makers, all traded options, trades that open a position; full-sample mode 0 on 60% of intervals	Cboe	none	$\text{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)\dagger}$

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Table 13 (continued)

Variable	Description	Source	Diurnal adjustment	Transform $g(v)$	Predictors, $k \in \mathcal{K}$
FOMC (3; scheduled calendar)					
`fomc_day`	FOMC announcement-day dummy; full-sample mode 0 on 95% of intervals	FOMC calendar	none	v, not winsorized	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)\dagger}$
`fomc_until_inv`	$1/(1+n^{+}_{t,i})$, $n^{+}_{t,i}$ intervals to the next announcement	FOMC calendar	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
`fomc_since_inv`	$1/(1+n^{-}_{t,i})$, $n^{-}_{t,i}$ intervals since the last announcement	FOMC calendar	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$

Notes: Each series u enters as $u^{(k)}$, its adjusted, transformed and winsorized value averaged over the previous k intervals, and as $\bar{a}^{(k)}(u)$, the share of those intervals on which it was observed. †Closing-interval models only: $\bar{\chi}^{(k)}(u)$, the share of those intervals off the series' mode, for the 6 series whose full-sample mode accounts for at least 20% of intervals (mode and share in the description), the rule of the closing-interval forecasts reported, which predate the causal rule of Section 4. The pooled intraday models of Table 1 carry no $\bar{\chi}^{(k)}$. Under the causal rule, the specification, 5 series are mode-inflated at some interval: `numobs`, `stocktwits_attention`, `stocktwits_sentiment`, `stocktwits_sentcount` and `fomc_day`. $\mathcal{K} = \{1, 5, 25, 125, 625, 3125\}$ in the closing-interval model and in the intraday model; both models use every group here (Section 3). Diurnal adjustment: division by the trailing same-interval mean (mean) or standard deviation (s.d.); g: the variance-stabilizing map; the volatility-demand series take $\operatorname{asinh}(u_{t,i}/\kappa_{t,i})$ in place of both where the routing rule of Section 4 applies, on 99.4% or more of their intervals after the first print. A series is signed when the quantity it measures can be negative, not when its sample has a negative value. The columns give the chain of the pooled intraday forecasts of Table 1, in which the standard-deviation floor and the inverse hyperbolic sine routing use earlier intervals only. The closing-interval forecasts predate that chain: they take both over the full sample, call a series signed when any value in its full sample is negative, and log and winsorize the FOMC day dummy. In this table that changes two rows: `spread_vwstock`, signed by a single negative print (2018-11-12 11:00 ET) and so given $g(v) = v$ and the s.d. adjustment, and `fomc_day`, given $\log v$. Unnormalized and unscaled: without the factors $\sqrt{N_{t,i}}/\mathrm{RV}_{t,i}^{3/2}$ (skewness), $N_{t,i}/3$ (quarticity) and $\pi/2$ (bipower variation).

I Statistical tests

- **Diebold–Mariano test.** To compare the predictive accuracy of two competing forecasts A and B, we apply the Diebold and Mariano (1995) test to the daily QLIKE loss differentials, incorporating the small-sample adjustment of Harvey et al. (1997) for one-step-ahead forecasts:

$$d_t = L_t^A - L_t^B, \qquad \mathrm{DM} = \sqrt{\frac{|\mathcal{O}|-1}{|\mathcal{O}|}}\,\frac{\bar{d}}{\sqrt{\hat{\Omega}/|\mathcal{O}|}}.$$

  - Here A and B denote competing forecast series $f_{m,\mathcal{X},\mathcal{I}}$, indexed by the model class m, the predictor set $\mathcal{X} \in \{\mathrm{HAR}, \mathrm{Full}\}$, and the estimation interval set $\mathcal{I} \in \{\{c\}, \{1,\dots,M\}\}$.
  - The evaluation set $\mathcal{O}$, indexed chronologically by $t = 1,\dots,|\mathcal{O}|$, equals the full evaluation span $\mathcal{O} = \mathcal{T}_{\mathrm{oos}}$ ($|\mathcal{T}_{\mathrm{oos}}| = T_{\mathrm{oos}}$) for statistical accuracy comparisons and the option-data subsample $\mathcal{T}_{\mathrm{opt}} \subset \mathcal{T}_{\mathrm{oos}}$ for straddle return comparisons (Section 3).
  - We denote by $\bar{d}$ the sample mean of d_t over $\mathcal{O}$ and compute two-sided p-values under the standard normal distribution as $p = 2\{1 - \Phi(|\mathrm{DM}|)\}$.
  - Under this sign convention, $\mathrm{DM} < 0$ indicates that forecast A achieves a lower average loss than forecast B. Throughout the paper, "statistically significant" refers to significance at the 5% level ($p < 0.05$).
- **Newey–West long-run variance.** Following Newey and West (1987), we estimate the long-run variance $\hat{\Omega}$ using Bartlett kernel weights with lag truncation parameter q_{NW}:

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{q_{\mathrm{NW}}}\Big(1 - \frac{j}{q_{\mathrm{NW}}+1}\Big)\hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{|\mathcal{O}|}\sum_{t=j+1}^{|\mathcal{O}|}(d_t - \bar{d})(d_{t-j} - \bar{d}).$$

E13. Decision: regenerate the remaining DM statistics.
Problem: `dm_test` now divides $\hat{\gamma}_j$ by $|\mathcal{O}|$, as displayed. The accuracy tables and macros were regenerated on 2026-10-18, but the DM statistics and significance stars in the appendix tuning and study tables are read from stored study outputs computed with the old divisor $|\mathcal{O}| - j$, so they predate the change and must be regenerated.

  - For the Diebold–Mariano test, we set the bandwidth according to the plug-in rule $q_{\mathrm{NW}} = \lfloor 4(|\mathcal{O}|/100)^{2/9} \rfloor$ (its numerical value on $\mathcal{T}_{\mathrm{oos}}$ is reported in Section 6.1).
- **Test of a mean daily return.** For any daily return series r_t—either the return of an individual trading strategy or the return differential Δ_t between two strategies—observed on the $T = 614$ post-warmup days $\mathcal{T}_{\mathrm{opt}}^{+}$ (Section 7.2), we test whether its mean is zero with the procedure that Lazarus et al. (2018) recommend for the Newey–West estimator:

$$t = \frac{\bar{r}}{\sqrt{\hat{\Omega}_r/T}}, \qquad \hat{\Omega}_r = \sum_{|j|<S}\Big(1 - \frac{|j|}{S}\Big)\hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{T}\sum_{t=|j|+1}^{T}(r_t - \bar{r})(r_{t-|j|} - \bar{r}),$$

  with the Bartlett kernel, bandwidth $S = \lceil 1.3\,T^{1/2} \rceil = 33$, and fixed-$b$ critical values for $b = S/T$ (Kiefer and Vogelsang, 2005).
  - The autocovariances divide by T, with no small-sample factor; at lag truncation $S-1$ the estimator equals the Newey–West variance of the previous item.
  - This bandwidth is larger than that of conventional rules such as $0.75\,T^{1/3}$, which Lazarus et al. (2018) obtain from the formula of Andrews (1991) for a first-order autoregression with coefficient 0.25. Combined with fixed-b critical values, it reduces the size distortion of the test when the returns are serially correlated (Lazarus et al., 2018).

  - Under the null hypothesis and fixed b, the limiting distribution of t is a functional of Brownian motion that depends only on the kernel and on b (Kiefer and Vogelsang, 2005). We obtain its quantiles by simulation, computing the same statistic on 1,000,000 independent series of 1,000 i.i.d. standard normal draws with bandwidth $1{,}000\,b$. This gives the two-sided 5% critical value 2.11 and the p-values reported in the text, in place of the standard normal value 1.96.

- **Test of equal Sharpe ratios.** A test of $\bar{\Delta} = 0$ concerns the mean return differential, not the difference between two Sharpe ratios, which divides each strategy's mean return by its own standard deviation. For every statement about a Sharpe ratio difference $\Delta\mathrm{SR} = \mathrm{SR}(r^A) - \mathrm{SR}(r^B)$, with SR the ratio of Section 7.2, we therefore use a paired circular block bootstrap of $\Delta\mathrm{SR}$.
  - Each of 4,999 bootstrap samples joins blocks of 10 consecutive days, drawn with replacement from the T post-warmup days arranged on a circle; both strategies are resampled on the same days. Each drawn day keeps its own calendar year, and A_y stays at its value in the original sample, so each sample gives $\Delta\mathrm{SR}^*$ computed exactly as $\Delta\mathrm{SR}$.
  - The p-value is two-sided, the share of samples with $|\Delta\mathrm{SR}^* - \Delta\mathrm{SR}| \geq |\Delta\mathrm{SR}|$. The block length is fixed rather than calibrated, and the random seed is fixed.
  - The bootstrap is not studentized: SR averages ratios over calendar years, and no closed-form standard error of that average is used.
- **Test of equal Sharpe ratios of the pooled daily returns** (robustness). The studentized circular block bootstrap test of Ledoit and Wolf (2008) tests the equality of the Sharpe ratios of the daily returns pooled over the T days, the ratio of the mean to the standard deviation of all T returns. It is the test of the second alternative definition in Table 8 and is reported beside the bootstrap test above in Table 14.
  - With $u = (\mu_A, \mu_B, \mathrm{E}[(r_t^A)^2], \mathrm{E}[(r_t^B)^2])$, the difference of the Sharpe ratios of the daily returns, $\mu_A/\sigma_A - \mu_B/\sigma_B$, is a smooth function $f(u)$; its standard error follows from the delta method and the long-run covariance matrix Ψ of $(r_t^A, r_t^B, (r_t^A)^2, (r_t^B)^2)$.
  - The test statistic is $d = |f(\hat{u})|/\mathrm{se}(\hat{u})$. Each of 4,999 bootstrap samples joins blocks of 10 consecutive days, drawn with replacement from the T post-warmup days arranged on a circle, and gives $d^* = |f(\hat{u}^*) - f(\hat{u})|/\mathrm{se}(\hat{u}^*)$, with Ψ estimated in that sample by the block-based estimator of Ledoit and Wolf (2008). The p-value is $\big(\#\{d^* \geq d\} + 1\big)/(4{,}999 + 1)$.
  - In the original sample, Ψ is estimated with the Bartlett kernel at bandwidth equal to the block length, whose expectation is approximately that of the block-based estimator; Ledoit and Wolf (2008) use a prewhitened quadratic-spectral kernel at this step. The block length is fixed at 10 days rather than calibrated; the next item reports the number of rejections at block lengths 5 and 20. The random seed is fixed.
  - The test concerns the Sharpe ratios of daily returns over the T post-warmup days, without the year-by-year annualization of Section 7.2.
- **Robustness to the long-run variance estimator.** Table 14 lists every comparison of strategy returns whose statistical significance Section 7 or Appendix G.5 states. For each comparison it reports the t-statistic of the mean return differential with the Newey–West variance at lag truncation $q_{\mathrm{NW}} = \lfloor 4(T/100)^{2/9} \rfloor = 5$ (Newey and West, 1994) and at $q_{\mathrm{NW}} = \lfloor 1.5\,T^{1/3} \rfloor = 12$, the lag truncation of an earlier version of this paper, both with standard normal p-values; the fixed-b test above; the p-value of the bootstrap test of equal Sharpe ratios; and that of the Ledoit and Wolf (2008) test. Statements in the text about mean returns use the fixed-b test, and statements about Sharpe ratios use the bootstrap test.

  - Of the 34 tests of a mean, 3 are statistically significant at the 5% level under the fixed-b test, 7 at lag truncation 5 and 8 at lag truncation 12 with standard normal critical values.
  - Of the 33 tests of equal Sharpe ratios, 9 reject at the 5% level under the bootstrap test; under the Ledoit and Wolf (2008) test of the pooled ratios, 8 reject with block length 10, 8 with block length 5 and 4 with block length 20.

- **Multiple-comparison procedures.** For joint comparisons across predictor sets and model families, one may also consider the model confidence set of Hansen et al. (2011), the superior predictive ability test of Hansen (2005), and the StepM procedure of Romano and Wolf (2005).

  E14. Decision: drop the sentence, or report the set.
  Problem: the model confidence set, the superior predictive ability test, and StepM are named, and the paper reports none of their outputs.
  This bullet names three procedures and says they are for the comparison of predictor sets: the model confidence set, which returns the set of models not beaten at a stated confidence level; the superior predictive ability test, which returns a p-value against a benchmark; and StepM, which steps through the models and rejects some of them. No table in the paper states an MCS membership, an SPA p-value, or a StepM rejection. The next bullet says the main tables are point estimates and contain no bootstrap intervals, which is the kind of output those three procedures produce. The comparisons that are printed are Diebold–Mariano statistics on QLIKE and on squared error.

- **Reporting convention.** All main tables report point estimates accompanied by pairwise Diebold–Mariano statistics, t-statistics of mean returns with fixed-b critical values, or block-bootstrap p-values for Sharpe ratio differences, without bootstrap confidence intervals.

J Notes to the authors (not part of the paper)

- **Final rerun (23 October 2026): the pooled numbers in this draft are final** for the linear models (OLS, ridge, lasso, elastic net), LightGBM, XGBoost and the random forest (Intel's oneDAL engine), on the final design at the main window and the robustness window (Table 1, Sections 6 and 7, Table 7). The generators read the final runs under the switch REHAUL_FINAL_RUN=1 (writeup/final_run.py, which lists the regeneration order); a model whose final forecasts are missing prints *pending* and is left out of every count (writeup/tests/check_pending.py).
  - **Variable importance** (Section 6.5): the linear models' importance and the trees' SHAP attributions on the final design at the main window are in, and so is the trees' window learning curve (LightGBM, XGBoost and the forest at 1,000–3,500 days, 24 of 24 walks; the full set's extra gain over the HAR benchmark, Table 5, writeup/make_rehaul_fip_lc.py). Still to come: the ablations (below), the MinGRU, a linear SHAP on the same forecast rows (so both model families share one measure), and the trees' SHAP values at the robustness window.
  - **Ablations:** leave one bucket out, then one variable out within the buckets that matter. The linear preview on the earlier design exists; the final run on the final design is not run; trees to be decided.
  - **MinGRU network:** parked in the paper. Its final rerun is running (results/pooled_mingru_final_2014start_w3000 and _w3500, RULE_gru.txt with addenda 1–3: refit at every bar, tuned on the last pre-sample fold, grid extended while a pick sits at an edge, with no compute cap per James's "max tuning"). It enters when its walks are scored.
  - **Standardization** (STANDARDIZE_PENALTY, merged 2430543): the penalized

Table 14: Tests of strategy returns under alternative long-run variance estimators

			NW, $q_{\mathrm{NW}} = 5$		NW, $q_{\mathrm{NW}} = 12$		NW, $S = 33$, fixed-b		Boot.	LW
	$\bar{\Delta}$ (%)	ΔSR	t	p	t	p	t	p	p	p
Panel A: Pooled intraday model minus short-straddle benchmark										
Ridge, HAR	4.76	1.10	0.93	0.350	0.95	0.344	0.90	0.388	0.178	0.350
Ridge, full	10.85	1.96	2.08	0.037	2.02	0.043	1.89	0.076	0.004	0.034
Lasso, HAR	4.76	1.10	0.93	0.350	0.95	0.344	0.90	0.388	0.178	0.350
Lasso, full	10.50	1.84	2.04	0.041	2.02	0.043	1.97	0.066	0.010	0.040
Elastic net, HAR	4.76	1.10	0.93	0.350	0.95	0.344	0.90	0.388	0.178	0.350
Elastic net, full	11.47	2.06	2.22	0.026	2.12	0.034	1.94	0.069	0.003	0.026
LightGBM, HAR	2.50	0.73	0.48	0.634	0.52	0.606	0.53	0.610	0.347	0.614
LightGBM, full	9.48	1.87	1.83	0.067	1.83	0.068	1.69	0.112	0.004	0.055
XGBoost, HAR	4.96	1.02	0.92	0.356	1.02	0.308	1.08	0.304	0.189	0.315
XGBoost, full	8.63	1.78	1.66	0.096	1.64	0.101	1.51	0.153	0.006	0.090
Random forest, HAR	−0.03	0.13	−0.01	0.994	−0.01	0.993	−0.01	0.993	0.816	0.992
Random forest, full	11.67	2.07	2.29	0.022	2.35	0.019	2.17	0.043	0.002	0.019
Panel B: Pooled intraday model, full predictor set minus HAR benchmark										
Ridge	6.09	0.86	1.70	0.089	1.65	0.100	1.60	0.130	0.161	0.106
Lasso	5.74	0.74	1.59	0.111	1.56	0.119	1.59	0.132	0.227	0.119
Elastic net	6.71	0.96	1.91	0.056	1.77	0.076	1.65	0.119	0.135	0.077
LightGBM	6.98	1.14	2.09	0.036	2.01	0.045	1.92	0.072	0.051	0.040
XGBoost	3.67	0.76	1.10	0.270	1.10	0.272	1.05	0.318	0.141	0.269
Random forest	11.70	1.95	2.52	0.012	2.58	0.010	2.42	0.026	0.007	0.020
Panel C: Pooled intraday minus closing-interval model										
Ridge, HAR	−4.43	−0.49	−1.52	0.128	−1.55	0.121	−1.36	0.197	0.284	0.134
Ridge, full	2.57	0.37	0.85	0.398	0.89	0.373	0.92	0.380	0.342	0.395
Lasso, HAR	−4.43	−0.49	−1.52	0.128	−1.55	0.121	−1.36	0.197	0.284	0.134
Lasso, full	0.42	0.28	0.12	0.907	0.12	0.904	0.12	0.906	0.529	0.904
Elastic net, HAR	−4.43	−0.49	−1.52	0.128	−1.55	0.121	−1.36	0.197	0.284	0.134
Elastic net, full	0.23	0.26	0.07	0.941	0.07	0.941	0.07	0.945	0.514	0.941
LightGBM, HAR	1.10	0.29	0.26	0.792	0.27	0.790	0.26	0.804	0.666	0.813
LightGBM, full	2.55	0.56	0.59	0.552	0.60	0.551	0.60	0.567	0.387	0.580
XGBoost, HAR	1.59	0.15	0.64	0.522	0.67	0.502	0.67	0.519	0.700	0.519
XGBoost, full	2.78	0.58	0.67	0.503	0.69	0.488	0.68	0.512	0.289	0.532
Random forest, HAR	−1.01	−0.25	−0.40	0.690	−0.41	0.680	−0.46	0.659	0.531	0.678
Random forest, full	8.11	1.11	2.71	0.007	2.96	0.003	2.72	0.013	0.014	0.005
Panel D: Closing-interval model, full predictor set minus HAR benchmark										
Elastic net	2.05	0.21	0.78	0.436	0.78	0.437	0.74	0.476	0.607	0.450
Lasso	0.89	−0.03	0.35	0.727	0.34	0.733	0.35	0.739	0.942	0.737
Panel E: Closing-interval model minus short-straddle benchmark										
Elastic net, full	11.24	1.80	1.95	0.051	2.00	0.046	1.92	0.072	0.018	0.035
Panel F: Short-straddle benchmark, mean return										
Short straddle	2.43	–	0.58	0.559	0.62	0.536	0.67	0.522	–	–

Notes: Each row is a comparison whose statistical significance the text states, on the 614 post-warmup days of $\mathcal{T}_{\mathrm{opt}}$. $\bar{\Delta}$ is the mean daily return differential at mid-quotes in percent of the straddle premium (the mean return itself in the last panel), and ΔSR the difference of the annualized Sharpe ratios of Section 7.2. The t-statistics test $\bar{\Delta} = 0$ with the Newey–West long-run variance: at lag truncation $q_{\mathrm{NW}} = \lfloor 4(T/100)^{2/9} \rfloor = 5$ (Newey and West, 1994) and at $q_{\mathrm{NW}} = \lfloor 1.5\,T^{1/3} \rfloor = 12$, both with standard normal p-values, and at bandwidth $S = \lceil 1.3\,T^{1/2} \rceil = 33$ with fixed-b p-values (Lazarus et al., 2018; Kiefer and Vogelsang, 2005), $T = 614$. Boot. is the p-value of the paired block-bootstrap test of equal Sharpe ratios SR (Section 7.2; block length 10, 4,999 resamples), and LW that of the Ledoit and Wolf (2008) test of equal Sharpe ratios of the daily returns pooled over the T days (block length 10, 4,999 resamples). The text uses the fixed-b test for statements about mean returns and the bootstrap test for statements about Sharpe ratios. All p-values are two-sided.

pooled models are refit on columns standardized within each estimation window (Section 5.2).

  - **Window and sample** (decided 23 October 2026): first evaluation day 2 January 2014 (2012–13 leave the accuracy sample; the option-trading sample is unchanged), penalties and configurations tuned on the last pre-sample fold (validation year 2013), main window 3,000 trading days, 3,500 days as robustness. The learning curves: the linear models' closing-interval accuracy stops improving near 3,000 days and their every-interval accuracy does not improve beyond 1,000; the trees' pre-sample fits improve until about 3,500. Out of sample, 3,500 days against 3,000 change no tree's accuracy significantly and move only the linear models (Table 7). Below the main window the trees do improve: from 1,000 to 2,000 days the full set's every-bar QLIKE falls significantly in all three trees, and its lead over the HAR benchmark widens until 3,000 days (Table 5; 24 of 24 tree walks finished). The dense-and-weak tests across windows (the linear models' coefficients, the trees' extra gain) are in Section 6.5.
  - **Closing-interval models:** the linear models are not rerun (James, 22 October 2026); their full-sample rules and the data fixes they predate are disclosed (notes S4a, S4c, S4d, E28; Section 3). The closing-interval *trees* are being rerun with a refit at every trading session instead of every 10 (James, 23 October 2026, note E32; branch insurance-selling-2026-09-19-e34-e32-daily), hyperparameters kept per each specification's rule. Their numbers, the appendix's "every 10 trading days" and the E32 box change when that rerun lands.

- **Accuracy against the benchmark:** 0 of the 12 pooled intraday forecasts have lower QLIKE than the daily HAR benchmark, 0 significantly (Diebold–Mariano, 5% level); closing-interval tests are left out of this count (Appendix G).
- **Closing-interval forecasts, as fitted (disclosed, not a fix):**
  - the activity flags are computed after the forward fill, so the FOMC flags read 1 on every trade day after November 2023;
  - the VIX, volatility-demand and FOMC feeds end early and are carried forward without limit into 2024 (the pooled final design caps the fill and splices the VIX family instead).
- **Missing outputs:** paired-comparison tables (axis 1, axis 2), DM against the OLS in the master table, and the model confidence set over models (note E14).
- **Open decisions for James** (23 October 2026):
  - E21: the code counts n_{gap} in trading days, which reverses the 17 October reading (Appendix B); please confirm.
  - E2: the vendor of each file, the price field and the roll rule are not documented in the repository; the core file also carries stamps inside CME's daily halt. Needs the data provider.
  - The pooled interval-shift test (Section 7): which days, the noise-band formula under the Sharpe ratio's yearly weights, and how to read the earlier-interval pattern.
  - The random forest's storyline (RF_STORY_OPTIONS.md): at the main window it has the largest pooled Sharpe gain from the full set, against a near-zero Sharpe ratio for its HAR-only arm that is higher at the robustness window (Table 7).
  - The Sharpe ratio's yearly weights give the short final year (2024) a quarter of the weight; one sentence of disclosure, or a sensitivity check.
  - Still open from the notes: E13 (old-divisor DM statistics in the closing study), E14 (MCS / SPA / StepM named but not reported), E23's sensitivity check, E30 (sentiment

formula: the professor), E33 (MAE and calibration: compute or drop), and the reach of the late-start rule over the evaluation span.

- **The full outline,** with source and macro pointers, is `writeup/PAPER_REHAUL_OUTLINE_2026-10-15.md`.